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Disease Prediction and flare-ups in Rheumatoid Arthritis using Machine Learning

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SCHOOL OF SCIENCE & TECHNOLOGY

A thesis submitted for the degree of

Master of Science (MSc) in Data Science

JULY 2026

THESSALONIKI – GREECE



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Abstract

Background: Rheumatoid arthritis flares are episodic exacerbations that accelerate joint damage and frequently occur between visits, making clinical management reactive. Existing predictive models rely heavily on narrow clinical trial data, with none validated for Greek and Southern European populations. This study evaluates an interpretable machine learning (ML) framework to predict next-visit flare occurrences using real-world, multi-modal longitudinal outpatient data. Additionally, this study investigates direct comparison between task-specific supervised classifiers and large language model (LLM) zero-shot and few-shot inference, examining whether pretrained medical knowledge can substitute patient-specific learning.

Methods: A retrospective longitudinal dataset of 106 patients was analyzed from a dual-setting rheumatology cohort in Thessaloniki, Greece. Four tuned classifiers, Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost), were trained within a patient-aware stratified group nested cross-validation scheme to prevent leakage. Models were evaluated using average precision (AP) and Area Under the Receiver Operating Characteristic Curve (ROC-AUC). Concurrently, Claude Haiku was evaluated across cross-validation partitions using zero- and few-shot prompting paradigms across two data serialization formats, establishing a benchmark against the ML models.

Results: LR performed best (AP: 0.508, ROC-AUC: 0.695, Brier score: 0.201), outperforming baselines (AP: 0.331, ROC-AUC: 0.500). Comparative AP values were 0.470, 0.450, and 0.414 for SVM, XGBoost, and RF, respectively. At its F1-optimal threshold (0.295), LR achieved 83% sensitivity (108/130 flares) and ~48% specificity. Elastic Net regularization produced a sparse five-predictor equation dominated by Disease Activity Score 28 (odds ratio: 1.667 per standard deviation). LLM configurations consistently underperformed ML baselines, yielding APs of 0.40–0.42 and ROC-AUCs of 0.61–0.64. Notably, the impact of context exemplars was layout-dependent; few-shot prompting marginally improved performance for the structured List template (ROC-AUC from 0.63 to 0.64) but degraded performance for the narrative Text template (ROC-AUC from 0.62 to 0.61).

Conclusions: Real-world outpatient data demonstrates the feasibility of interpretable flare prediction, where a linear model outperformed ensemble techniques. Although LLMs did not match the discriminative performance of patient-specific classifiers in this evaluation, it achieved higher overall accuracy due to a conservative prediction strategy, offering potential utility as a training-free baseline in data-scarce environments. Constraints include sample size and label subjectivity, but learning-curve trends justify multi-center expansion and prospective validation.

Keywords: rheumatoid arthritis; flare prediction; machine learning; large language models; longitudinal clinical data

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List of Abbreviations

AI	Artificial Intelligence
Anti-CCP	Antibodies Against Cyclic Citrullinated Peptides
AP	Average Precision
bDMARD	Biological Disease Modifying Antirheumatic Drug
CDAI	Clinical Disease Activity Index
CRP	C-Reactive Protein
csDMARD	Conventional Synthetic Disease Modifying Antirheumatic Drug
CV	Cross-Validation
DAS28	Disease Activity Score 28
DMARD	Disease Modifying Antirheumatic Drug
EHR	Electronic Health Record
ESR	Erythrocyte Sedimentation Rate
EULAR	European League Against Rheumatism
FN	False Negative
FP	False Positive
FPR	False Positive Rate
GH	Global Health
KNN	k-Nearest Neighbors
LLM	Large Language Model
LR	Logistic Regression
MCP	Metacarpophalangeal
MICE	Multiple Imputation by Chained Equations
ML	Machine Learning
MTX	Methotrexate
PR	Precision-Recall
RA	Rheumatoid Arthritis
RBF	Radial Basis Function

RF	Random Forest
RF_a	Rheumatoid Factor
RNN	Recurrent Neural Network
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
RQ	Research Question
SHAP	Shapley Additive exPlanations
SJC	Swollen Joint Count
SVM	Support Vector Machines
TJC	Tender Joint Count
TN	True Negative
TP	True Positive
TPR	True Positive Rate
tsDMARD	Targeted Synthetic Disease Modifying Antirheumatic Drug
VAS	Visual Analogue Scale
XGBoost	eXtreme Gradient Boosting

1 Introduction

Rheumatoid arthritis (RA) is a chronic, systemic autoimmune disease characterized by persistent joint inflammation. It typically begins with the small joints of the hands and feet, progressing symmetrically to involve multiple joints, and may also affect extra-articular organs such as the skin, lungs, and heart [1]. RA affects about 0.5% to 1% worldwide, with women two to three times more affected than men and usually presents between the ages of 35 and 60 [2]. Clinical features include joint pain, morning stiffness, swelling, progressive cartilage and bone erosion, which can ultimately lead to irreversible deformities and disability [3]. The cause of RA remains unknown but arises between genetic susceptibility (e.g., HLA-DR alleles) and environmental factors such as smoking and poor dental health. RA is a dynamic disease that typically worsens over time. Early diagnosis and intervention are therefore critical, as up to 90% of patients can prevent or significantly slow joint damage when treatment is initiated promptly. Consequently, accurate prognosis is essential for preserving joint integrity and improving quality of life [2].

Among the most clinically significant manifestations of RA's fluctuating course is the disease flare which is a temporary but clinical meaningful exacerbation of disease activity, characterized by increased joint pain, swelling, and systemic inflammation, occurring against a background of previously controlled or stable disease. Flares are associated with accelerated joint damage, reduced quality of life, and increased healthcare utilization, and their unpredictable occurrence represents one of the principal challenges in long-term RA management [4]. Unlike gradual disease progression, flares often develop between scheduled clinical visits, meaning that by the time a patient is formally assessed, the window for early therapeutic intervention may have already narrowed.

Current clinical practice is mostly reactive. Because disease activity is only assessed every few months, clinicians are forced to respond to flares after they occur rather than preventing them. A predictive tool that uses routine clinical data to forecast flare risk at the next visit has the potential to shift RA management from reactive to proactive, enabling earlier interventions, targeted monitoring, and better allocation of hospital resources. However, predicting disease activity and flare-ups remain a major challenge because of the disease's inherent heterogeneity and the vast amount of data involved. These complexities

have highlighted the potential of Machine Learning (ML) which can manage high-dimensional datasets and uncover patterns that traditional statistical methods often miss. ML approaches can integrate omics, imaging, and clinical information to improve diagnostic accuracy, monitor disease activity and support flare prediction [5].

Despite this growing body of work, a critical gap remains between the predictive framework reported in the literature and their applicability to routine clinical settings. As discussed in detail in Chapter 2, existing high-performing models have been developed on demographically narrow cohorts from specific geographic regions, with a notable absence of validated frameworks for Greek and broader Southern European RA populations. This gap is clinically meaningful given that RA disease presentation, treatment patterns, and healthcare infrastructure vary across geographic regions, potentially limiting the direct transferability of models developed elsewhere. Furthermore, many existing approaches rely on highly controlled clinical trial data and specialized data sources that are not available in standard rheumatological practice, potentially restricting the real-world feasibility of these models in resource-constrained clinical environments.

The aim of this thesis is to address these limitations by developing and evaluating an interpretable ML framework capable of predicting RA flares at the next clinical visit, using routinely available clinical, laboratory, and treatment data, collected from a Greek rheumatology outpatient setting. A core objective of this analytical design is to stage a direct, comparative evaluation between custom-trained statistical ML models and foundation large language model (LLM) reasoning to systematically assess their respective performance, inherent trade-offs, and clinical utility. To achieve this, the study used a longitudinal dataset of 106 patients. The methodology encompassed preprocessing irregular real-world data, isolating a robust feature set via feature selection and developing four ML classifiers: Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost). To strictly mitigate the risk of data leakage and ensure a robust performance estimation, all hyperparameter tuning, imputation, and feature selection were conducted within a patient-aware Stratified Group K-fold nested cross-validation (CV) framework. Given the moderate imbalance of flare events, models were primarily evaluated using Average Precision (AP), supplemented by Area Under the Receiver Operating Characteristic Curve (ROC-AUC). Finally, this work directly extends the

benchmark evaluation to an LLM-based classification paradigm utilizing Anthropic's Claude Haiku, evaluated across the identical CV partitions to ensure a rigorous and direct comparison.

This aim is operationalized through three research questions (RQs):

- *RQ1*: Can routinely collected clinical, laboratory, and treatment variables from real-world rheumatology outpatient visits predict the occurrence of a disease flare at a patient's next clinical encounter?
- *RQ2*: Which ML classifier achieves the optimal balance of predictive performance and clinical interpretability for next-visit flare prediction in a small longitudinal cohort?
- *RQ3*: Does general medical knowledge encoded in a LLM during pretraining provide competitive flare prediction performance compared to task-specific supervised classifiers trained on patient-specific longitudinal data?

The remainder of this thesis is organized as follows. Chapter 2 provides the theoretical background of RA clinical management, reviews existing ML applications in rheumatology with a particular focus on flare prediction, and identifies the research gap this study addresses. Chapter 3 details the methodological framework, encompassing data acquisition, preprocessing, feature engineering, target variable definition, the ML pipeline (including CV strategy, imputation evaluation, model development, and hyperparameter optimization), and the structural implementation of the zero-shot and few-shot LLM inference pipelines. Chapter 4 presents empirical results, covering exploratory data analysis, feature selection outcomes, cross-model performance comparisons, and the interpretability analysis of the selected model. Chapter 5 concludes the thesis with a summary of principal findings, study limitations, and directions for future research.

2 Literature Review and Theoretical Background

This chapter establishes the clinical and computational foundations of the thesis. It is divided into three main sections. First, it outlines the medical background of RA, focusing on disease activity, biomarkers, and the clinical significance of flares. Second, it reviews the current landscape of ML applications in rheumatology. Finally, it identifies the specific limitations in the existing literature that this study aims to address.

2.1 Medical background of RA

2.1.1 Biomarkers and Serology

To effectively monitor and predict disease activity, clinicians use a combination of clinical measures and biomarkers, which act as biological fingerprints, providing measurable indicators of a patient's physiological state. In RA, serological assessment relies primarily on two key autoantibodies: Rheumatoid Factor (RF_a) and antibodies against cyclic citrullinated peptides (anti-CCP). Autoantibodies are immune proteins that mistakenly target the body's own cells and tissues, leading to autoimmune diseases. These markers are essential not only for diagnosis but also for classifying patients into distinct prognostic subsets.

RF_a is considered the hallmark of the disease and is detectable in approximately 60-80% of RA patients. However, its utility as a standalone biomarker is limited by its lack of specificity. RF_a can be present in other autoimmune disorders, such as Systemic Lupus Erythematosus, chronic infections, and even in a significant proportion of the healthy elderly population, which complicates its diagnostic precision [6]. In contrast, anti-CCP antibodies are directed against peptides containing citrulline, an amino acid generated by the post-translational modification of arginine residues (the procedure of citrullination). The presence of these antibodies defines a distinct clinical phenotype of RA with specific genetic and environmental risk factors. While exhibiting a sensitivity comparable to RF_a, anti-CCP demonstrates a superior specificity ranging from 96% to 98% [6], [7].

For predictive modeling and clinical management, anti-CCP is considered the more informative biomarker. Importantly, these antibodies can be detected in serum years before the onset of clinical symptoms, aiding early diagnosis. Furthermore, anti-CCP positivity is

strongly associated with a more severe disease course, predicting higher rates of radiographic progression and erosive joint damage compared to seronegative patients [6]. Consequently, anti-CCP status serves as a critical predictive feature for identifying high-risk patients who may require early aggressive therapy to prevent functional disability.

Beyond autoantibodies, monitoring disease activity requires markers of systemic inflammation. The Erythrocyte Sedimentation Rate (ESR) and C-Reactive Protein (CRP) are the most widely used acute-phase reactants in clinical practice, reflecting the patient's current inflammatory status. ESR measures the speed at which red blood cells settle in plasma. This process depends heavily on acute-phase proteins, especially fibrinogen, but is also shaped by red blood cell characteristics and other plasma components. However, it changes slowly and can be affected by factors such as age, gender, and anemia.

In contrast, CRP is a direct measure of inflammation produced by the liver, primarily in response to interleukin-6 (IL-6) secretion. It is more sensitive to short-term changes, rising and falling rapidly with the onset and resolution of inflammation, and is generally unaffected by the confounding factors that limit ESR. While both markers are integrated into disease activity scores like Disease Activity Score 28 (DAS28), discussed thoroughly in *Section 2.1.2*, CRP is often preferred for assessing immediate therapeutic response due to its dynamic nature and higher specificity for acute inflammation [8].

2.1.2 Assessing Disease Activity

Given the inherent heterogeneity of RA, disease activity cannot be adequately captured by a single clinical or laboratory variable. To address this complexity, the Disease Activity Score was introduced in the early 1990s. However, it required an extensive assessment of 44 joints, making it difficult for routine clinical practice. To improve feasibility without compromising validity, the score was simplified to the DAS28, which focuses on a reduced set of 28 distinct joints (bilateral shoulders, elbows, wrists, metacarpophalangeal (MCP) joints, proximal interphalangeal joints and knees). Although the DAS28 has since become the globally accepted standard for monitoring RA activity and guiding treatment response, it is important to recognize that it complements and not replaces a thorough patient history and physical

examination [9]. The standard formula using ESR lab test (DAS28-ESR) is calculated as follows in Equation 2.1:

$$DAS28_{ESR} = 0.56 \cdot \sqrt{TJC28} + 0.28 \cdot \sqrt{SJC28} + 0.70 \cdot ln(ESR) + 0.014 \cdot GH \quad (1)$$

where:

- TJC28 is the Tender Joint Count (range 0–28), which represents the patient's subjective pain response upon palpation.
- SJC28 is the Swollen Joint Count (range 0–28), which represents objective soft tissue swelling (synovitis).
- ESR is the acute-phase reactant measured in mm/h.
- GH is global health, a subjective assessment of disease activity measured on a 100 mm Visual Analogue Scale (VAS), where 0 represents "no pain/best health" and 100 represents "worst pain/ worst health".

Similarly, for the DAS28-CRP formula, CRP is measured in mg/L and replaces ESR as the acute phase marker (Equation 2.2):

$$DAS28_{CRP} = 0.56 \cdot \sqrt{TJC28} + 0.28 \cdot \sqrt{SJC28} + 0.36 \cdot ln(CRP + 1) + 0.014 \cdot GH + 0.96 \quad (2)$$

Alternative formulas utilizing only three variables (excluding GH) are also available where patient-reported outcomes may be missing. Based on the DAS28 score, the European League Against Rheumatism (EULAR) has defined specific cut-off points to classify disease states [10], as depicted in Table 1:

Table 1: EULAR disease activity classification.

State	DAS28 range
Remission	DAS28 < 2.6
Low Disease Activity	2.6 ≤ DAS28 ≤ 3.2
Moderate Disease Activity	3.2 < DAS28 ≤ 5.1
High Disease Activity	DAS28 > 5.1

Although DAS28 remains one of the most used metrics in RA, alternative indices have been developed to simplify assessment. For example, the Clinical Disease Activity Index (CDAI) is widely used and is calculated as the sum of the 28-joint tender and SJCs, both patient and physician global assessment, yielding a score from 0 to 72. Unlike DAS28, CDAI does not require markers such as CRP and ESR. Scores are categorized into remission (≤ 2.8), low (> 2.8-10), moderate ($>$ 10-22) and high ($>$ 22) disease activity, and can be further dichotomized into controlled (≤ 10) and uncontrolled (> 10) states, making CDAI a practical measure for predictive modeling when laboratory data are not available [11].

2.1.3 Defining Disease Flares and Remission

In RA, a flare refers to a sudden increase in disease activity that causes significant pain and functional impairment. Flares often occur when physicians reduce the dosage of treatments such as Disease Modifying Antirheumatic Drugs (DMARDs) after remission is achieved, as these therapies can be costly and increase the risk of infections. Although flare-ups can usually be controlled by increasing the medication dose, this leads to more hospital visits, greater treatment burden, and a reduced quality of life for patients [4].

To manage the disease effectively, a Treat-to-Target strategy aims to achieve and maintain clinical remission, which is a very low level of disease activity. According to EULAR criteria, remission is defined as a DAS28 score < 2.6 [10]. However, this definition can be subjective, as DAS28 has components such as patient's GH and TJC, making it challenging for data analysis. Moreover, increase in DAS28 does not always reflect true inflammation, as conditions such as fibromyalgia, depression, structural joint damage can elevate the subjective factor in the score. This discordance introduces label noise, and may lead to patient's misclassification and false interpretation [9].

In contrast to remission, there is no 'gold standard' for defining a disease flare in RA. Flare definitions vary substantially across studies, reflecting differences in clinical priorities and tolerance for subjectivity. One approach, particularly in studies of drug tapering, defines flares using absolute DAS28-CRP thresholds. For example, flare may be defined as a loss of low disease activity (DAS28-CRP ≥ 3.2) at any visit, a sustained increase in DAS28-CRP across consecutive visits, or the need for therapeutic escalation. While these criteria are quite

straightforward, they depend heavily on DAS28 scores and remain vulnerable to inflation by subjective symptoms [12].

Another proposal is to impose stricter objective constraints by requiring evidence of physical disease worsening. In that case, a flare is characterized by a clinically significant increase in DAS28 (≥ 1.2) accompanied by a rise in SJC, often alongside a minimum absolute DAS28 threshold. This approach acknowledges that changes in swollen joints represent a more objective marker than tender joints or patient-reported outcomes alone. Some of these criteria have proposed the addition of treatment escalation as another criterion, based on the clinician's assessment of the degree of worsening of the disease. While these criteria improve the specificity for inflammatory flares, they may reduce sensitivity by excluding milder or early disease worsening that has not yet resulted in swollen joints or treatment changes [4].

Accurate classification of remission or flare is vital in making decisions regarding treatment. A confirmed flare will prompt treatment intensification, while remission will allow drug tapering. Knowing how treatment options are structured and work is essential, since medication changes are central to both prediction models and the clinical intervention studied.

2.1.4 Therapeutic Management

The therapeutic management of RA has evolved significantly over the last decades. The cornerstone of modern therapy is DMARDs. Unlike nonsteroidal anti-inflammatory drugs or corticosteroids, which provide temporary symptomatic relief, DMARDs are designed to target the immune system, reduce inflammation, and prevent radiographic joint damage [1].

Current guidelines [13] describe a line of therapy strategy to align with the Treat-to-Target principle. First line is the conventional synthetic DMARDs (csDMARDs) which should be started, once the diagnosis of RA has been made [14]. The most used drug of this category is Methotrexate (MTX), universally recognized as the 'anchor drug'. It is cost effective and has high efficacy. Other agents such as Leflunomide or Sulfasalazine are recommended. Because csDMARDs are slow-acting, glucocorticoids (for short-term treatment) are often used as the 'bridge' to suppress inflammation faster, but they should be tapered as quickly as possible to avoid long-term toxicity.

If the treatment target is not achieved, typically within 3 to 6 months and poor prognostic factors are present, the therapy moves on to the second line of therapy which includes biological DMARDs (bDMARDs) and targeted synthetic DMARDs (tsDMARDs). Firstly, bDMARDs are highly specific, genetically engineered proteins that target specific cytokines or cell surface molecules. The most used are Tumor Necrosis Factor (TNF) inhibitors such as Adalimumab, Etanercept, Infliximab. TsDMARDs are a newer class of drugs, known as Janus Kinase (JAK) inhibitors. Examples are Tofacitinib, and Baricitinib, often combined with MTX [2]. While highly effective, recent guidelines suggest using them with caution, particularly with patients with other comorbidities, such as cardiovascular risks.

Consequently, these therapies have revolutionized patient outcomes, but still the selection of the right drug remains a challenge. There is no definitive way to predict which medication is the optimal for each patient, emphasizing the heterogeneity of the disease. The treatment decision often relies on a 'trial and error' approach (as in ML), cycling through different biological agents until a response is observed. This delay can prolong disease activity and cause a financial burden on healthcare systems due to excessive cost of these therapies.

2.2 Application of ML in the Clinical Management of RA

Advancing the management of RA requires a comprehensive approach that integrates early detection with the precise forecasting of disease behavior. Due to high heterogeneity of the condition, universal treatment is not often sufficient, leading to a growing reliance on objective tools. Therefore, recent literature has focused on leveraging ML to enhance the clinical pipeline, starting from diagnostic studies and progressing to the longitudinal prediction of disease activities and inflammatory flares. This section explores these methodologies, highlighting the shift from traditional methods to advanced ML frameworks. An overview of the literature is provided in Table 2:.

2.2.1 Disease Diagnosis

ML methods are more employed to enhance diagnostic accuracy by identifying complex, multidimensional patterns inside molecular and phenotypic data. For example, Luan et al.

[15] developed a ML multivariate classification model integrating serum metabolomic and lipidomic profiles, finding diagnostic biomarkers capable of distinguishing both seropositive and seronegative RA patients with over 95% ROC-AUC. Extending beyond molecular profiling, Alarcón-Paredes et al. [16] introduced a computer aided diagnosis screening framework that combined hand thermal, RGB (Red, Green, Blue) imaging and grip force measurements, showing the adaptability of Artificial Intelligence (AI) in leveraging multimodal physical assessments for RA detection.

Furthermore, deep learning algorithms are utilized in RA diagnosis. Wu et al. [17] implemented a Dense-Net based architecture to analyze ultrasound images, targeting to detect synovial proliferation in the MCP joints and to classify joint conditions, achieving ROC-AUC values closely to 90%. Complementing this analysis, deep learning has also showed good performance in assessing structural joint damage on conventional radiographs. Üreten et al. [18] employed a YOLOv4 object detection framework in combination with a pre trained VGG-16 network to classify hand radiographs into RA, osteoarthritis, or normal, scoring an overall accuracy of 90.7%.

Together, these papers underscore the growing integration of AI-driven methodologies across molecular, physical, and radiographic domains, giving space for the development of predictive models in RA management.

2.2.2 Disease Activity Prediction

Historically, prediction of RA progression relied on statistical approaches such as Linear Regression. These methods were instrumental in identifying population-level risk factors, but they exhibit limitations in personalized forecasting. Traditional statistical models are hypothesis-driven and depend on assumptions such as linearity, normality, and independence, which are rare in complex clinical cases. As a result, these methods often struggle to capture non-linear relationships among biomarkers, treatment modifications, and patient's characteristics. In response to these limitations, ML approaches are data-driven and optimized for predictive performance. By modeling high-dimensional data, ML methods aim to improve generalization and enable more accurate individualized predictions in multifactorial diseases such as RA [19].

A foundational study in this transition was conducted by Norgeot et al. [11], in which they developed a Recurrent Neural Network (RNN), a deep learning model, to forecast disease activity at the next clinical visit using longitudinal Electronic Health Record (EHR) data. The disease state was defined using the CDAI and predicted if it would be controlled or uncontrolled at the next visit. In the primary cohort (n=578) coming from a University Hospital in San Francisco, the model achieved an ROC-AUC of 0.91, outperforming probabilistic baseline classifiers (ROC-AUC ~ 0.54). External validation in a separate public safety-net hospital cohort (n=242), achieved an ROC-AUC of 0.74 indicating a moderate generalizability. Regarding feature importance, the study identified that the combination of disease activity history, laboratory values (ESR, CRP), and medication data (corticosteroids) are the strongest predictors. While this study demonstrated the possibility of using deep learning algorithms into small RA datasets, its reliance on binary CDAI outcomes and EHR-derived scores may limit granularity and increase label noise.

Kalweit et al. [20] further advanced the application of deep learning in this domain by introducing AdaptiveNet, a RNN designed to handle irregular time intervals and missing values in longitudinal clinical data. This study used the Swiss Clinical Quality Management Registry with over 9500 patients and 65.000 visits. The objective was to predict DAS28-ESR both as a binary classification task (using a threshold of active disease > 2.6) and as a continuous outcome (regression). In the classification task, AdaptiveNet achieved an accuracy of 75.6% and an ROC-AUC of 0.72, slightly outperforming classical ML models. However, RF performed comparably (accuracy:75%, ROC-AUC:0.71), proving that ensemble methods remain competitive for structured clinical data. In contrast, in the regression analysis, deep learning method showed a clearer advantage, achieving a Mean Squared Error of 0.9, outperforming Linear Regression and SVM. Feature importance analysis revealed that painful joint count, disease duration, and age were the strongest predictors of disease activity, whereas medication history contributed minimally, contradicting the findings of Norgeot et al. [11]. Despite overall performance, low scores of specificity were observed for all models, highlighting the challenge of identifying remission states in RA.

More recently, Alsaber et al. [21] concluded a comprehensive comparative analysis of multiple ML algorithms (Lasso, Ridge, SVM, RF, and XGBoost) using longitudinal data

coming from the Kuwait Registry for Rheumatic Diseases to predict remission in patients receiving bDMARDs at a 1-year follow-up. The 'black box' limitation of complex systems was addressed via the application of Explainable AI techniques. Tree based ensemble methods demonstrated superior performance, with RF and XGBoost consistently outperforming linear models and SVM, achieving high sensitivity (up to 95%) and accuracy across treatment subgroups. Using Shapley Additive exPlanations (SHAP) values, the authors revealed that patient reported outcomes for pain and patient global assessment, alongside biomarkers such as CRP, were the most influential predictors of remission. The findings highlight a broader shift in the literature toward interpretable ML approaches, emphasizing that transparency and clinical explainability are as important as predictive performance for real-world scenarios.

Collectively, these studies demonstrate the effectiveness of ML predicting overall disease trajectories, particularly remission versus active disease. However, forecasting acute flare represents a distinct and clinically critical challenge, with unique definitional and modeling complexities discussed in the following section.

2.2.3 Inflammatory Flares Prediction

Vodencarevic et al [22] addressed the challenge of predicting individual flare probability in RA patients in sustained remission who were reducing their bDMARD dosage (tapering). The dataset was derived from the Reduction of Therapy in patients with RA in Ongoing remission (RETRO) controlled trial [23] (n=41, 135 visits), and the study aimed to support clinical decision-making by estimating the risk of worsening (flares), which was defined by the threshold DAS28 $\geq$ 2.6, within a 14-week window. To maximize predictive accuracy, the authors employed a nested CV framework to evaluate four classifiers (LR, K-nearest Neighbors (KNN), Naïve Bayes, and RF) and integrate them into a Stacking Meta Classifier, an ensemble technique that achieved superior performance with an ROC-AUC of 0.81. Feature importance analysis identified dose percentage as the most important factor, following by DAS28, disease duration, CRP/ESR. This study highlights the advantage of using standardized clinical trial data over "messy" EHRs to train robust models. However, the authors noted significant limitations, including the small sample size which constrained

model generalizability, and a high sensitivity to missing data, where performance dropped sharply with even minor data unavailability.

Complementing the clinical modeling approach, O'Neil et al. [24] investigated the biological heterogeneity [21] of the same RETRO trial dataset (n=130 patients) to predict flares using serum proteomics (analysis of all the proteins in a cell or a tissue). The authors quantified 1307 serum proteins using the SOMAscan platform and compared the performance of a linear model (LASSO) against a nonlinear one (XGBoost). Consistent with findings of Alsaber et al. [21] regarding the superiority of tree-based models, XGBoost significantly outperformed LASSO (AUC=0.80). Feature importance analysis via SHAP values identified inflammatory markers such as IL-17F, as key predictors of future flare. This study demonstrates that while single biomarkers lack predictive power, ML can integrate complex proteomic signals to identify patients at risk of relapse.

A further step toward medical implementation was demonstrated with the AI-based Clinical Decision Support System (CDSS) Reuma Care Manager developed to predict short-term flare risk within a treat-to-target framework [25]. The training was done in longitudinal data incorporating ten variables (e.g., DAS28-ESR, CRP, joint counts), and the model achieved an ROC-AUC of 0.80. The system integrated SHAP-based explainability and was evaluated in a pilot study in which physicians assessed cases with and without AI support. When using the tool, there were less discrepancies between physicians and the model with a slightly increased confidence, while usability was rated high. However, acceptance remained moderate due to concerns regarding transparency and overreliance. This work illustrates the progression from predictive modeling toward clinically integrated, interpretable AI systems in RA care.

Finally, a rapidly emerging trend in flare prediction involves the use of data generated by the patients, through wearables and smartphones to capture continuous disease fluctuations between clinical visits. Gossec et al. [26] utilized physical activity monitoring in 155 patients (82 with RA and 73 with Axial Spondyloarthritis), applying a Naïve Bayes classifier to minute-level step data and resulted in 96% sensitivity for weekly flare detection. However, the method depended heavily on device adherence and could miss flares that do not affect mobility. Complementing this with subjective data, Gandrup et al. [27] used daily symptom questionnaires via the Remote Monitoring of RA (REMORA) smartphone app [28].

Comparing multiple algorithms, they found that Regularized LR (AUC 0.82) outperformed RF in this small cohort (n=20), indicating the limitation of ensemble methods in very small sample sizes with noisy self-reported data.

Collectively, these studies highlight future predictive frameworks will require a hybrid model, merging continuous sensor-derived data with patient-reported outcomes, to enable 'just-in-time' interventions.

***Table 2**: Summary of ML approaches for RA diagnosis and prediction.*

Reference	Algorithms applied	No. of data	Type of the primary data	Objective	Key findings	Limitations
Luan et al. *(2021)* [15]	Binary LR, RF, SVM, PLS-DA for feature selection	n = 325 total (225 RA patients, 100 healthy controls)	Serum metabolomic & lipidomic profiles + clinical markers (Affiliated Hospital of Nanjing University of Chinese Medicine)	Identify serum biomarkers for seropositive and seronegative RA and develop ML diagnostic model	26 metabolites/lipids discriminated RA vs. controls LR: 90.2% accuracy (89.7% sensitivity) Successfully classified seronegative RA Markers correlated with disease activity	Single-center cohort Discovery-stage (associative) Not tested against other inflammatory arthritides Limited generalizability
Alarcón-Paredes et al. *(2021)* [16]	RF, Multilayer Perceptron, LR, k-NN; wrapper forward feature selection	n = 278 women (240 main dataset: 120 RA, 120 controls; 38 external test set)	Hand thermal and RGB images + grip force (digital dynamometer) + age	Develop a low-cost computer aided diagnosis screening tool for early RA detection in women using thermal/RGB imaging and grip force	RF outperformed other models RGB: 94.5% accuracy (AUC 0.962) Thermal: 90% (AUC 0.954) External validation: 94.7% accuracy (36/38 correctly classified) Grip force and age were key predictors	Female-only cohort Single-region dataset No comparison with gold-standard imaging Did not explore deep learning approaches

Reference	Algorithms applied	No. of data	Type of the primary data	Objective	Key findings	Limitations
Wu et al. *(2022)* [17]	DenseNet-121 with transfer learning and data augmentation	n = 208 RA patients (1,337 ultrasound images)	Grayscale ultrasound images of MCP joints using the OMERACT-EULAR Synovitis Scoring system	Evaluate automated deep-learning classification of RA MCP joint conditions on 2D grayscale ultrasound	Scenario 1 (absence vs. presence of synovial proliferation): ROC-AUC 0.886, 82.1% accuracy Scenario 2 (healthy vs. diseased): ROC-AUC 0.916, 80.4% accuracy Highest performance using ROI with annotated masks	Small dataset Limited to binary classification Excluded Doppler ultrasound data
Norgeot et al. (2019) [11]	Deep Learning (RNN)	n= 578 patients (Internal) n=242 patients (External)	EHR University of California, San Francisco (internal) Zuckerberg San Francisco General Hospital (external)	Predict disease activity (CDAI) on the next visit (controlled vs uncontrolled)	AUC 0.91 (Internal), 0.74 (External) Proved Deep Learning feasibility on longitudinal EHR data	Limited transferability to different hospital settings Lack of granularity due to binary classification outcomes
Kalweit et al. (2021) [20]	AdaptiveNet (RNN), RF, SVM, Linear Regression	n>9500 patients, >65000 visits	Clinical Registry (Swiss Clinical Quality Management)	Predict DAS28-ESR score (regression) and active disease defined as DAS28 > 2.6 (Classification)	Classification: AdaptiveNet ROC-AUC 0.73, accuracy 75.6% Regression: Mean Squared Error: 0.90 (outperformed classical ML) Top predictors: Painful joints, disease duration, age	Low specificity (61.5%) in classification Retrospective registry data with missing values Lack of external validation

Reference	Algorithms applied	No. of data	Type of the primary data	Objective	Key findings	Limitations
Alsaber et al. (2024) [21]	XGBoost, RF, SVM, Lasso, Ridge, SHAP (for feature importance)	n=1968 patients, 11.195 follow-up visits	Clinical Registry (Kuwait Registry for Rheumatic Diseases)	Predict remission at 1-year follow-up	RF & XGBoost showed high sensitivity (~95%) and accuracy SHAP identified patient's assessment & CRP as top predictors	Retrospective design (human data errors, missing records) Lack of standardized data representation No genetic or physical activity features
Vodencarevic et al. (2021) [22]	LR, KNN, Naïve Bayes, RF, Stacking Meta-classifier (Ensemble)	41 patients, 135 visits (31 flare events)	Clinical Trial (RETRO)	Predict individual RA flare risk (DAS-28 ESR ≥ 2.6) within 14 weeks for patients tapering bDMARDs	Stacking Meta-classifier best (AUC 0.81) Top predictors: % dose change of bDMARDs, DAS-28 ESR, ESR, disease duration, CRP, remission duration	Small sample size High sensitivity to missing data (performance dropped) Imperfect calibration (overestimates flare risk at lower levels)
O'Neil et al. (2021) [24]	XGBoost, LASSO, SHAP	n=130 patients	Clinical trial data (RETRO) Serum Proteomics (SOMAscan)	Identify biological markers of remission and predict relapse risk in RA patients undergoing therapy withdrawal	XGBoost (AUC 0.80) significantly outperformed LASSO (0.58) Key markers for flares: IL-17F, MYC	No external validation Only baseline samples used 1,307 proteins represent a subset of human proteome

Reference	Algorithms applied	No. of data	Type of the primary data	Objective	Key findings	Limitations
Labinsky et al. (2023) [25]	ML with SHAP	258 patients/557 visits (73 routine + 40 RETRO, oversampled); Pilot: 50 patients, 10 vignettes, 5 physicians	CDSS study, RETRO	Evaluate accuracy, usability, and physician acceptance of AI-powered CDSS for RA flare prediction and its impact on tapering decisions	ROC-AUC 0.80, NPV 93% CDSS improved physician-AI agreement and decision confidence Tapering recommendations increased from 20% to 50%	Small pilot sample (5 physicians, 10 vignettes) evaluated on simulated cases Lacked imaging/comorbidity data Potential anchoring bias from viewing AI probabilities
Gossec et al. (2019) [26]	Naïve Bayes	n=155 patients (82 RA, 73 Axial Spondyloarthritis), 1,339 weekly flare assessments and 224,952 hours of physical activity assessments	Continuous physical activity data (steps per minute) recorded via wearables (Withings Activité Pop)	Detect flares using physical activity patterns	Sensitivity 96%, specificity 97% Best with 1-hour aggregation Flares linked to decreased activity intensity	Single-region cohort No external validation Confounders (mood, weather, illness) not accounted for Models complex for clinicians
Gandrup et al. (2024) [27]	LR, RF, Naïve Bayes	n=20 patients (1,325 daily reports and 213 weekly reports over 3 months)	REMORA Smartphone App (self-reported symptoms)	Predict flares using daily self-reported symptoms	AUC 0.82 (Regularized LR) Simpler models performed better than RF on small, noisy data	Very small sample High noise in self-reports No clinical/lab data (CRP, ESR) for correlation

2.3 Limitations and Research Gap

Despite the accelerating integration of ML within rheumatology, a critical evaluation of the existing literature reveals methodological and demographic, and technological limitations that restrict the real-world clinical deployment of current predictive frameworks.

A consistent geographic and demographic bias characterizes the foundational datasets used for longitudinal modeling. High-performing predictive frameworks have been predominantly trained and validated on distinct populations, such as in United States [11], and Kuwait [21]. Therefore, there is a notable absence of predictive longitudinal modeling applied to Greek and Southern European demographics. Because RA outcomes are influenced by genetics and environmental factors [2], the lack of localized data limits the broader generalizability of current global models.

A second limitation concerns the nature of the data on which existing models have been trained. A substantial proportion of the reviewed studies relied heavily on highly controlled clinical trial datasets or specialized data streams (e.g., comprehensive serum proteomics, specialized imaging, or wearable physical activity trackers), that are not routinely available in usual rheumatological practice [22], [24], [26], [27]. While such data sources offer standardized and complete records, they do not reflect the irregular visit intervals, missing laboratory values, and heterogeneous documentation characteristic of routine clinical care.

A third limitation, which represents an open area of investigation within the existing RA literature, concerns the comparative evaluation of task-specific ML approaches against LLM reasoning. The surge of LLMs in clinical workflows has sparked interest in whether their pretrained medical knowledge can replace task-specific supervised learning. To date, the utility of LLM-based zero-shot or few-shot inference for RA flare prediction has not been thoroughly documented, leaving an opportunity to systematically evaluate the distinct strengths of patient-specific longitudinal training alongside the generalized clinical reasoning offered by modern language models.

This study is designed to address all three limitations simultaneously. It is developed and evaluated on a longitudinal cohort derived from routine rheumatological outpatient visits and uses extensively standard clinical variables available in everyday practice including

DAS28 scores, routine inflammatory markers, treatment records, and patient demographics. The resulting pipeline is designed to remain scalable and applicable in resource-constrained environments. Additionally, this study extends the comparative evaluation to include LLM-based zero-shot and few-shot classification, providing a comparison between ML classifiers trained on patient data and general-purpose LLM reasoning for RA flare prediction. The following chapter details the comprehensive analytical framework designed to achieve these objectives, encompassing the data acquisition process, longitudinal preprocessing strategies, feature selection methodology, ML pipelines, and LLM classification approach employed to model individual disease trajectories.

3 Methodology and Data Description

This is a retrospective and observational study using longitudinal clinical data from patients with RA. The dataset consists of routinely collected information from clinical follow-up visits, including demographic characteristics, serological markers, laboratory tests, disease activity indicators, treatment-related variables, recorded comorbidities, and physician-recorded flare status. After preprocessing, each observation corresponded to a patient visit, while the prediction target represented whether the same patient experienced a flare at the next recorded visit. The main objective of this study is to formulate RA flare prediction as a supervised binary classification problem and evaluate whether routinely available clinical data can be used to predict next-visit flare occurrence. In this case, the methodology focused on structuring the longitudinal dataset, defining the prediction target, engineering clinically meaningful features, and comparing ML algorithms.

3.1 Data Source and Dataset Structure

The dataset used in this study consists of pseudonymized clinical records from 106 patients diagnosed with RA. Pseudonymization refers to the removal of direct identifying information and the assignment of a unique code to each patient, allowing records to be analyzed without revealing patient identity. Only patients with more than one recorded visit were included, as the study required longitudinal information to examine changes over time and define next-visit flare occurrence. The clinical data for this retrospective study was collected from a dual-setting outpatient cohort in Thessaloniki, Greece. To ensure a diverse, representative, and generalizable sample of routine clinical care, the dataset aggregates patient records from two distinct clinical environments: the Rheumatology Department of the Ippokrateio General Hospital and an affiliated private rheumatology outpatient practice directed by Dr. Theodoros Dimitroulas, Professor of Rheumatology at the Aristotle University of Thessaloniki and clinical faculty member at the hospital.

The data was recorded in Microsoft Excel and organized into two complementary components: patient-level information and visit-level information. For several categorical variables, predefined dropdown lists were used to reduce inconsistencies caused by spelling

differences and free-text variation. Also, date fields were stored in a standardized format to reduce inconsistencies during data entry. Patient-level variables describe characteristics that are stable or recorded once for each patient, while visit-level variables describe clinical status, laboratory findings, treatment information, and flare status at each recorded follow-up visit. This structure allows the dataset to capture both baseline patient characteristics and changes over time, which is essential for formulating the prediction of flare occurrence at the next clinical visit.

3.1.1 Patient-level Variables

Patient-level variables included information that was recorded once for each patient and remained stable across the follow-up visits. These variables contained demographic information such as gender and date of birth, diagnostic information, such as the year of RA diagnosis, serological characteristics, including patient's RF_a and anti-CCP status, and lifestyle related variables such as smoking status and recorded comorbidities. Date of birth was used only during preprocessing to derive the patient's age at each clinical visit and was not retained in the final modeling dataset, to reduce the use of directly identifiable information.

The year of diagnosis was used to estimate the disease duration over time, while RF and anti-CCP status were included as they are commonly used in the clinical characterization of RA. Smoking status and comorbidities were recorded to capture additional lifestyle-related and clinical factors that may influence disease activity, laboratory findings, treatment decisions, or the interpretation of flare events. The patient-level variables included in the dataset are summarized in Table 3.

***Table 3**: Patient-level variables included in the RA dataset.*

Variable	Description	Values / Format
patient_id	Unique pseudonymized patient identifier	Unique code, e.g., RA001
date_of_birth	Patient date of birth	Date format: DD/MM/YYYY
gender	Patient gender	Female / Male
year_of_diagnosis	Year of first RA diagnosis	Year format: YYYY

smoking	Smoking status	Current / Never / Ex-smoker
rf_status	Rheumatoid factor status	Negative / Positive
anti_ccp_status	Anti-cyclic citrullinated peptide antibody status	Negative / Positive
comorbidities	Recorded coexisting conditions	Free text, e.g., hypertension, dyslipidemia

3.1.2 Visit-level Variables

Visit-level variables included information recorded at each clinical visit and reflected the patient's clinical status, laboratory findings, disease activity, treatment decision, and flare occurrence at the specific time point. Each visit was linked to the corresponding patient through the unique patient id and included the visit date and type, allowing the longitudinal ordering of observations for each patient. Clinical variables recorded at the visit level included the TJC28, SJC28, and VAS. Laboratory variables comprised CRP, the corresponding upper limit of normal for CRP, and ESR. The CRP upper limit was recorded because reference ranges may differ between labs and was later used to normalize CRP values.

Treatment-related variables included the names of csDMARDs, MTX use and dose, the names of bDMARDs and tsDMARDs, and corticosteroid dose. Additional variables described the physician's treatment decision and, where applicable, the reason for treatment switching. These fields were left empty for baseline visits, since no previous treatment decision or switch had yet been recorded for the patient at the first observed visit. Finally, a flare indicator variable captured the physician's clinical assessment of flare presence at each visit. A summary of the visit-level variables is presented in Table 4:

***Table 4**: Visit-level variables included in the RA dataset.*

Variable	Description	Values / Format
patient_id	Unique pseudonymized patient identifier	Unique code, e.g., RA001
visit_date	Date of clinical visit	Date format, DD/MM/YYYY
tjc_28	28 tender joint count	Numeric count: 0-28

sjc_28	28 swollen joint count	Numeric count: 0-28
crp	C-reactive protein value	Numeric value, mg/L
crp_upper_limit	Laboratory-specific upper normal limit for CRP	Numeric value, mg/L
esr	Erythrocyte sedimentation rate	Numeric value, mm/h
vas	Patient visual analogue scale	Numeric score: 0-100
das28_score	Disease Activity Score based on 28 joints	Numeric score
csdmard_name	Conventional synthetic DMARD recorded at visit	Predefined drug list / None
mtx_use	Methotrexate use	Yes / No
mtx_dose	Methotrexate dose	Numeric dose, mg/week
b_ts_dmard_name	Biological or targeted synthetic DMARD recorded at visit	Predefined drug list / None
steroid_dose	Recorded steroid dose	Numeric dose, mg/day
treatment_decision	Clinical treatment decision at visit	Escalation / Tapering / Stable / Switch / Stop
switch_reason	Reason for treatment switch	Inefficacy / Remission / Adverse event / Patient preference / Pregnancy / None
flare_event	Physician-recorded flare status	Yes / No

Biological and targeted synthetic DMARDs were recorded in a combined variable (*b_ts_dmard_name*), because tsDMARDs use was limited in the dataset.

3.2 Data Preprocessing and Feature Engineering

3.2.1 Data Integration and Cleaning

The initial dataset was maintained in Microsoft Excel and divided into two distinct sheets. One sheet, labeled Patient_Static, contained information specific to each patient, while the other, Visits_Longitudinal, included details pertaining to individual clinical visits. To unify these data sources, the sheets were merged based on a pseudonymized patient identifier,

which was present in both sheets as a common column. The resulting dataset was organized around visit-level records, meaning that each row in the combined dataset represented a single clinical visit. To enrich the visit-level data, patient-level variables were linked to each visit, thereby providing a comprehensive profile of the patient for every recorded visit.

After data integration, basic cleaning steps were applied to improve consistency and prepare the dataset for further preprocessing. Date variables such as *visit_date* and *date_of_birth* were converted to appropriate standardized date format. The dataset was ordered by *patient_id* and *visit_date* to ensure the correct chronological sequence of visits for each patient. Additional cleaning included the encoding of categorical features into numeric binary values such as *gender, rf_status, and flare_event.* Treatment decision categories were also grouped into broader clinically meaningful categories, to reduce variability and improve interpretability. Since the comorbidities column was recorded as free text, further standardization was needed. This included correcting spelling inconsistencies, removing placeholder values, and standardizing separators so that multiple comorbidities were presented in a consistent comma-separated format.

After the initial cleaning steps, missing values were examined and handled according to the type and clinical meaning of each variable. Variables with a very high percentage of missing values were dropped from the dataset, using a threshold of 0.5, meaning if a variable had more than 50% missingness, it was removed. However, clinically important variables for RA assessment were retained, despite missingness. These variables were DAS28 score, and inflammation-related variables, such as CRP and ESR, due to the relevance to flare prediction. TJC28, SJC28, and VAS were excluded from this list as they are already reflected in the DAS28 score. For categorical variables, where missingness represented the absence of a specific condition or treatment, replacement values were used. For example, missing values in treatment-related variables were replaced with values such as ‘None’ and ‘No switch’ when no drug use or treatment switch took place. In that way, the missing values reflected the clinical context of the dataset, rather than applying a single uniform rule to all variables.

3.2.2 Feature Engineering and Derived Variables

Feature engineering was performed to augment the original clinical records, generating derived and transformed variables that complement the longitudinal structure of the dataset

and the clinical characteristics relevant to RA flare prediction. These variables were designed to capture both the patient's baseline clinical profile and changes occurring across follow-up visits.

Firstly, the patient's birth year was extracted from the *date_of_birth* and used to calculate *age_at_visit* for each clinical visit. This approach helped preserve data privacy by allowing removal of the original *date_of_birth* variable after processing. The variable *disease_duration* was calculated using the year of RA diagnosis and the year of the visit, to estimate how long the patient had been diagnosed at each time point. To capture potential non-linear effects of disease duration on flare risk (both early and long-standing disease may carry elevated risk through different mechanisms), a squared transformation of this variable (*disease_duration_squared*) was additionally derived. Furthermore, *days_since_last_visit* was calculated as the time interval between consecutive visits for the same patient. A value of zero was assigned to the first recorded visit of each patient, where no previous visit was available. These variables were included because age, disease duration, and visit spacing may carry valuable information about flare risk.

Disease activity and flare-history features were also derived from the longitudinal visit records. The variable *das28_change* was calculated to represent the change in DAS28 score compared with the previous visit of the same patient. Positive values denote an increase in disease activity and negative values denote a decrease. In addition, *previous_flare* was created to indicate whether a flare had been recorded at the preceding visit. The prediction target, named *flare_next_visit* was also generated during this process by shifting the physician-recorded flare status one visit forward within each patient. This variable is described in more detail in Section 3.3.

Additional laboratory and inflammation-related features were created from CRP and ESR values. CRP values were normalized using the laboratory-specific upper limit of normal, by dividing the recorded CRP value by *crp_upper_limit*, resulting in the feature *crp_normalized*. Values of *crp_normalized* below or equal to 1 indicate CRP values within the normal range, while values above 1 indicate elevated CRP relative to the corresponding laboratory limit. Also, lagged ESR from the preceding visit (*lag_esr1*) was derived to represent the patient's recent inflammatory state. A lagged CRP feature was not created as its values spike incredibly fast and go back to normal just as quickly, so it carries almost zero

clinical relevance [8]. Furthermore, a combined inflammation feature was made (*inflammation_flag)* showing if there was evidence of inflammation based on either CRP or ESR levels. To account for the non-normal distributions frequently observed in these continuous variables, mathematical transformations (particularly logarithmic scaling) were applied to highly skewed features to stabilize variance and optimize them for the ML algorithms (further discussed in Section 4.1.3).

Comorbidity-related features were created from the free-text comorbidities field. Recorded comorbidities were cleaned and grouped into broader clinical categories to improve interpretability. The comorbidity groups consisted of cardiovascular (*comorb_cardiovascular*), metabolic/endocrine (*comorb_metabolic_endocrine*), respiratory (*comorb_respiratory*), renal/urological (*comorb_renal_urological*), gastrointestinal/hepatic (*comorb_gastrointestinal_hepatic*), malignancy (*comorb_malignancy*), musculoskeletal (comorb_musculoskeletal), neurologic/psychiatric (*comorb_neurologic_psychiatric*), and other comorbidities (*comorb_autoimmune_other*). In addition, *comorbidity_count* was calculated to represent the total number of recorded comorbidities for each patient.

Finally, treatment-related features were engineered to summarize medication usage at each visit. These included binary indicators for csDMARD use, b/tsDMARD use, and corticosteroid use (*csdmard_use*, *b_ts_dmard_use*, *steroid_use* respectively), as well a *treatment_count* variable, representing the total number of active treatment categories at each visit. Also, continuous dosages for MTX and corticosteroids were engineered into binned ordinal categories to manage zero-inflation (*mtx_dose_category* and *steroid_dose_category*). Furthermore, to mitigate class sparsity in the existing categorical variable for physician interventions (*treatment_decision*), clinically analogous actions were aggregated into a broader feature (*treatment_decision_grouped*). Specifically, 'escalation' and 'switch' decisions were combined into an escalation_or_switch category, while 'tapering' and 'stop' actions were unified as tapering_or_stop. Lastly, missingness indicators, such as *das28_missing*, *crp_missing*, and *esr_missing*, were made to check whether these laboratory measurements were available at each visit, and if this missingness carries additional information. Together, these features provided a more structured representation of the patient's clinical status and profile, for further ML analysis. The variables derived from feature engineering process are presented in Table 5.

***Table 5**: Engineered features created during preprocessing.*

Feature	Derived from	Description	Values/Format
birth_year	date_of_birth	Extracted year of birth	Year format, YYYY
age_at_visit	birth_year, visit_date	Patient age at each visit	Numeric value, years
disease_duration	year_of_diagnosis, visit_date	Approximate disease duration at each visit	Numeric value, years
disease_duration_squared	disease_duration	Squared transformation of disease duration	Numeric value, years^2
days_since_last_visit	visit_date	Time interval between consecutive visits	Numeric value, days
das28_change	das28_score	Change in DAS28 compared with previous visit	Numeric value
previous_flare	flare_event	Flare status at previous visit	Binary: 0/1
flare_next_visit	flare_event		Binary: 0/1
crp_normalized	crp, crp_upper_limit	CRP value adjusted for the laboratory-specific upper limit of normal	Numeric ratio
esr_lag1	esr	Lagged ESR value from the preceding visit	Numeric value, mm/h
inflammation_flag	crp_normalized, esr	Indicator of elevated	Binary: 0/1

		inflammatory markers	
comorb_group_name	comorbidities	Individual binary flags for 9 distinct clinical categories (e.g., cardiovascular, respiratory)	Binary: 0/1 per group
comorbidity_count	comorbidities	Number of recorded comorbidities	Numeric count
csdmard_use	csdmard_name	Indicator of csDMARD use	Binary: 0/1
b_ts_dmard_use	b_ts_dmard_name	Indicator of b/tsDMARD use	Binary: 0/1
steroid_use	steroid_dose	Indicator of corticosteroid use	Binary: 0/1
treatment_count	csdmard_use, b_ts_dmard_use, steroid_use	Number of active treatment categories at the visit	Numeric count
mtx_dose_category	mtx_dose	Ordinal dosage categories	Categorical (0 mg, 1-10 mg, 11-20 mg, 21-25 mg)
steroid_dose_category	steroid_dose	Ordinal dosage categories	Categorical (0 mg, 1-10 mg, >10 mg)
treatment_decision_grouped	treatment_decision	Broader aggregation of physician's actions	Categorical (escalation_or_switch, tapering_or_stop)

das28_missing	das28_score	Indicator of missing das28 score value	Binary: 0/1
crp_missing	crp	Indicator of missing CRP value	Binary: 0/1
esr_missing	esr	Indicator of missing ESR value	Binary: 0/1

3.3 Target Variable Definition

The outcome variable of this study was defined as flare occurrence at the next recorded clinical visit. The original visit-level variable, *flare_event*, indicated whether a flare was recorded by the physician at the specific present visit. To transform this information into a prediction target, the flare status was shifted one visit forward within each patient's visit history. As a result, each visit was assigned a target variable indicating whether the patient experienced a flare at the next visit.

Because this study relies on real-world clinical practice data, the next-visit prediction architecture does not correspond to a fixed, standardized chronological forecasting period. The temporal intervals between consecutive longitudinal appointments vary dynamically based on clinical necessity or patient scheduling adherence. Therefore, the distribution of days between consecutive visits was evaluated. To verify whether this irregularity introduced an underlying bias into the prediction target, a Chi-square test for independence was conducted across tracking intervals (under 6 months, 6-12 months, and greater than 12 months), the results of which are detailed in Section 4.1.2.

Formally, the derived binary target variable was named *flare_next_visit*. A value of 1 indicated that a flare was recorded at the patient's next visit, while a value of 0 indicated that no flare was recorded. In this way, the ML task was formulated as a supervised binary classification problem, where information available at visit t was used to predict flare status at visit t+1. Since the target required information from a following visit, the final visit of each patient could not be used as a prediction instance and was excluded from the modeling

dataset. Furthermore, patients with only one recorded visit were excluded from this study as they offer no longitudinal information. With this approach, the target reflects future flare occurrence while preserving the longitudinal structure of the dataset.

It is important to note that flare prediction was based on the overall physician-assessed flare status rather than a purely score-based definition, such as a strict change in DAS28. Because flare occurrence was not explicitly coded as a simple binary variable in the medical files, it was derived retrospectively via manual chart review. The unstructured clinical notes were evaluated to extract the *flare_event* variable, determining the presence of a flare based exclusively on the attending physicians' recorded observations. The criteria utilized by the physicians to establish a flare event were grounded in standard rheumatology practices. These criteria included a multi-dimensional assessment: a meaningful increase in TJCs and SJCs, worsening patient-reported pain or morning stiffness, and laboratory evidence of systemic inflammation via CRP or ESR. Critically, this clinical definition was applied uniformly across both clinical settings involved in this study. This approach reflects the real-world clinical nature of the dataset, where flare assessment is not a single metric, but a holistic decision dependent on multiple factors, including patient symptoms, comorbidities, disease activity measures, and expert clinical judgment.

3.4 Final Modeling Dataset

The final modeling dataset was created after data integration, preprocessing, feature engineering, and target variable creation. Although the initial integrated dataset included all recorded visits from the patients, not all visit records could be used for the prediction task. Since the objective of this study was to predict flare occurrence at the next visit, the final visit of each patient was excluded because it did not carry any information to define the outcome variable. Therefore, only visits with a valid value for the next-visit flare target were retained for exploratory analysis and model development.

In addition, variables with high proportion of missing values were removed according to the predefined missingness threshold (more than 50%), except for clinical important variables that were kept due to their relevance to RA assessment. The detailed missingness pattern and removed variables are presented during the exploratory analysis in Chapter 4.

The final dataset represents the analysis-ready dataset for exploratory data analysis and visualization, feature selection, model training, and model evaluation. The characteristics of the final dataset are described in Table 6.

Table 6 *Summary of the final modeling dataset.*

Dataset characteristic	Value
Number of patients	106
Total number of recorded visits	499
Visits excluded due to absence of next visit	106
Visits with next-visit target	393
No flare at next visit	263
Flare at next visit	130

3.5 Machine Learning Methodology

3.5.1 Train Test Splitting and Data Leakage Prevention

The final dataset contained repeated observations from the same patients since each patient contributed more than one clinical visit. Therefore, the splitting strategy had to account for the longitudinal nature of the data. A simple random train test split or standard K-Fold CV would divide the dataset at the visit level. This would not be appropriate because visits from the same patient could appear in both the training and test sets. Such a split would introduce patient-level data leakage, as the model could indirectly learn patient-specific patterns during training and then be evaluated on other visits from patients already represented in the training data. As a result, model performance could be overestimated.

To address this issue, a group-based splitting strategy was considered. In Group K-Fold CV, the patient identifier (*patient_id*) can be used as the grouping variable, ensuring that all visits from the same patient remain within the same fold. In this way, each patient appears either in the training set or in the test set for a given split, but not in both. This approach is more appropriate for longitudinal data, as it evaluates the model in patients that were not seen during training.

Furthermore, the target variable (*flare_next_visit*) in this study was moderately imbalanced, with fewer next-visit flare cases than non-flare cases. For this reason, Stratified Group K-Fold CV was selected as the final validation strategy [29]. This method combines two important requirements: it keeps all records from the same patient in the same fold, while also attempting to preserve the proportion of flare and non-flare cases across folds. This allowed the validation process to reduce patient-level leakage while maintaining a more balanced representation of the target classes in each fold. In each CV iteration, the model was trained on a subset of patients and validated on a separate subset of patients. This ensured that model performance was estimated on unseen patients rather than on unseen visits from patients already represented in the training data. Any preprocessing steps that depended on the data distribution, such as imputation, scaling and feature selection, were performed within the training folds to prevent information from the validation data influencing model development.

When hyperparameter tuning was introduced, an additional source of leakage risk emerged. Selecting hyperparameters via CV and then evaluating the tuned model on the same folds produces optimistically biased performance estimates, as the chosen parameters have already been implicitly adapted to the full data distribution. To eliminate this risk, nested CV was applied to all four classifiers during the hyperparameter tuning phase. The outer loop consisted of the 5-fold Stratified Group K-Fold described above, with each outer fold serving exclusively as the held-out evaluation set. Within each outer training fold, a separate 3-fold Stratified Group K-Fold inner loop was used for hyperparameter optimization via GridSearchCV or RandomizedSearchCV, ensuring that the test fold remained entirely invisible during tuning. Crucially, to prevent data leakage during threshold optimization, the classification thresholds were selected completely independent of the outer evaluation folds. The optimal operating cutoff for each classifier was determined out-of-fold using only the training and inner validation data, and only then applied to the completely unseen outer evaluation folds.

Evaluation metrics were computed individually within each outer fold and then aggregated to yield robust performance estimates that reflect true generalization to unseen patients. Following the completion of the nested CV pipeline, a final model was refit on the complete longitudinal dataset using the optimized hyperparameter search procedure, solely

for the purposes of structural model inspection, coefficient stability analysis, and feature importance extraction.

3.5.2 Feature Selection Strategy

The feature selection process followed a multi-criteria framework combining statistical hypothesis testing, effect size estimation, multivariate importance assessment, and clinical knowledge. Features were evaluated this way to ensure that both statistically significant and clinically plausible predictors were considered for inclusion. This process was done prior to any modeling and was applied exclusively to the training data within the CV framework to prevent information leakage.

The initial screening of the variables was performed using hypothesis tests appropriate for each feature type. For continuous values, the Mann-Whitney U test was conducted to assess whether the distribution of each feature differed between the patients who experienced a flare at their next visit and those who did not. This test was preferred from the t-test given the right-distributions characteristic of inflammatory markers such as CRP and ESR, and the relatively small sample size. The point-biserial correlation coefficient was additionally computed as an effect size measure, enabling the ranking of features by the magnitude rather than merely the statistical significance of their association with the target variable. For binary and categorical variables, the chi-square test of independence was applied, with Cramer's V reported as the corresponding effect size. Cramer's V provides a measure of association bounded between 0 and 1, giving direct comparison across variables with different numbers of categories. Categorical and binary variables with very low prevalence in the dataset were identified and assessed for statistical reliability. For example, if a binary variable's positive category contains a very small number of observations, statistical tests assumptions are violated, and effect size becomes unreliable. Such variables were excluded from the final feature as they do not offer any significant information for the target variable [30].

Following the univariate screening, pairwise correlations among continuous features were examined to identify redundant variable pairs. Pearson, Spearman, and point-biserial coefficients were calculated depending on data types, and one variable from any pair

exceeding r = 0.85 was excluded. Attention was paid to features derived from the same source variable, as they are mathematically dependent by construction and may introduce multicollinearity that destabilizes coefficient estimates in linear models and inflates the apparent importance of correlated features in tree-based models.

Furthermore, a RF classifier was trained on the full candidate feature set as screening tool. Feature importances were derived by averaging the mean decrease in impurity across five stratified group CV folds using 500-tree RF models, producing a stability-adjusted importance ranking. The coefficient of variation of importance across folds (the ratio of the standard deviation to the mean importance, or CV ratio) was also computed as a measure of cross-fold reliability, with lower values indicating more consistent importance estimates across different patient subsets.

Of course, all these criteria were applied within the constraints of clinical domain knowledge. Some variables were retained despite borderline or non-significant univariate associations where strong theoretical justification existed for their inclusion. The final feature set was determined by the convergence of statistical evidence, multivariate importance, cross-fold stability, and clinical reasoning. The resulting feature set comprised continuous, binary, and categorical variables spanning disease activity, inflammatory markers, treatment characteristics, comorbidity burden, and patient demographics, processed through a unified preprocessing pipeline as described in the preceding subchapter.

3.5.3 Modeling Pipeline and Missing Data Imputation

A unified preprocessing pipeline was constructed using ColumnTransformer and applied consistently across all classifiers. Continuous variables were imputed using median imputation via SimpleImputer and standardized using StandardScaler. Binary and categorical variables were imputed using the most frequent value strategy, with categorical variables additionally encoded using OneHotEncoder to transform nominal categories into a numerically appropriate representation compatible with all classifiers. Three continuous variables (*das28_score, esr, crp_normalized*) were treated separately from the standard preprocessing pipeline due to their elevated missingness rate (37%, 8%, and 17% respectively) and the likelihood that their inter-variable relationships would not be adequately

captured by simple imputation strategies. Four imputation techniques were explored for these variables as a part of a comparison described in Section 4.3.

As a simple reference baseline, median imputation was evaluated first, which replaced missing clinical values with the median of the observed values for the corresponding feature. Moving beyond the baseline, KNN imputation was implemented. This technique replaces each missing value with the weighted average of the k most similar observed cases, where similarity is defined by Euclidecraman distance across the non-missing features [31]. While both median and KNN imputation operate in a single pass, the remaining two approaches utilized the Multiple Imputation by Chained Equations (MICE). This method is an iterative imputation framework that models each incomplete variable as a function of all other variables in the dataset, cycling through each variable with missing data in sequence and updating imputations until convergence [32]. This approach preserves the multivariate relationships between variables and produces statistically principled imputations under the missing at random assumption. Two MICE variants were evaluated. MICE with Bayesian Ridge Regression (MICE-Bayesian) and MICE with RF Regressor (MICE-RF). MICE-Bayesian assumes approximately linear relationship between predictors and incorporates regularization by assigning a Gaussian prior distribution over the regression coefficients, effectively shrinking weights to counter multicollinearity [33]. MICE-RF makes no distributional assumptions and captures non-linear inter-variable relationships through ensemble tree splitting [34].

All four imputation strategies were fitted exclusively on training fold data and applied to the corresponding held-out fold within the CV loop, ensuring a fair and leakage free comparison. As DAS28 change and lagged ESR are derived features from the imputed DAS28 score and ESR values respectively, dedicated recalculation functions were integrated into the pipeline immediately following the imputation step. These functions recomputed the derived variables from the newly imputed source values at each fold, ensuring internal consistency between the source variables and their temporal derivatives throughout training and evaluation. This step was executed prior to standardization, ensuring the derived features were computed on the imputed scale rather than on standardized values.

3.5.4 Classification Models

Four supervised classification algorithms were selected for this study: LR, SVM, RF, and XGBoost. These models where chosen to represent a spectrum of algorithmic complexity and interpretability, ranging from linear parametric approaches to non-linear ensemble methods. The inclusion of both interpretable models and high-performing ensemble methods was deliberate, as clinical prediction tasks require not only predictive accuracy but also a degree of transparency in decision making, particularly in rheumatological contexts where model outputs may inform treatment decisions [35]. All four algorithms have established precedent in structured clinical data tasks, and their combined evaluation allows for a comparison of baseline and advanced modeling approaches.

LR is a parametric, linear classification model that estimates the probability of a binary outcome as a function of a set of input features [36]. Despite its relative simplicity, it remains one of the most widely used algorithms in clinical prediction modeling due to its interpretability and computational efficiency [37]. The model estimates the log-odds of the positive class (the occurrence of a flare at the next visit) as a linear combination of the input features:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \qquad (3)$$

where p denotes the probability of a flare, β_0 is the intercept, and $\beta_1..\beta_n$ are the learned feature coefficients. The predicted probability is recovered via the sigmoid transformation:

$$p = \frac{1}{1+e^{-(\beta_0+\beta^T x)}} \qquad (4)$$

A key advantage of LR in a clinical context is the direct interpretability of its coefficients: each β_j represents the change in log-odds associated with a unit increase in feature x_j, and the corresponding odds ratio e^{β_j} quantifies the multiplicative change in the odds of flare [36]. In this study, an Elastic Net penalty was applied to the LR objective, combining L1 (Lasso) and L2 (Ridge) regularization to simultaneously perform feature selection and coefficient shrinkage [38]. This is particularly appropriate given the moderate dimensionality and limited sample size of the dataset, where standard maximum likelihood estimation risks overfitting. The mixing parameter between L1 and L2 penalties and the regularization strength ($C = 1/\lambda$) were treated as hyperparameters and optimized via nested CV.

SVM is a kernel-based discriminative classifier that identifies the optimal decision boundary (hyperplane) that maximizes the margin between classes in the feature space [39]. For a linearly separable binary problem, the decision function takes the form:

$$f(x) = w^T x + b \qquad (5)$$

where w is the weight vector normal to the hyperplane and b is the bias term. The optimization objective maximizes the geometric margin 2/||w|| subject to correct classification constraints. For non-linearly separable data, the kernel trick implicitly maps input into a higher dimensional feature space where linear separation becomes possible, without requiring the explicit computation of that mapping. The soft margin parameter C controls the trade-off between maximizing the margin and tolerating misclassifications on the training set and was tuned as part of the hyperparameter optimization procedure.

RF is a bagging-based ensemble method that constructs a large collection of decorrelated decision trees during training and aggregates their individual predictions to form a final output. Each tree is trained on a bootstrap resample of the training data, and at each internal node, only a randomly selected subset of features is considered as candidates for splitting (feature bagging). This dual source of randomization in both samples and features reduces the variance of the ensemble relative to any individual tree, while preserving low bias, and avoids overfitting without ant explicit regularization. The predicted class probability for a new observation is computed as the mean of the probabilistic outputs across all T trees:

$$\hat{p}(y = 1|x) = \frac{1}{T}\sum_{t=1}^{T} \hat{p}_t(y = 1|x) \qquad (6)$$

RF handles non-linear relationships and feature interactions naturally, is robust to irrelevant and redundant features, and provides feature importance estimates based on the mean decrease in impurity across all trees, supporting clinical interpretability [40]. Its robustness to overfitting relative to single decision trees makes it particularly well-suited for small clinical datasets with heterogeneous variable types present in this study.

XGBoost is a scalable, regularized gradient boosting framework that constructs an ensemble of decision trees sequentially, with each successive tree trained to correct the residual errors of the current ensemble. Unlike RF, which constructs trees independently in

parallel, XGBoost employs an additive strategy that minimizes a regularized objective function at each boosting round:

$$L^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \quad (7)$$

where l is the differentiable loss function (binary cross entropy in this study), $\hat{y}_i^{(t-1)}$ is the prediction from the previous boosting round, f_t is the new tree being added, and $\Omega(f_t)$ is a regularization term penalizing model complexity through both the number of leaves and the magnitude of the leaf weights [41]. This explicit regularization distinguishes XGBoost from earlier gradient boosting implementations and contributes to its strong generalization performance on structured data. Several properties of XGBoost are directly relevant to the characteristics of this dataset: it incorporates built-in L1 and L2 regularization, handles missing values natively by learning optimal default split directions during training, and supports class imbalance correction via the scale_pos_weight parameter, which was computed from the class distribution of each training fold independently during nested CV. XGBoost has consistently achieved state-of-the-art performance on structured clinical prediction tasks [41] and was included in this thesis as the high-performance benchmark which the interpretable models were compared.

3.5.5 Model Evaluation Metrics

Model performance was evaluated using a combination of metrics. Since the prediction task was shaped into a binary classification problem, the positive class (1) was defined as the occurrence of a flare at the next recorded visit, while the negative class (0) represented no flare at the next visit.

The confusion matrix was used to summarize correct and incorrect predictions, including true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Accuracy was reported as a general measure of overall correct classification. However, because the dataset showed moderate class imbalance it was not interpreted as the only indicator of performance. Precision, recall, and F1-score were calculated to provide a more detailed assessment of model behavior for the flare class. In this context, recall was the most important metric as it reflected the ability of the model to identify patients who later experienced a flare, while F1-score was also emphasized because it provided a balance

between recall and precision, making it useful when evaluating performance on the flare class. Additionally, specificity was evaluated as a secondary metric to monitor the model's ability to correctly identify stable, non-flare visits.

Beyond threshold-dependent metrics, the Precision-Recall (PR) curve was used as the primary visual and quantitative tool for model comparison. Unlike the ROC curve, which evaluates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR), the PR curve directly captures model performance on the positive class across all possible decision thresholds, making it more informative rather than optimistic for class imbalance. The area under the PR curve, reported as AP, was therefore adopted as the primary scalar metric for ranking and comparing models, as it summarized the trade-off between Precision and Recall in a single value that is sensitive to the class distribution. ROC-AUC curve was additionally reported as it remains a widely reported benchmark. To quantify the uncertainty and stability of the predictive metrics, 95% Confidence Intervals (CIs) were estimated for both ROC-AUC and AP using bootstrap resampling. The optimal classification threshold for each model was determined by identifying the point on the PR curve that maximized the F1-score, rather than applying the default threshold of 0.5. This approach is more appropriate for imbalanced datasets where the 0.5 threshold tends to favor the majority class and under detect the flare events. Additionally, probability calibration was assessed via calibration plots, which map predicted probabilities against observed flare frequencies, where perfect calibration tracks the diagonal reference line. The Brier score defined by the mean squared error between predicted probabilities and true binary outcomes, served as the scalar summary metric, where lower values indicate superior probability accuracy. This evaluation is critical for clinical utility, ensuring that a predicted probability (e.g., 40%) directly mirrors the true empirical flare rate in the patient population, allowing clinicians to reliably act on individual risk scores.

Finally, a learning curve analysis was conducted for the best model to evaluate the relationship between sample size scaling and generalization performance. Plotting training and CV AP against the number of training instances diagnoses whether the algorithm operates in a high-bias or high-variance regime, demonstrating if the model has hit its performance ceiling or if a larger patient cohort would yield further predictive gains.

3.5.6 Hyperparameter Optimization Framework

Each classifier underwent hyperparameter optimization to improve predictive performance beyond the default configuration assessed in the baseline phase. As described in 3.5.1, nested CV was adopted as the evaluation framework for all tuned models. All models were optimized using AP as the primary scoring criterion, consistent with its adoption as the primary evaluation metric, as discussed in 3.5.5.

LR was configured with the Elastic Net penalty, which combines L1 and L2 regularization. The solver was set to SAGA, which is the sklearn solver supporting Elastic Net regularization, with a maximum of 5000 iterations to ensure convergence across all fold configurations. Three hyperparameters were tuned: the regularization strength C, which controls the penalty intensity, the mixing l1_ratio which governs the balance between L1 strict regularization and L2 coefficient shrinkage, and class_weight, which determines whether the loss function applies equal or inverse-frequency weighing to address class imbalance. The search space was bounded toward stronger regularization values, as the small dataset size makes overfitting a primary concern. Given the compact grid, GridSearchCV was used for the inner loop rather than RandomizedSearchCV, ensuring complete coverage of all parameter combinations [42]. The LR grid was comprised of:

- l1_ratio ∈ [0, 0.15, 0.35, 0.45, 0.5]
- C ∈ [0.05, 0.1, 0.5, 1]
- class_weight ∈ [balanced, None]

The SVM was configured with probability calibration enabled to allow threshold-based classification and probability score output. Three kernel functions were evaluated in this study: the linear kernel, the Radial Basis Function (RBF) kernel, and the polynomial kernel, with the optimal kernel selected independently at each outer fold of the nested CV [43]. An additional feature selection step using SelectKBest with mutual information scoring was incorporated into the pipeline, as it is sensitive to irrelevant features in high-dimensional or noisy settings. The number of features selected (k) was included as a tunable parameter within each kernel grid. The tolerance parameter (tol), which defines the convergence criterion for the optimization solver, was additionally included in the linear and RBF kernel grids but it was excluded from the polynomial, to avoid computational stagnation. Smaller tolerance values enforce stricter convergence at the cost of increased computation time, while

larger values allow earlier termination. Giving the distinct parameter spaces across kernels, RandomizedSearchCV was applied within each kernel's grid independently per outer fold, with 50 random samples drawn per kernel. To train the final production model, a RandomizedSearchCV was executed for 50 iterations across the full dataset as well, combining all three kernel grids as a list of parameter dictionaries [44]. The tuning grids are described in Table 7:

***Table 7**: Tuning grids for SVM kernels.*

Parameter	Linear kernel	RBF kernel	Polynomial kernel
selector__k	5, 7, 10, all	5, 7, 10	5, 7, 10
C	0.01, 0.1, 1, 10, 100	0.01, 0.1, 1, 10, 100	0.01, 0.1, 1, 10
class_weight	balanced, None	balanced, None	balanced, None
tol	1e-4, 1e-3, 1e-2	1e-4, 1e-3	-
gamma	-	scale, auto, 0.01, 0.1, 1	scale, auto, 0.01, 0.1
degree	-	-	2, 3, 4

RF was configured with n_jobs=1 at the classifier level, with parallelism delegated entirely to RandomizedSearchCV (n_jobs=-1), to prevent nested parallelism conflicts that can cause performance degradation. The scale of the ensemble was evaluated by varying the number of decision trees (n_estimators), while the hyperparameter search space was designed to constrain tree complexity in response to the small dataset size, where deep trees are prone to overfitting individual patient fold compositions. Minimum samples per leaf and minimum samples per split were included to enforce minimum node sizes, providing an additional regularization mechanism. The min_impurity_decrease parameter was introduced to require that each split achieve a minimum reduction in node impurity before being executed. Both Gini and entropy criteria were considered as splitting criteria, and two feature sampling strategies (square root and log2) were evaluated as the maximum feature per split. Two class weighting strategies were evaluated: balanced and balanced_subsample. The latter recomputes class weights independently within the bootstrap sample drawn for each tree rather than from the full training set and it was hypothesized to be more appropriate for a longitudinal dataset where bootstrap samples may not perfectly reflect the global class distribution. Given the large, combined search space, RandomizedSearchCV with 50 iterations was also used for the inner loop and for the final model refit [45].

The tuning grid comprised:

- n_estimators ∈ [100, 300, 500]
- max_depth ∈ [3, 4, 5]
- min_samples_leaf ∈ [3, 5, 7, 10]
- min_samples_split ∈ [10, 15, 20]
- min_impurity_decrease ∈ [0.0, 0.001, 0.005, 0.1]
- criterion ∈ [gini, entropy]
- max_features ∈ [sqrt, log2]
- class_weight ∈ [balanced, balanced_subsample]

XGBoost was configured with a binary logistic objective function and log-loss as the evaluation metric, consistent with the binary classification task. The histogram-based tree construction method (tree_method='hist') was specified for computational efficiency, which approximates the optimal split point search using binned feature values. As with RF, n_jobs=1 was also set at the classifier level with parallelism handled by the search procedure. A key design decision specific to XGBoost was the fold-aware computation of the scale_pos_weight. Rather than computing this weight from the full dataset, the ratio of negative to positive training samples was computed independently within each outer fold's training partition. The XGBoost search space included both tree-structural and regularization hyperparameters. Maximum depth was highly restricted, reflecting the consistent selection of shallow trees across baseline experiments on this small dataset. Learning rate was searched over slow values to favor gradual boosting convergence, paired with higher tree counts to compensate. Three regularization parameters were explicitly included. Firstly, gamma, the minimum loss reduction required to execute a split. Secondly reg_alpha, the L1 regularization term on leaf weights, and reg_lambda, the L2 regularization term. Subsampling parameters subsample and colsample_bytree were included to introduce stochasticity and reduce variance through random row and feature sampling per tree. Lastly, the identical randomized optimization framework utilized for the RF model was applied to the XGBoost classifier, with the search budget extended to 60 iterations for both the inner validation loops and the final global refit [46].

The tuning grid comprised:

- n_estimators ∈ [200, 300, 500]
- max_depth ∈ [2, 3]
- learning_rate ∈ [0.005, 0.01, 0.05]
- min_child_weight ∈ [6, 10]

- gamma ∈ [0.1, 0.3, 0.5]
- reg_alpha ∈ [0, 0.1, 0.5, 1]
- reg_lambda ∈ [3, 5]
- subsample ∈ [0.6, 0.7, 0.8]
- colsample_bytree ∈ [0.5, 0.6, 0.8]
- scale_pos_weight ∈ [fold_ratio × 0.75, fold_ratio, fold_ratio × 1.25]

3.6 Large Language Model Methodology

3.6.1 Data Serialization

Standard ML operates on numerical feature matrices where each patient visit is represented as a vector of encoded and transformed values. LLM models, by contrast, process natural language text, requiring a transformation of the tabular clinical dataset into coherent textual descriptions. This process known as serialization, determines which clinical information is conveyed to the model and how it is linguistically framed, both of which may influence prediction quality [47].

Two serialization formats were implemented and compared. The first referred to as the List Template [47], presents each clinical variable as a labeled bullet point, producing a structured enumeration of the patient's clinical profile (e.g. 'Age:54, Gender: Female, DAS28 score: 4.10, CRP: 0.40 mg/L'). The second, referred to as the Text Template [48], renders the same information as a narrative clinical summary, written in the natural language style of a medical case description (e.g. 'A 54-year-old female patient presents with a DAS28 score of 4.10, indicating moderate disease activity…'). Both approaches are presented in Appendix.

A deliberate distinction was maintained between the variables used for LLM serialization and those used in the ML feature pipeline. The latter operated on engineered and transformed features (log scaling, squared variables, binary indicators) optimized for numerical processing. The LLM serialization used the original, clinically interpretable values: raw CRP, raw ESR, free-text comorbidity descriptions. Missing values were handled through explicit linguistic representation rather than imputation. When a clinical value was absent, the serialization function described it as 'not recorded at this visit'. This approach is methodologically more transparent than imputation, as it does not introduce artificial values, and allows the LLM to reason about the clinical context of missing data.

The serialization selected a subset of variables from the original dataset chosen for their direct clinical interpretability: age at visit, disease duration, gender, RF_a status, anti-CCP status, DAS28 score, DAS28 change from the previous visit, CRP, CRP's laboratory-specific upper limit, ESR, MTX dose, steroid dose, csDMARD use, b/tsDMARD use, treatment decision, previous flare, days since last visit, and free-text comorbidity description.

3.6.2 Prompting Strategies

Two prompting strategies were evaluated, each representing a different degree of task specific information provided to the model at inference time. To ensure a scientifically rigorous and balanced benchmark, the LLM evaluation protocol was built to directly mirror the ML pipeline. The entire evaluation was executed using a 5-fold Stratified Group CV protocol across the total cohort of 106 patients.

In the zero-shot condition [49], the model received a system prompt establishing its clinical role and output format, followed by a single patient visit description and a prediction request, with no prior examples of correctly classified cases. This condition tests the model's ability to apply pretrained medical knowledge to individual flare prediction without any calibration to the specific patient cohort. Both List and Text template were evaluated to assess whether the presentation format of clinical information influences prediction quality. The system prompt specified the model's role as an expert rheumatologist, defined a disease flare as a clinically meaningful increase in disease activity at the subsequent visit, characterized by worsening joint symptoms and elevated inflammatory markers. Crucially, the prompt integrated the baseline class distribution of the specific clinical cohort (31% flare vs. 69% stable) to ground the model's priors, explicitly instructing it to rigorously balance risk and protective factors to prevent default over-prediction. Finally, it imposed a structured output format using XML tags, requiring binary prediction within <prediction> tags. Explicit XML-tagged output was specified as a technique recommended in Anthropic's prompt engineering guidelines for improving output consistency [50]. The comprehensive system and user prompt structures utilized for this zero-shot architecture are detailed in Appendix.

In the few-shot condition [49], six representative clinical examples with known outcomes were prepended to the patient description before requesting a prediction. Crucially, to prevent patient-level data leakage, these few-shot examples were harvested exclusively

from the active training split of each specific CV fold, completely isolating the test fold patients. Three examples were drawn from confirmed flare visits and three from confirmed non-flare visits. More specifically, flare examples required DAS28 scores to be between 4.0 and 6.5, a history of previous flare, and elevated CRP or ESR values. The no-flare examples required DAS28 scores to be between 2.6 and 4.0, no prior flare history, and stable CRP and ESR values. This selection strategy was designed to maximize the calibration signal provided by each example. By partitioning data at the patient level prior to sampling, this framework guaranteed that no longitudinal records from any patient represented in the few-shot context ever appeared in the corresponding fold's evaluation set, ensuring the LLM was tested exclusively on completely unseen individuals. The few-shot prompt presented each example with its serialized clinical description followed immediately by the known prediction outcome. Historical reasoning was omitted to prevent the model from generating superficial placeholder text instead of genuine clinical deliberation. The structural framework for these few-shot prompts is documented in the Appendix.

3.6.3 Model Selection, Pipeline, and Execution

Claude Haiku (claude-haiku-4-5-20251001), developed by Anthropic [51], was selected as the LLM for this evaluation based on three practical criteria: computational cost, inference speed, and documented clinical reasoning capability. As a lightweight model in the Claude family, Haiku offers substantially lower per-token cost than larger frontier models while retaining sufficient medical reasoning capacity for structured clinical classification tasks. Given that the evaluation required evaluating the complete longitudinal visit sequence across the 106 patients, cost and latency were non-trivial considerations. Each strategy incurred an estimated API cost of $0.1-$0.5 depending on the output token length, and runtime of approximately 9 minutes per strategy.

All API calls were made via the Anthropic Messages API using the Python anthropic library. The model was queried under strictly controlled execution settings, utilizing a fixed temperature of 0 to eliminate stochastic variance and ensure completely deterministic, reproducible outputs. The inference payload was further constrained with a max_tokens threshold of 30 to limit response length to the requested classification tags. To maintain pipeline stability, the execution script integrated an automatic rate-limit exception handler

featuring a 15-second retry delay alongside a standard 0.5-second sleep duration between consecutive iterations. The evaluation pipeline operated programmatically. For each test visit, the script compiled the system and user prompts, submitted the payload to the API, and processed the raw text response through a multi-tiered regular expression parser to map text outputs back into standard binary labels (1 for Flare, 0 for No Flare). Results for each strategy were saved to a dedicated CSV file containing the visit index, patient identifier, actual outcome, predicted outcome, parsed prediction, DAS28 score, serialization template used, and strategy name.

Unlike the ML classifiers, for which an F1-optimal decision threshold was identified from the PR curve, the LLM produces a hard binary prediction without an adjustable probability threshold. Consequently, no threshold optimization was performed for LLM strategies, and the reported metrics reflect the model's default classification behavior.

3.7 Ethical Considerations and Data Privacy

The dataset used in this retrospective study consisted of pseudonymized clinical records from patients with RA. Direct identifying information was removed before analysis, and each patient was represented by a unique code (e.g., RA001). This allowed the longitudinal records of each patient to be linked across visits while reducing the risk of patient identification. Date of birth was only used during the preprocessing part to derive the age-related variable (*age_at_visit*) and was not retained in the final modelling dataset.

The data was used for research purposes within the scope of this thesis. Since the dataset contained clinical information, particular attention was given to data privacy and confidentiality through the analysis. The data was stored and processed securely and was not uploaded to cloud-based storage services. No patient names, contact details or other direct identifiers were included in the dataset. Results were reported only in aggregated form, so that individual patients could not be identified from tables, figures, or model outputs. Furthermore, the LLM addition was performed via an external commercial API, introducing additional data privacy considerations. Only de-identified data were included in the serialized patient descriptions transmitted to the API.

The ML models and LLM classification approach developed in this thesis were intended for exploratory and research purposes. They were not designed to replace clinical judgement or to be used as standalone clinical decision-making tools. Instead, the aim was to investigate whether routinely collected clinical variables could provide useful information for next-visit flare prediction and to evaluate the potential and limitations of this approach in a real-world dataset.

3.8 Methodological Limitations

Several methodological limitations were considered when interpreting the findings of this study. First, the dataset was relatively small, which may restrict the generalizability of the results to broader RA populations. Although the data reflected real-world clinical practice, the number of patients and prediction instances was limited, especially for training and evaluating ML/LLM models.

Second, the dataset was retrospective, as it used already existing clinical records collected during previous routine follow-up visits, without any direct patient or intervention being involved. As a result, the timing of visits was not standardized, and patients had different follow-up intervals, limiting exact calendar precision. However, this approach directly mirrors real workflows, where decisions are made during actual visits rather than on fixed calendar dates. In addition, missing values were present in several clinically important variables, including disease activity and laboratory measurements. Although preprocessing strategies were applied to handle missingness, data may still introduce bias or reduce the available information for model development.

Third, it is important to acknowledge the inherent subjectivity of the target variable compared to structured clinical metrics. A physician's diagnosis of a flare naturally includes a human, subjective element based on clinical intuition and patient communication. In contrast, the DAS28 score serves as a calculated, semi-objective clinical index. While a patient's DAS28 trajectory is highly correlated with disease exacerbations, doctor-labeled flares capture real-world clinical decision-making that a mathematical formula may overlook. Consequently, this study deliberately targeted the physician's diagnostic label rather than a strict DAS28 cutoff to ensure that the models predict actionable clinical

interventions. While this choice renders the pipeline far more applicable to actual healthcare workflows, it introduces an unavoidable baseline of inter-physician subjectivity and potential label noise into the model training process.

Fourth, the evaluation pipelines differ slightly because the ML models required an extra inner loop to tune their settings without cheating. Because the LLM was not trained or tuned, it only required a standard CV loop. Crucially, the final test groups were kept identical between both approaches, ensuring both were tested on the exact same patient cases for a fair comparison. Additionally, the LLM evaluation was conducted using a single commercial model version (Claude-Haiku-4-5-20251001), and the results may differ across different LLM families, model sizes, or open-source alternatives.

Fifth, the models developed in this thesis were exploratory and were evaluated using the available dataset. External validation on an independent cohort was not performed, and therefore the ability of the models to generalize to other clinical setting remain uncertain. This limitation applies to both ML and LLM classifiers. All these constraints should be considered when interpreting the performance of the proposed approaches and highlight the need for larger, more standardized, and externally validated datasets in future work.

4 Implementation and Results

This chapter presents the execution of the ML framework and evaluates its predictive performance. Firstly, exploratory data analysis maps the cohort's clinical characteristics, missingness patterns, and target distributions. It then details the feature selection procedure used to finalize the predictive features. Finally, it evaluates the preprocessing pipeline's optimization before presenting the hyperparameter tuning configurations and performance metrics of the four classifiers. The chapter concludes with an interpretation of the feature patterns driving the model's clinical decisions.

4.1 Exploratory Data Analysis

4.1.1 Dataset Overview and Descriptive Characteristics

The final dataset used for exploratory analysis and model development was derived from longitudinal clinical records of 106 patients with RA. In total, 499 clinical visits were initially recorded. After defining the *flare_next_visit* variable, the final visit of each patient was excluded, since no subsequent visit was available to determine the outcome. As a result, 393 visits were kept for analysis.

Baseline patient characteristics, calculated from each patient's first recorded visit, are summarized in Table 8. All values were derived from the complete processed dataset, prior to excluding each patient's final visit. The table includes demographic characteristics, serological status, disease duration, disease activity and inflammation-related measurements at the first visit, as well as comorbidity information. This baseline table gives the clinical profile of the cohort before examining the longitudinal dataset. Variables removed due to high missingness are not included in this table and are presented separately in the missing data analysis. Continuous variables were summarized using either mean ± standard deviation or median with interquartile range, depending on their distribution. Categorical variables were reported as counts and percentages.

Table 8: *Baseline patient characteristics.*

Patient characteristics (n=106)	
Age, years	60.5 ± 11.5
Female, n (%)	83 (78.3%)
Male, n (%)	23 (21.7%)
Disease duration, years	4.0 (0.0-14.0)
RF_a positive, n (%)	54 (51.9%)
Anti-CCP positive, n (%)	64 (61.5%)
DAS28 score, units	3.9 ± 1.3
CRP, mg/L	0.9 (0.4-2.1)
ESR, mm/h	25.0 (16.0-37.0)
Patients with ≥ 1 comorbidity, n (%)	84 (79.2%)
Number of visits per patient	4.0 (3.0-6.0)

Note: Continuous variables are shown as mean ± SD or median (IQR); categorical variables are shown as n (%)

The study population included 83 females (78.3%) and 23 males (21.7%), with a mean age of 60.5 ± 11.5 years at their first recorded visit. At baseline, the average disease duration was 4 years with IQR 0.0-14.0, indicating that patients were at varying stages of RA follow-up. In terms of serological status, 54 patients (51.9%) were RF_a positive, and 64 (61.5%) were anti-CCP positive. The mean DAS28 score was 3.9, which is above the threshold for low disease activity (>2.6). Laboratory values showed a median CRP of 0.9 and a median ESR of 25.0. CRP values were normalized to account for differing laboratory upper limits of normal. Additionally, 84 patients (79.2%) had at least one recorded comorbidity, underscoring the prevalence of concurrent clinical conditions in the cohort.

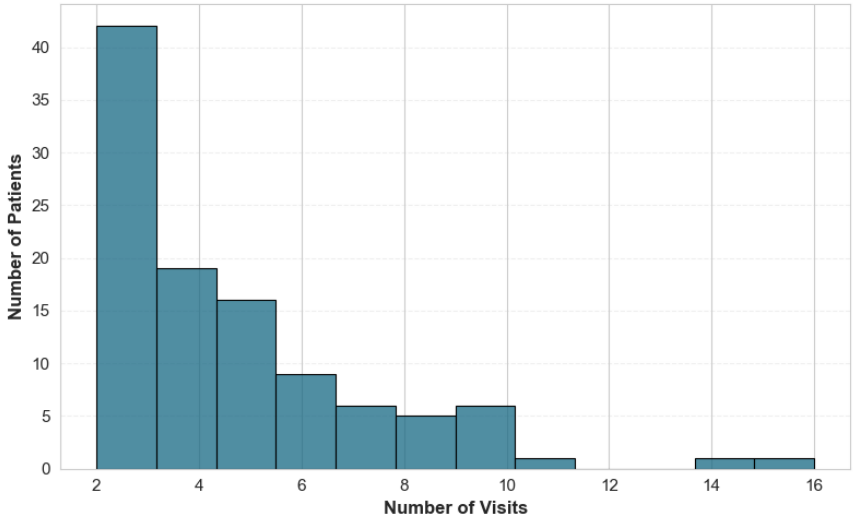


***Figure 1**: Distribution of visits per patient.*

The median number of visits was 4 and the distribution is shown in Figure 1. It demonstrates that patients contributed varying numbers of follow-up visits, which is expected in a real-world retrospective dataset. Most patients had a relatively small number of visits, while a smaller number contributed to longer follow-up visits. This variability supports the need for patient-level grouping during model validation.

Time-varying clinical characteristics across the 393 visits are presented in Table 9. The median interval between consecutive visits was 127.0 days, although the interval varied across visits, reflecting the non-standardized nature of routine clinical follow-up. The mean DAS28 score across visits was 3.4, while the mean DAS28 change from the previous visit was -0.1. These values suggest that, on average, disease activity was in moderate range, with relatively small average changes between consecutive visits.

***Table 9:** Summary of time-varying clinical characteristics across visits.*

Characteristic	Value
Number of visits with next-visit target	393
Days since last visit, days	127.0 (0.0-238.0)
DAS28 score, units	3.4 ± 1.2
DAS28 change from previous visit	-0.1 ± 0.9
CRP, mg/L	0.63 (0.26-1.60)

ESR, mm/h	20.0 (11.0-35.0)
Elevated CRP, n (%)	116 (29.5%)
Elevated ESR, n (%)	173 (44.0%)
MTX use, n (%)	219 (55.7%)
csDMARD use, n (%)	223 (56.7%)
b/tsDMARD use, n (%)	160 (40.7%)
Corticosteroid use, n (%)	148 (37.7%)
Treatment count	2.0 (1.0-3.0)
Previous flare, n (%)	136 (34.6%)

Note: Continuous variables are shown as mean ± SD or median (IQR); categorical variables are shown as n (%)

The median normalized CRP ratio was 0.63, while the median ESR was 20.0. Elevated CRP was observed in 116 visits (29.5%), and elevated ESR was observed in 173 visits (44.0%), indicating the inflammation activity was present in a substantial proportion of patient's visits. Treatment related variables showed that MTX was used in 219 visits (55.7%), csDMARD use was recorded in 223 visits (56.7%), b/tsDMARD use was recorded in 160 visits (40.7%), and corticosteroid use was recorded in 148 visits (37.7%). The median treatment count was 2.0, indicating that treatment combination was common across visits. Lastly, previous flare was seen in 136 visits (34.6%), suggesting that a considerable proportion of visits occurred in patients with recent flare history.

4.1.2 Missing Data and Target Distribution

Missing data was first examined to understand the completeness of the dataset and identify variables that could affect subsequent analysis and model development. Since the dataset included routinely collected clinical records, missingness was expected, particularly for clinical examination and disease activity variables that may not have been recorded at every visit. Before doing the main exploratory analysis, variables with more than 50% missingness were excluded from the final dataset, except for clinically important variables that were kept due to their relevance to the disease assessment.

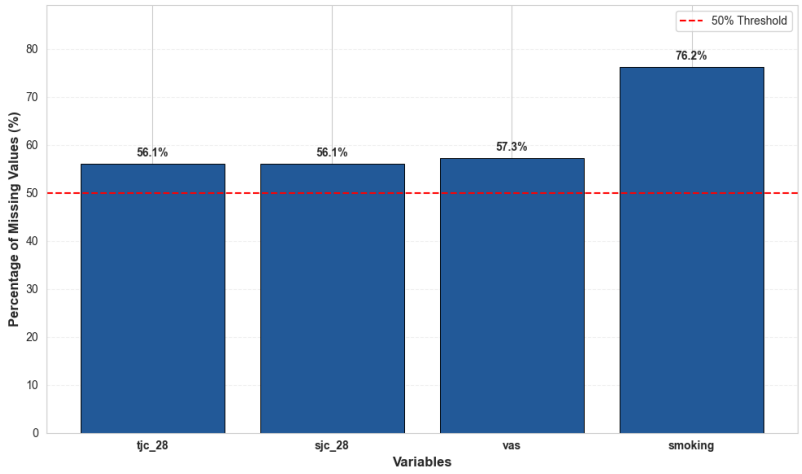


***Figure 2**: Variables excluded due to high missingness.*

Figure 2 shows the variables that exceeded the missingness threshold and were removed from the analysis. The variables that did not contribute to the final dataset were TJC28 (56.1%), SJC28 (56.1%), VAS (57.3%), and smoking status (76.2%). It was better not to keep them to reduce instability during model development. Although TJC28, SJC28, and VAS are significant for flare assessment, their substantial proportion of missing values raised concerns regarding their reliability. Since these variables are components of the DAS28 score, their exclusion was deemed acceptable for this analysis.

After removing highly sparse variables, missingness was further explored among the retained variables, with particular attention to clinically important disease activity and inflammation-related variables. The percentage of missing values among the retained variables is presented in Figure 3.

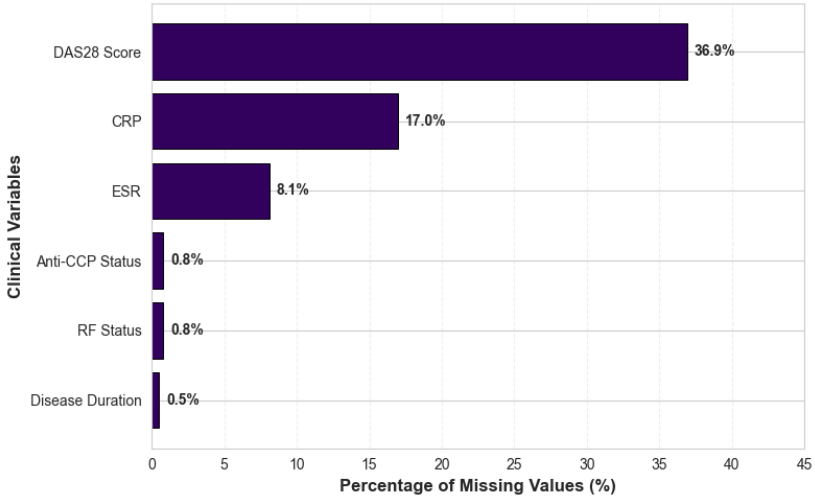


***Figure 3**: Percentage of missing values among clinical variables.*

The highest missingness was observed in the DAS28 score (36.9 %), followed by the CRP (17 %) and ESR (8.1 %) ones. This pattern reflects the real-world nature of the dataset, where not all clinical and laboratory measurements were available on every visit. In contrast, variables such as RF_a and anti-CCP status (both 0.8 %), and disease duration (0.5 %) showed extremely low missingness (below 1%), and therefore required only minimal handling. Of course, the feature engineered variables, as they originated from these variables, they carry missingness. Specifically, *disease_duration_squared* inherited the exact missingness of its root variable (0.5%), while the composite inflammation_flag exhibited a nominal missingness rate of 5.6%. In contrast, the longitudinal tracking features demonstrated substantially higher sparsity: *das28_change* stood at 34.1%, and *esr_lag1* reached 32.3% (a notable increase from the 8.1% observed in the ESR feature). This elevated missingness is caused probably by chronological patient-visit transitions and initial baseline enrollments.

To examine whether missingness in clinically important variables was associated with the prediction target, missingness indicators for CRP, ESR, and DAS28 were compared with next-visit flare status using chi-square tests of independence. The results are presented in

Table **10**. No statistically significant association was observed for any of the three variables, as all p-values were greater than 0.05. Although flare proportions differed slightly between available and missing categories, these differences were small and did not suggest a strong relationship between missingness and next-visit flare occurrence.

***Table 10**: Association between missingness indicators and next-visit flare status.*

Variable missingness	Missing status	No flare, n (%)	Flare, n (%)	χ^2	p-value
CRP	Available	221 (68%)	105 (32%)	0.444	0.505
	Missing	42 (63%)	25 (37%)		
ESR	Available	240 (66%)	121 (34%)	0.181	0.671
	Missing	23 (72%)	9 (28%)		
DAS28	Available	172 (66%)	87 (34%)	0.035	0.852
	Missing	91 (68%)	43 (32%)		

To ensure that the variable time intervals between patient visits did not introduce an underlying chronological bias into the target distribution, a Chi-square test for independence was conducted. **Table 11** outlines the distribution of clinical visits and subsequent flare occurrences stratified by the tracking timeframes. Across the longitudinal dataset, the temporal intervals between consecutive visits exhibited a median of 181.0 days (Mean: 234.3 days, IQR: 150.0-296.5 days), ranging from a minimum of 14 days to a maximum of 1293 days.

***Table 11**: Flare target distribution stratified by visit interval.*

Visit interval	Total visits (N)	Confirmed flares (n)	Flare rate (%)
< 6 months	142	45	31.7
6-12 months	94	28	29.8
> 12 months	51	16	31.4

The Chi-square test for independence yielded a statistic of χ^2=0.0995 with a p-value of 0.9515. Because the p-value is substantially greater than the significance threshold (a=0.05), the null hypothesis cannot be rejected. This mathematically demonstrates that flare rates do not differ significantly across varying visit intervals. The baseline probability of a flare event remains statistically uniform (~30–31%) regardless of whether the temporal gap

between clinical encounters is short or extended, validating the stability of the target definition for the ML and LLM tasks.

The distribution of the target variable was examined after the creation of the *flare_next_visit* feature and the exclusion of the final recorded visit for each patient. Among these, 263 visits were followed by no-flare corresponding to 66.9% of the dataset, while 130 visits were followed by a flare, corresponding to 33.1%. The target distribution (

Figure **4**) showed a moderate class imbalance, with no-flare cases representing approximately two thirds of the dataset and flare cases representing approximately one third. Although the imbalance was not extreme, it was important to take it into account during model development and evaluation.

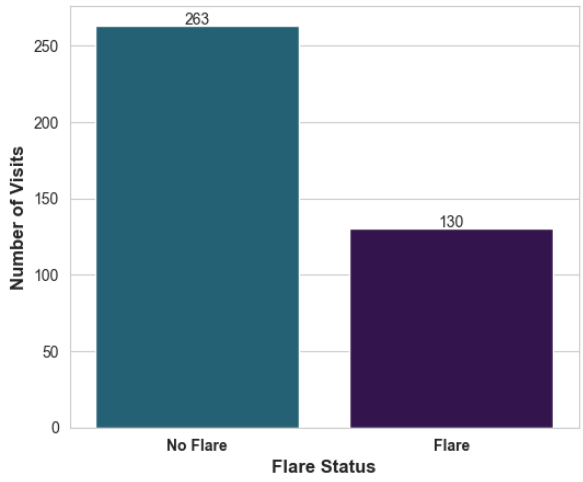


***Figure 4**: Distribution of the next-visit flare target.*

4.1.3 Feature Distributions and Cohort Characteristics

The distribution of key clinical, laboratory, treatment-related, and comorbidity variables was examined to better understand the structure of the final modeling dataset. The purpose of this section is to describe the main patterns observed across study visits before examining their relationship with the prediction target.

Disease activity was first explored using DAS28 score and the change in DAS28 score from the previous visit. As shown in Figure 5, DAS28 scores were distributed across a range of disease activity levels with most observations concentrated around moderate values

(mean = 3.4). To assess whether a variance stabilization technique was necessary for parametric modeling, a logarithmic transformation was tested. However, the Shapiro-Wilk test indicated a statistical deviation from normality for both the original and log-transformed scores (p=0.0069 and p=0.0065, respectively). Despite the formal rejection of the null hypothesis, visual inspection of the histogram and accompanying boxplot confirmed a relatively balanced distribution without severe distortions or high-leverage outliers. Therefore, the original DAS28 score was retained for the downstream modeling phase.

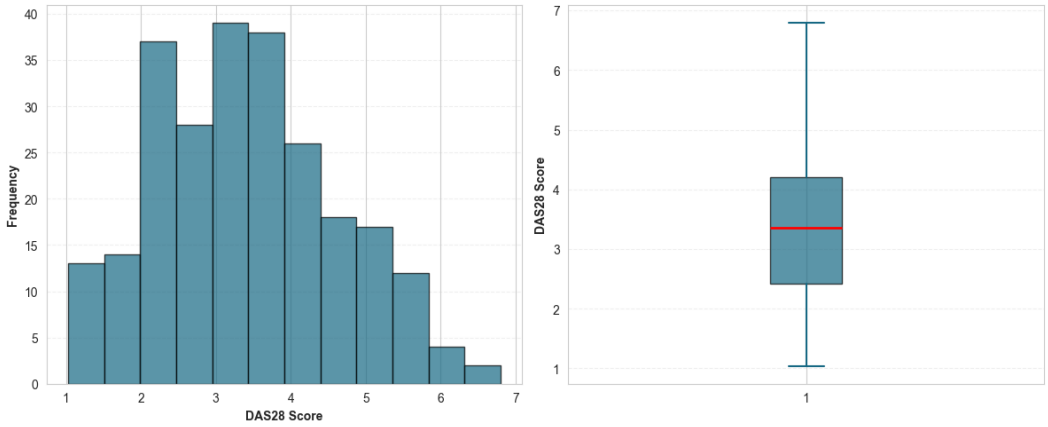


Figure 5: *Distribution of DAS28 score.*

The distribution of DAS28 change is presented in Figure 6 and is heavily leptokurtic, with most values concentrated sharply around zero. This indicates that for most observations, patient disease activity remained stable between consecutive visits. The mean change was -0.1, suggesting a negligible average decrease in cohort disease activity. However, the accompanying boxplot reveals heavy symmetrical tails containing both positive and negative outliers, reflecting sub-populations experiencing acute disease worsening or rapid improvement. Formal normality testing via the Shapiro-Wilk test indicated extreme deviations from normality ($p < 0.001$). Because the *das28_change* variable contains zero and negative values, attempting variance stabilization via logarithmic transformation would destroy clinical interpretability and without resolving the non-normality (log transformation $p < 0.001$). Consequently, the original *das28_change* was kept.

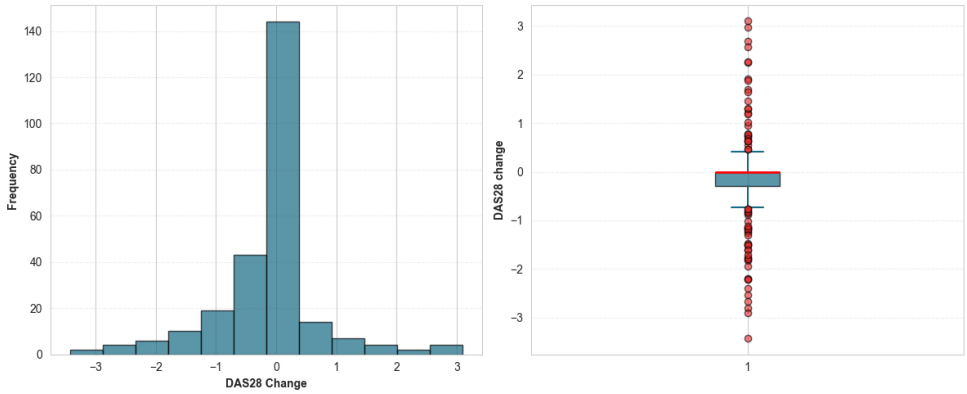


***Figure 6**: Distribution of DAS28 change from previous visit.*

Laboratory variables were then examined using CRP and ESR values. As shown in Figure 7(a-b) and Figure 8(a-b), both variables showed right-skewed distributions, with most observations concentrated at lower values and a smaller number of visits displayed elevated inflammatory markers. ESR demonstrated a slightly less extreme spread than normalized CRP, reflecting its nature as a slower, more stable inflammatory marker compared to the highly reactive CRP [8]. Because extreme right-skewness can introduce high-leverage outliers that destabilize parametric classifiers, variance stabilization techniques were evaluated. A log1p transformation (defined as log(1+x)) was selected over a standard logarithm to handle instances where the original laboratory measurement was zero. The effectiveness of this transformation was evaluated both visually and mathematically. While Shapiro-Wilk testing indicated that normality was not achieved for either the original or transformed variables (all $p < 0.001$), the transformation yielded massive improvements in the distribution geometry. For CRP, the Shapiro-Wilk test statistic (W) improved from 0.46 to 0.84, successfully compressing the extreme right tail (Figure 7(c-d)) and reducing the influence of outliers. Similarly, the ESR test statistic improved from 0.85 to 0.99 resulting in a highly balanced compact spread (Figure 8(c-d)). To preserve mathematical scale consistency, log transformation was also applied to *esr_lag1* yielding the new feature *log_esr_lag1* (W =0.99, p-value=0.01). Consequently, *log_crp_normalized*, *log_esr, and log_esr_lag1* were retained as candidate features for model development.

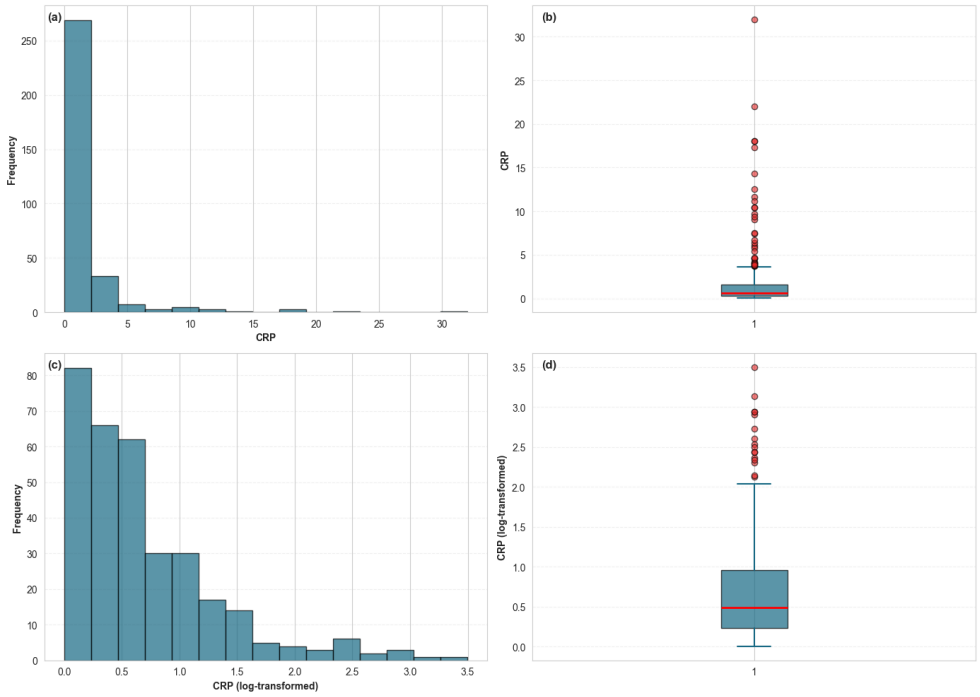


Figure 7: *Distribution of CRP before and after log1p transformation. (a) Histogram of CRP values. (b) Boxplot of CRP values. (c) Histogram of log-transformed CRP values. (d) Boxplot of log-transformed CRP values.*

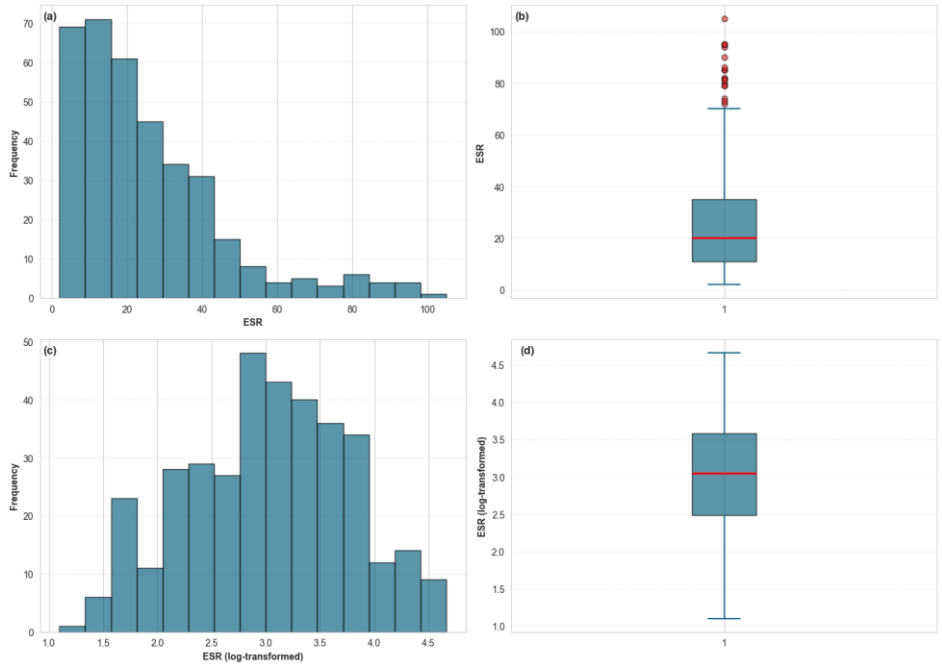


Figure 8: *Distribution of ESR before and after log1p transformation. (a) Histogram of ESR values. (b) Boxplot of ESR values. (c) Histogram of log-transformed ESR values. (d) Boxplot of log-transformed ESR values.*

Time-related variables were also examined to characterize the longitudinal nature of the dataset. Age at visit represented patient's age at each recorded clinical encounter. As seen

in Figure 9, age values were distributed across a broad adult age range, with most observations concentrated between approximately 50 and 75 years. The boxplot did not indicate outlying values, and the distribution did not show strong skewness. Therefore, age was retained in its original scale.

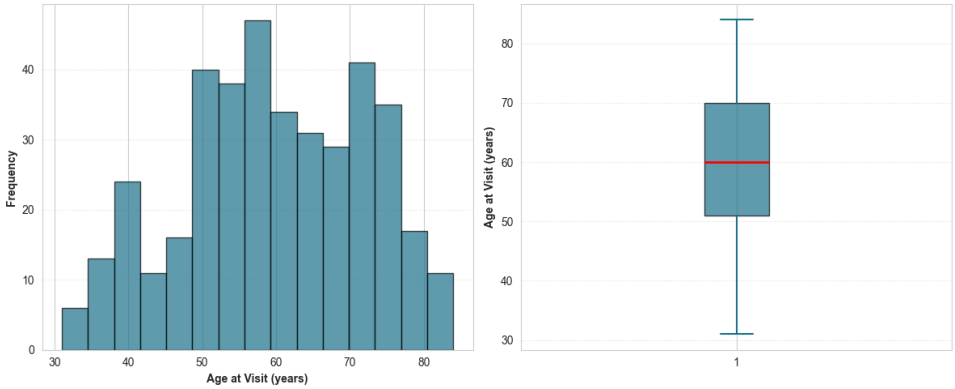


Figure 9: *Distribution of age at visit.*

The inter-visit interval of *days_since_last_visit* presented a unique structural distribution. As shown in Figure 10, the data features a massive spike exactly at zero, alongside secondary dense concentrations at specific intervals. The zero-inflation is a structural artifact representing baseline visits (where no prior visit exists to compare against), rather than a true measurement of zero elapsed time. The shape of the distribution shows patient appointments were routinely scheduled at fixed intervals (e.g., 3-month or 6-month follow-ups), creating structural spikes in the data with a tail of delayed visits. Because these spikes represent procedural reality rather than natural continuous variance, transformations are ineffective. Testing confirmed this, as the log transformation degraded the Shapiro-Wilk test statistic from 0.81 to 0.78. Therefore, the original *days_since_last_visi*t feature was maintained.

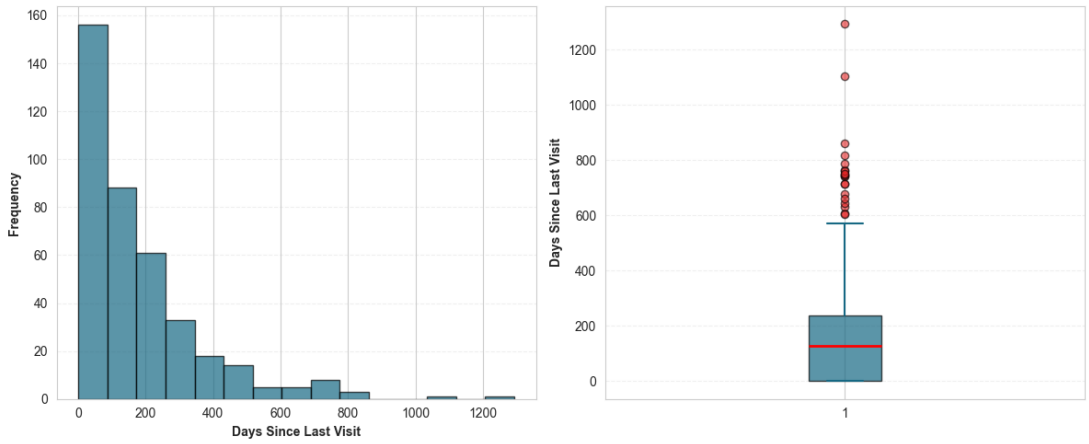


Figure 10: *Distribution of days since last visit.*

In addition, the disease duration variable exhibited a pronounced right skew (Figure 11 (a-b)). The cohort contained a high density of relatively recent diagnoses alongside a long right tail of patients with chronic disease. While a log1p transformation was evaluated (improved W=0.94 over original's W=0.82), the transformed plot (Figure 11 (c-d)) revealed some structural discontinuities. An isolated density bar appeared at zero, followed by discrete gaps before the main continuous curve begins. The zero-inflation represents new patients whose formal diagnosis was assessed at their baseline visit, while the subsequent discretization gaps are structural artifacts stemming from the differences between generalized diagnosis years and exact clinical visit dates. Thus, the log-transformed variant was rejected, and the original *disease_duration* feature was retained. The results from the Shapiro-Wilk test are displayed in detail in Table 12.

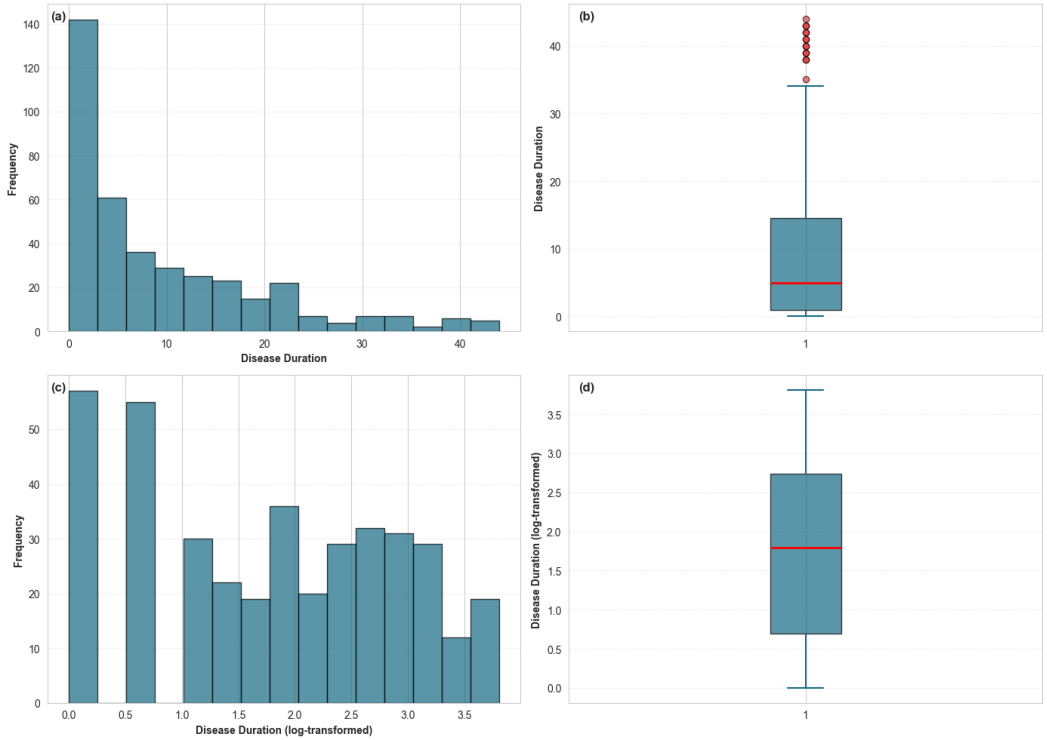


Figure 11: *Distribution of disease duration before and after log1p transformation. (a) Histogram of disease duration values. (b) Boxplot of disease duration values. (c) Histogram of log-transformed disease duration values. (d) Boxplot of log-transformed disease duration values.*

Table 12: *Shapiro-Wilk test results.*

Variable	Type	W statistic	p-value	Normal
das28_score	Original	0.984	0.0070	No
	Log-transformed	0.984	0.0065	No
das28_change	Original	0.835	6.567e-16	No
	Log-transformed	0.844	1.969e-14	No
esr	Original	0.855	8.598e-18	No
	Log-transformed	0.988	0.0037	No
esr_lag_1	Original	0.868	2.250e-14	No
	Log-transformed	0.986	0.0104	No
crp_normalized	Original	0.464	4.086e-30	No
	Log-transformed	0.837	7.241e-18	No
disease_duration	Original	0.821	1.546e-20	No
	Log-transformed	0.943	4.704e-11	No
days_since_last_visit	Original	0.808	2.949e-21	No

Variable	Type	W statistic	p-value	Normal
das28_score	Original	0.984	0.0070	No
	Log-transformed	0.984	0.0065	No
	Log-transformed	0.776	8.222e-23	No
age_at_visit	Original	0.980	2.839e-05	No
	Log-transformed	0.959	4.424e-09	No

Dosing regimens for MTX (*mtx_dose*) and corticosteroids (*steroid_dose*) were evaluated at the visit level. Because continuous medication data are characterized by zero-inflation (small dosages), these variables were analyzed via ordinal categories to better reflect therapeutic thresholds. For both medications, no use of these variables represented most of the observations, accurately reflecting clinical subsets managed on alternative therapies. Specifically, no use of MTX (0 mg) was recorded on 175 visits. Among the actively managed subsets, MTX was used in the moderate 11-20 mg category (150 visits), followed by lower 1-10 mg doses (42 visits) and aggressive 21-25 mg doses (26 visits). Corticosteroid dosing (*steroid_dose*) was similarly evaluated at the visit level, with an emphasis on ordinal categorization to reflect therapeutic use. The data indicated a predominance of zero active dose recordings, with no corticosteroid use documented in 245 clinical visits. Among visits with active corticosteroid therapy, the 1–10 mg dosage category was the most frequently utilized, observed in 139 visits. Escalations above 10 mg were uncommon (9 visits). This rare high-dose intervention likely captures an instance of acute 'bridge therapy' deployed to aggressively suppress a severe inflammatory flare.

Treatment-related variables were then examined. Firstly, the distribution of csDMARDs was evaluated. To assess drug preference without the distortion of unmedicated visits, the analysis was restricted to visits featuring at least one active csDMARD prescription (n=223 visits). As illustrated in Figure 12, MTX and Hydroxychloroquine were the dominant therapeutic agents in this cohort. Specifically, Hydroxychloroquine was utilized in 94.6% (n=211), while MTX was prescribed in 62.7% (n=140). Rare medications (such as Leflunomide and Azathioprine) were aggregated into a single other csDMARD category, representing 5.4% (n=12) of the active visits. The cumulative frequency of these agents significantly exceeds 100%, capturing the term of combination therapy or polypharmacy.

Rather than acting as mutually exclusive options, these drugs are frequently prescribed together to aggressively suppress joint inflammation and prevent disease flares [52]. It is important for predictive modeling, this high prevalence of overlapping co-treatment as it highlights the importance of evaluating these therapies as group and not as isolated, independent variables.

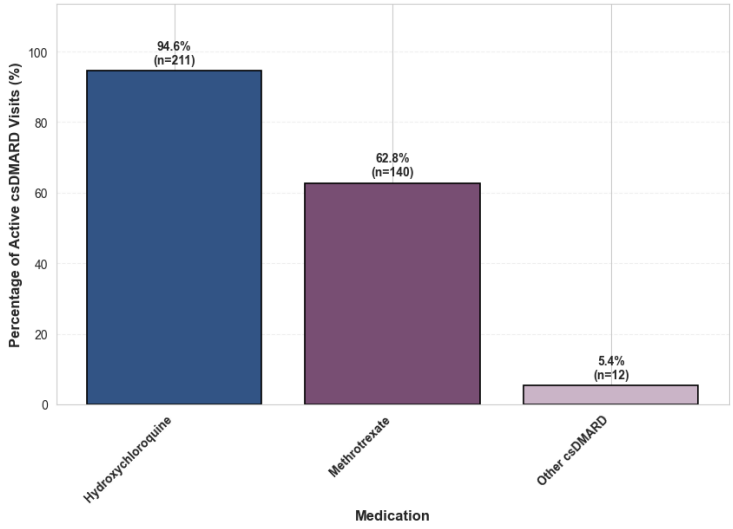


Figure 12: *Distribution of active csDMARD medications.*

Following the conventional therapies, the utilization of bDMARDs and tsDMARDs was analyzed. It was restricted to 160 encounters featuring an active b/tsDMARD prescription. As illustrated in Figure 13, anti-TNF therapies were the most frequently prescribed agents, led by Adalimumab (23.1%, n=37) and Etanercept (20.6%, n=33). The agent Rituximab was the third most common (17.5%, n=28) of the encounters. Upadacitinib and Tocilizumab each contributed 8.1% (n = 13) to the active prescriptions. Low frequency agents, like Abatacept and Infliximab, were grouped into a single category labeled as other b/tsDMARDs, which made up 22.5% (n = 36) of the cohort. Unlike the csDMARD cohort, where polypharmacy was regularly practiced, the cumulative frequency of b/tsDMARD utilization equals exactly 100%. This reflects the clinical contraindication against combining multiple biologic or targeted synthetic agents due to the severe risk of systemic immunosuppression [53]. Therefore b/tsDMARD utilization is entirely mutually exclusive at the visit level.

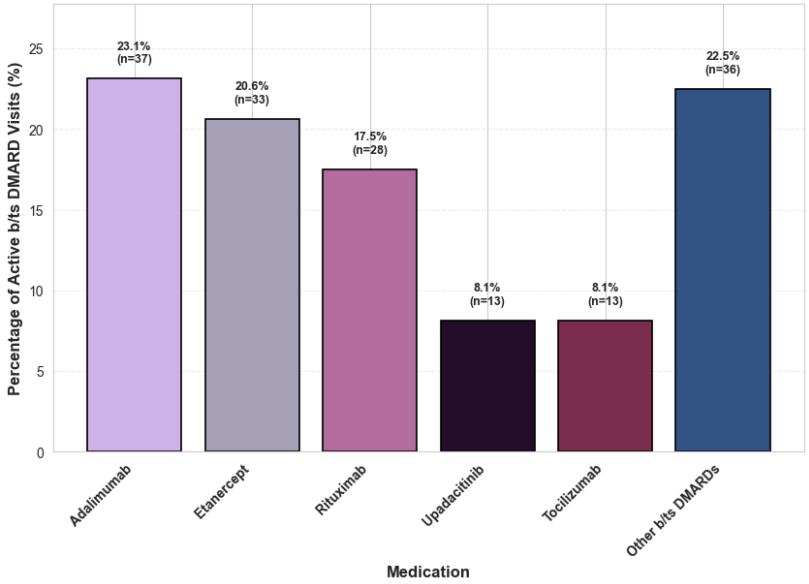


Figure 13: *Distribution of active b/tsDMARD medications.*

The evaluation of patient health histories revealed a considerable presence of systemic comorbidities within the cohort (Figure 14*)*. Metabolic/Endocrine disorders were the most prevalent, affecting 44.3% of the patients, closely followed by Cardiovascular conditions (40.6%). The high frequency of these specific categories aligns with established clinical literature, which identifies conditions such as hypertension and dyslipidemia as primary systemic complications associated with chronic RA [54]. Other notable comorbidities included Musculoskeletal issues (24.5%) and Gastrointestinal / Hepatic conditions (20.8%). Crucially, the cumulative percentages exceed 100%, suggesting a high degree of multimorbidity within the dataset. This is further supported by the engineered *comorbidity_count* feature, which indicates an average of 2.3 concurrent conditions per patient. So, it is demonstrated that patients in this cohort frequently present with complex, overlapping physiological vulnerabilities across multiple organ systems.

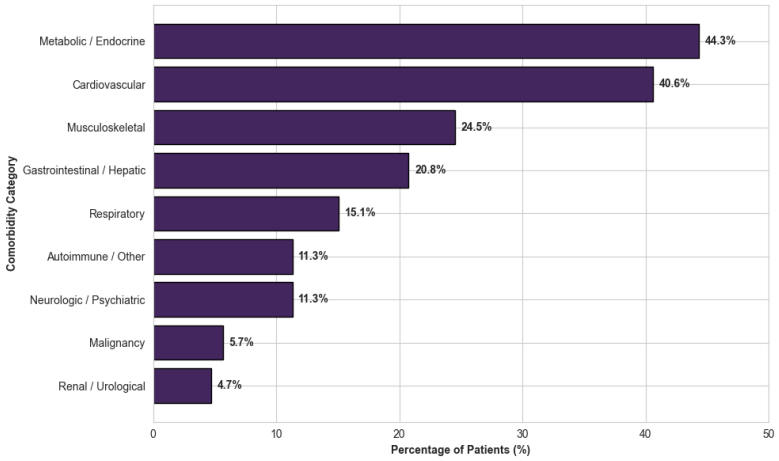


Figure 14: *Distribution of comorbidities.*

4.1.4 Feature Interactions and Target Associations

This section transitions from the univariate characterization of the cohort to a bivariate evaluation of the features against the primary predictive target, *flare_next_visit*. The objective is to identify which variables exhibit the strongest statistical and visual associations with subsequent disease exacerbation.

Figure 15 illustrates the continuous features demonstrating the most distinct variance between the managed (non-flare) and active (flare) states. To provide a comprehensive view of the data distributions, these variables are presented using composite box and line plots, simultaneously displaying medians/interquartile ranges, and the underlying density of individual patient encounters. Clinically the *das28_score* (Figure 15 (a)) showed the most marked visual distinction, with the flare group displaying a noticeably higher median and broader interquartile range compared to the non-flare group, suggesting substantial heterogeneity in disease activity among patients who subsequently experience flares. In addition, the longitudinal trajectory of disease activity (*das28_change*) showed that when a patient had a flare demonstrated a slight momentum toward worsening symptoms, in contrast to the stable or improving trajectories observed in the non-flare group (Figure 15(b)). Similarly, log-transformed inflammatory markers (CRP and ESR) display higher medians in the flare group, with ESR (Figure 15(c)) showing tighter distribution around elevated values and CRP (Figure 15(d)) exhibiting greater variability. Lagged ESR variable (Figure 15(e)) showed a similar pattern with ESR, as expected and *disease_duration* (Figure 15(f)) revealed

an interesting temporal pattern: while both groups share a high density of recently diagnosed patients near the zero-baseline, the flaring population displays an elevated median (~7.0 years vs. ~4.0 years) and a substantially larger upper quartile tail. This indicates that patients with long-standing, established disease chronic up to 40 years represent a higher-risk subgroup for subsequent flare episodes within this dataset.

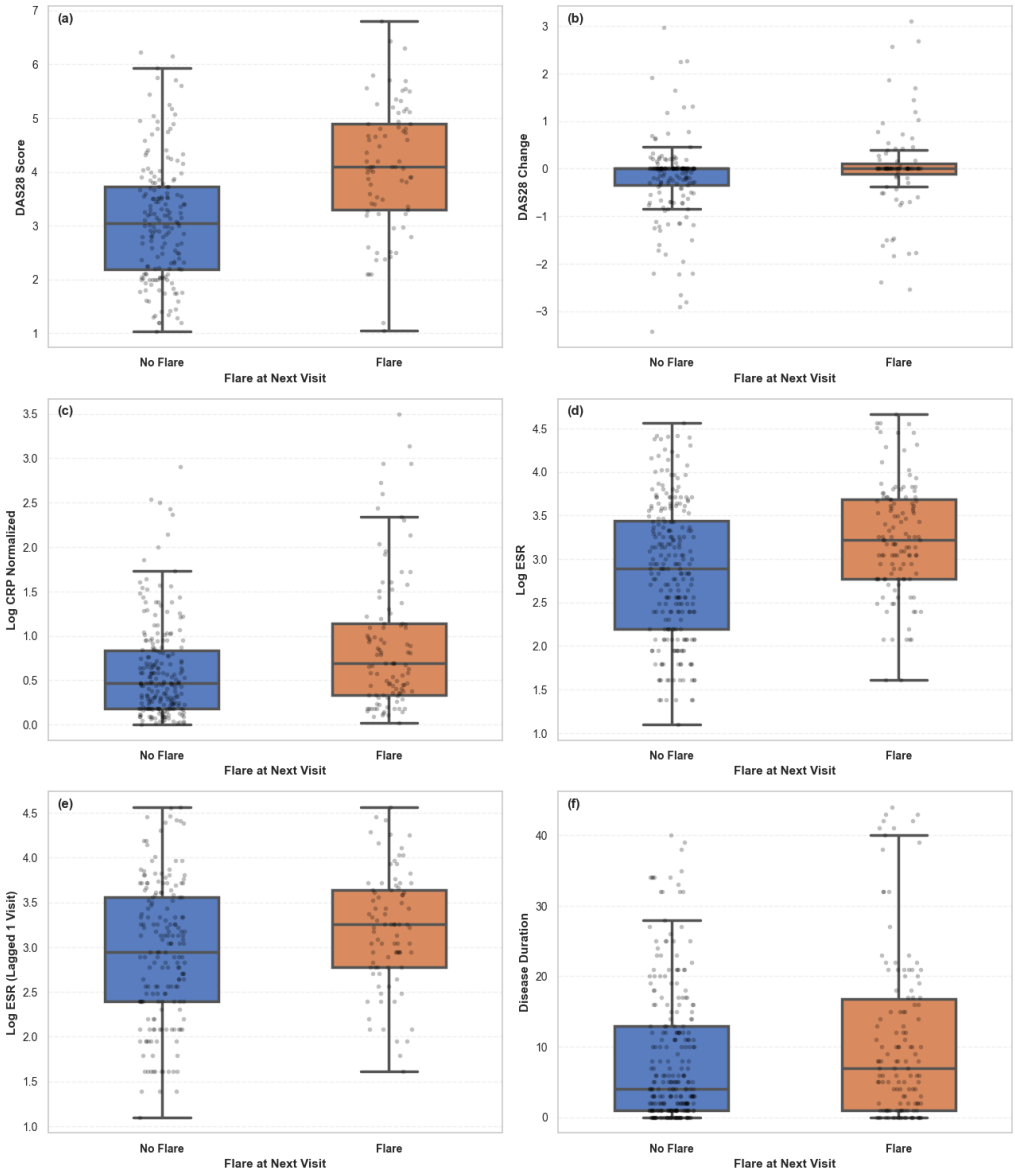


Figure 15: *Bivariate distributions of clinical and laboratory predictors stratified by next-visit flare status. (a) DAS28 score. (b) DAS28 change. (c) CRP levels. (d) ESR levels (e) Lag ESR (1 visit) (f) Disease duration.*

In contrast, *age_at_visit* and *days_since_last_visit* show considerable overlap between groups (Figure 16), with similar median values and comparable interquartile ranges, suggesting these variables may not visually distinguish flare risk. Other continuous variables, such as *treatment_count*, and *comorbidity_count*, exhibited similar uniformity and were consequently omitted from visual presentation.

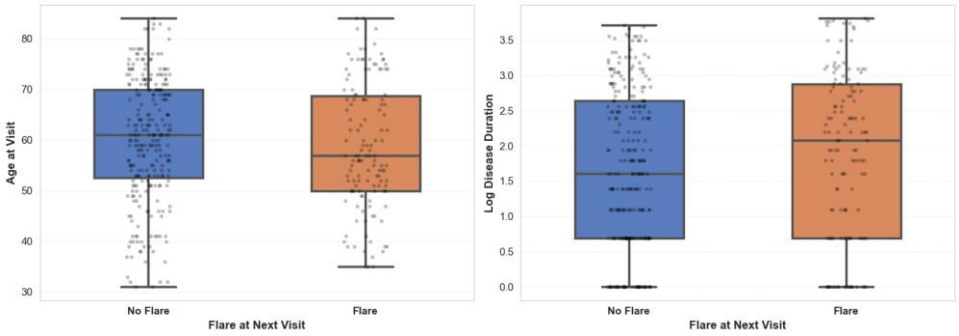


Figure 16: *Bivariate distributions of continuous variables stratified by next-visit flare status. (a) Age at visit. (b) Days since last visit.*

Building upon the analysis of continuous predictors, categorical and binned clinical features were evaluated to assess their proportional on next-visit flare. Stacked bar charts were generated to visualize the relative frequency of flares across different clinical subgroups. Figure 17 highlights two of the most informative binary predictors identified during exploratory analysis: previous flare history and the composite inflammation flag. Patients with a documented history of prior flares occupy a visibly higher proportion of the active flare class compared to those with stable histories. Similarly, the engineered inflammation flag (indicating either CRP or ESR elevations) demonstrates substantial predictive capacity, with flagged patients exhibiting a higher probability of future flares.

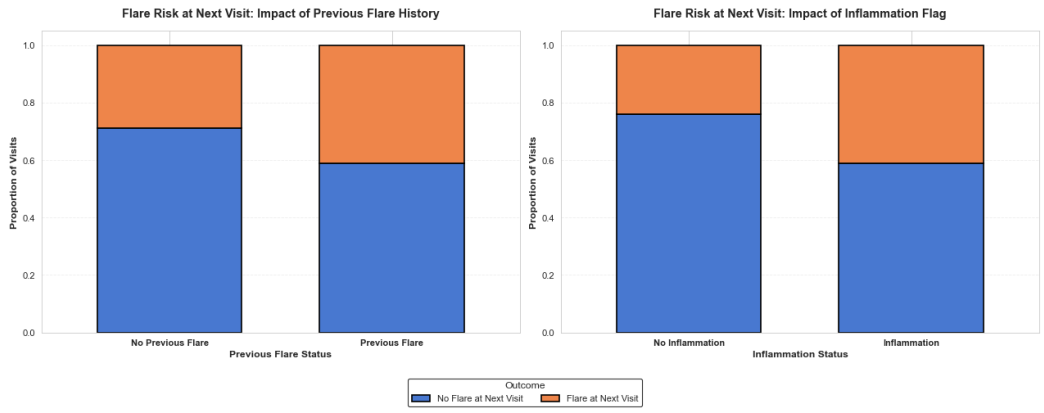


Figure 17: *Proportional risk of flare at the next clinical visit, stratified by previous flare (left) and composite inflammation flag (right).*

The visual analysis was extended to ordinal and binned categorical variables, specifically age groups and DAS28 disease activity categories (Figure 18). Examination of the binned age variable indicates that the risk distribution for flare is nonlinear and fluctuates across different age bands. Notably, the highest proportions of flare risk are observed within the 40-49 age group. Given this irregular pattern, modeling age as a continuous variable is preferable, as it captures the nuances in risk more effectively than arbitrary categorical groupings. The binned DAS28 scores reveal a strict, stepwise, dose-response relationship: as baseline disease activity escalates from Remission/Low to Moderate, and finally to High, the proportion of subsequent flares increases dramatically.

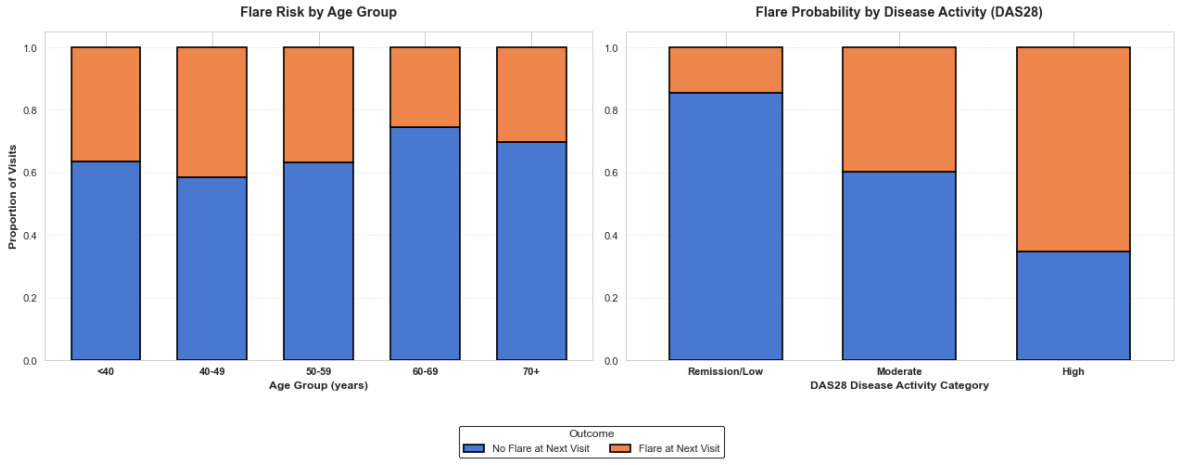


Figure 18*: Proportional risk of flare stratified by age groups (left) and binned DAS28 disease activity categories (right).*

In addition, a correlation matrix demonstrated the associations between the continuous features with each other, and with the target variable (Figure 19). The strongest correlation was naturally between *disease_duration* and *disease_duration_squared* (r=0.94). After that, lower correlations were observed including a positive association between DAS28 score and Log ESR (r = 0.55) and a more moderate with Log CRP (r=0.34) and lagged ESR (r=0.39), reflecting the shared inflammatory nature of these measures and the inclusion of ESR/CRP in DAS28 calculation. *Log_esr* and its antecedent, *log_esr_lag1*, demonstrated a robust temporal correlation (r=0.64), capturing a stable longitudinal trend in systemic inflammation across consecutive clinical intervals. Furthermore, the correlation between DAS28 score and DAS28 change was not very high (r=0.38), showing they do not introduce severe multicollinearity into the feature space. *Age_at_visit* demonstrated a strong positive correlation with comorbidity count (r = 0.53) while it remained weekly decoupled from treatment count (r = -0.11), consistent with expected disease progression patterns in older populations. MTX dose exhibits a moderate negative correlation with age (r = –0.33) and weak negative correlations with disease duration (r = –0.21) and *disease_duration_squared* (r=-0.20), suggesting that younger patients or those with more recent diagnoses receive higher MTX doses. Notable weak associations include the relationship between disease duration and comorbidity count (r = 0.29), and relatively weak correlations between flare occurrence at the next visit and baseline clinical variables (highest r = 0.38 with DAS28 score), suggesting that flare risk depends on complex interactions beyond individual baseline measures. *Days_since_last_visit* showed minimal correlation with disease activity measures, indicating that visit intervals are independent of inflammatory status at the time of assessment.

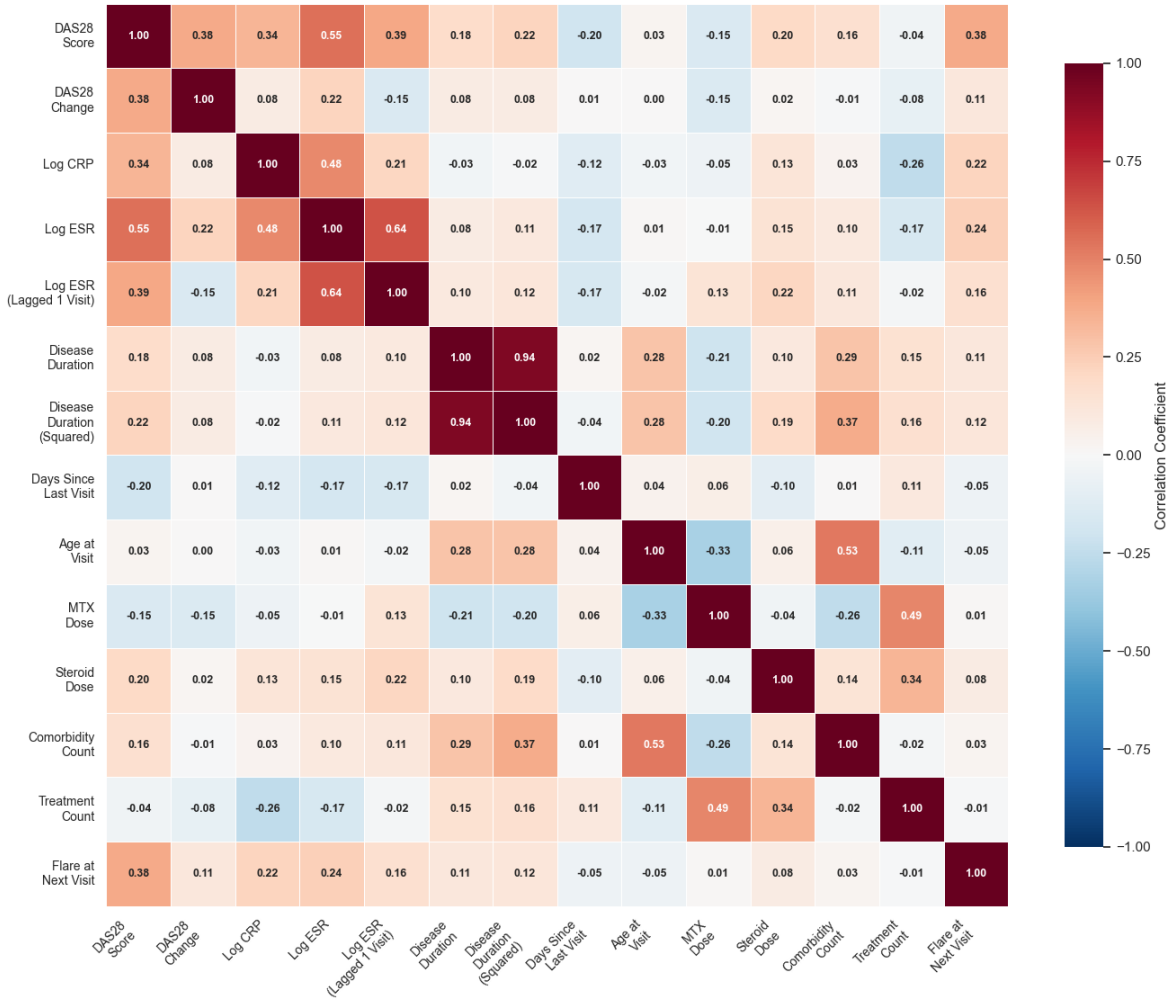


Figure 19: *Pearson correlation matrix for clinical and treatment variables.*

For investigating the correlations between categorical variables, the Cramer's V association matrix was used (Figure 19), revealing weaker associations across the dataset overall. Clinically, previous flare history demonstrates a strong relationship with treatment decisions (V=0.59), capturing the clinical practice of adjusting therapies in response to recent flares. Serological markers (RF_a/anti-CCP status) show the expected association, with V=0.57, confirming their biological relationship in autoimmune pathophysiology. Notably, most treatment variables (medication use, dosing categories, comorbidity presence) show weak associations (V<0.25). Critically, flare at next visit exhibits uniformly weak to moderate relationships with the categorical variables (highest V=0.36 with binned DAS28), indicating that categorical features alone provide limited predictive information for flare risk

and emphasizing the importance of continuous measures and temporal dynamics in flare prediction.

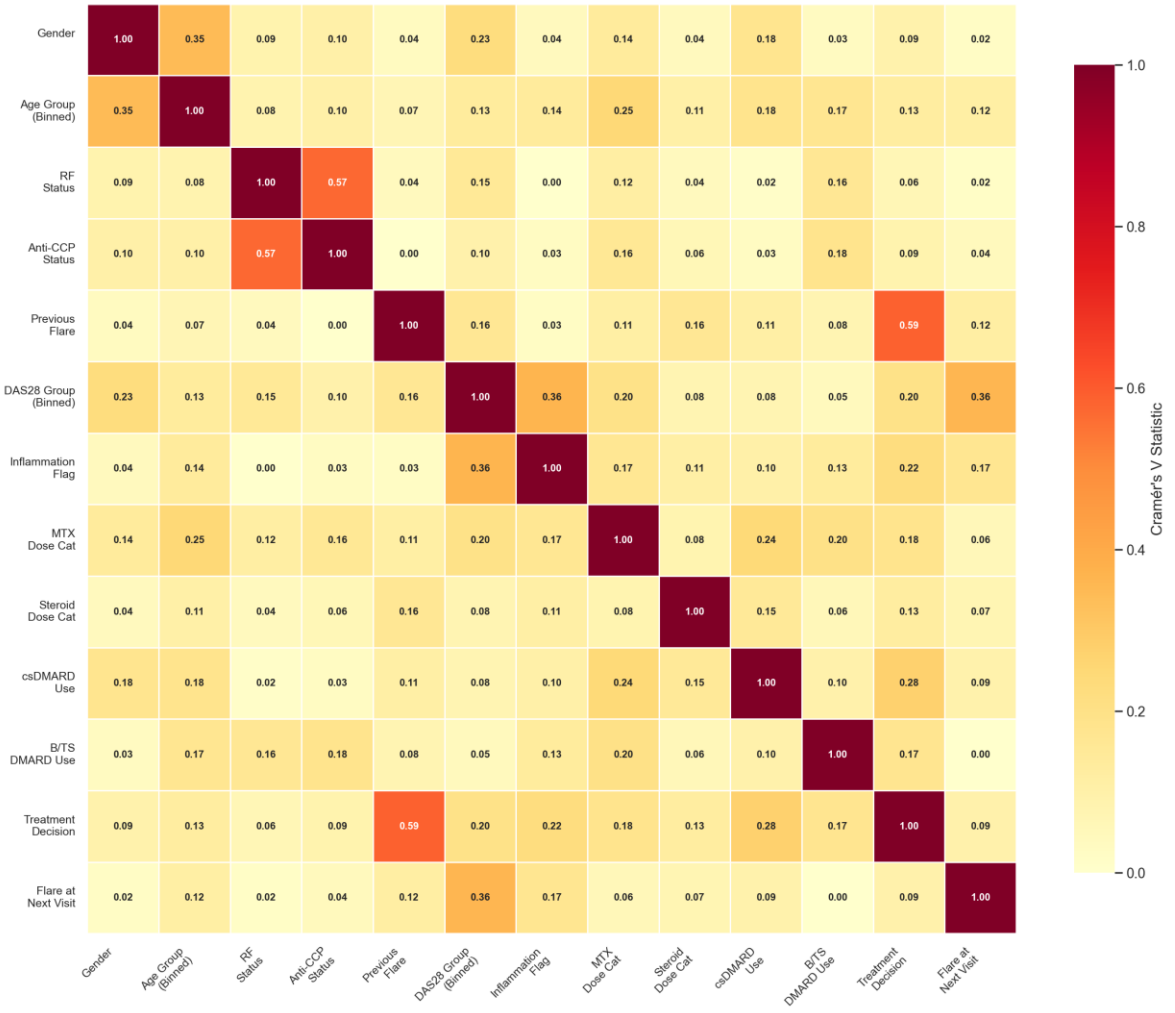


***Figure 20**: Cramer's V correlation heatmap for categorical clinical and treatment variables.*

Finally, to bridge the continuous and categorical feature spaces, an exploratory analysis was conducted to evaluate the relationship between continuous inflammatory markers and binary comorbidity subgroups. The most significant finding was a distinct increase in systemic inflammatory burden among patients with concurrent cardiovascular conditions, as evidenced by a strong statistical association with elevated ESR values ($p = 0.0002$; Figure 21). Clinically, this confirms that the dataset accurately captures the chronic, systemic inflammatory state typically observed in multi-morbid RA patients, where long-term vascular

stress fundamentally elevates baseline ESR. This systemic interaction was consistently reflected across other chronic domains, yielding highly significant associations e.g. between ESR and metabolic/endocrine (p = 0.0082) profiles. In contrast, CRP did not exhibit this same broad pattern of comorbidity association (p-value >0..05). This divergence clinically validates the distinct utility of both biomarkers: while CRP effectively captures acute, localized joint flares, ESR serves as a slower-moving marker that reflects the compounding, chronic inflammatory burden of secondary systemic diseases.

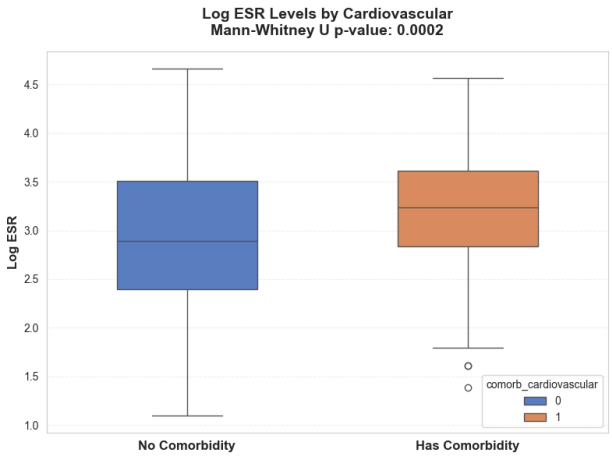


Figure 21: *Log ESR levels by cardiovascular comorbidity.*

Following the inflammatory markers, a second analysis evaluated disease activity (*das28_score*) against clinical actions (*treatment_decision_grouped*). As illustrated in Figure 22, the distribution of continuous DAS28 scores demonstrated a clear, logical, step-down stratification across the categorical treatment decisions (Kruskal-Wallis p=0.0011). Unmanaged baseline encounters exhibited the highest overall disease burden. Among active clinical decisions, a strict 'treat-to-target' staircase pattern emerged: median DAS28 scores were highest in patients requiring therapy escalation or switch in their medication, lower in patients maintained on stable regimens, and lowest in patients where therapy was successfully tapered or stopped. This result clearly shows that doctors followed recommended treatment guidelines, adjusting medications based on how active the disease was. This supports that the data and features are reliable and appropriate for building the ML models described in Chapter 4.2.

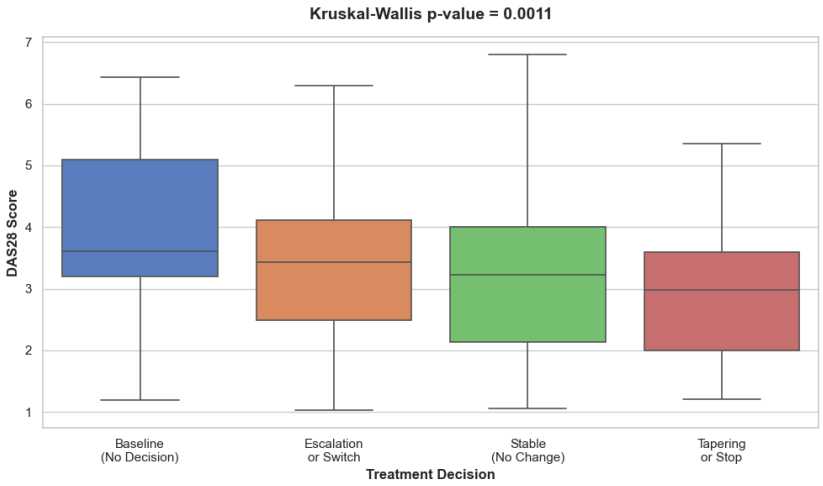


***Figure 22**: DAS28 score by treatment decision categories.*

4.2 Feature Selection Results

4.2.1 Univariate Statistical Screening

To define the final feature set for modeling, the strategy described in Section 3.5.2 was applied. To start with, the Mann-Whitney U test was utilized. The complete results are displayed in Table 13.

***Table 13:** Mann-Whitney U test results.*

Variable	No Flare (Mean±SD)	No flare (median[IQR])	Flare (Mean±SD)	Flare (Median[IQR])	p-value	Effect size |r|
DAS28 score	3.09±1.12	3.04[2.20-3.73]	4.09±1.20	4.09[3.30-4.90]	**<0.001**	**0.475**
DAS28 change	-0.21±0.82	0.00[-0.35-0.00]	0.00±0.92	0.00[-0.11-0.10]	**0.019**	**0.173**
Log ESR	2.87±0.78	2.89[2.20-3.44]	3.26±0.64	3.22[2.77-3.69]	**<0.001**	**0.281**
Log ESR (Lag 1 visit)	2.94±0.78	2.94[2.40-3.56]	3.20±0.65	3.26[2.77-3.64]	**0.0124**	**0.190**
Log CRP	0.60±0.53	0.47[0.18-0.83]	0.90±0.76	0.69[0.34-1.14]	**0.003**	**0.249**

Age at Visit	60.18±12.16	61.00[52.50-70.00]	58.85±12.44	57.00[50.00-68.75]	0.198	0.080
Disease Duration	8.45±9.57	4.00[1.00-13.00]	10.93±11.82	7.00[1.00-16.75]	0.0856	0.106
Disease Duration Squared	162.56±300.80	16.00[1.00-169.00]	258.08±469.78	49.00[1.00-280.75]	0.0856	0.106
Days since last visit	178.20±202.56	131.00[20.00-238.00]	156.85±175.55	111.00[0.00-236.00]	0.302	0.063
Treatment count	1.92±0.95	2.00[1.00-3.00]	1.89±1.07	2.00[1.00-3.00]	0.980	0.002
Comorbidity count	2.23±1.78	2.00[1.00-3.00]	2.34±2.03	2.00[0.25-3.00]	0.751	0.019
MTX dose	9.02±9.03	10.00[0.00-15.00]	9.25±8.82	10.00[0.00-15.00]	0.806	0.014
Steroid dose	1.92±3.33	0.00[0.00-4.00]	2.57±4.21	0.00[0.00-5.00]	0.120	0.084

DAS28 score emerged as the strongest predictor, with a medium-to-large effect size ($r=0.475$, $p<0.001$), with the flare group having a median of 4.09 [IQR 3.30–4.90] compared to 3.04 [IQR 2.20–3.73] in the no-flare group. Log ESR ($r=0.281$, $p<0.001$), log CRP ($r=0.249$, $p=0.003$), and the lagged ESR value from the preceding visit ($r = 0.190$, $p = 0.012$), showed significant but smaller effect sizes, with flare patients demonstrating elevated inflammatory markers at baseline. DAS28 change between visits was statistically significant ($r = 0.173$, $p = 0.019$), though its effect size was smaller than that of the absolute DAS28 score, indicating that current disease activity level is more informative than visit-to-visit trajectory in univariate terms. Several variables did not reach statistical significance, including *age_at_visit* ($r = 0.080$, $p = 0.198$), *disease_duration/disease_duration_squared* ($r = 0.106$, $p = 0.086$), *days_since_last_visit* ($r = 0.063$, $p = 0.302$), *steroid_dose* ($r = 0.084$, $p = 0.120$), *mtx_dose* ($r = 0.014$, $p = 0.806$), *treatment_count* ($r = 0.002$, $p = 0.980$), and *comorbidity_count* ($r = 0.019$, $p = 0.751$).

Chi-Square tests of independence and Cramer's V effect sizes were conducted across all binary and categorical features. The complete results are summarized in Table 14. Three variables demonstrated statistically significant associations with next-visit flare. The categorical DAS28 variable exhibited the strongest association among all binary and categorical features (χ^2 =32.49, p <0.001, V=0.362), with flare rates increasing across disease activity categories: 14.4% in the remission/low activity group, 39.6% in the moderate activity group, and 65.4% in the high activity group, reflecting a clear dose-response pattern between disease activity level and flare occurrence. The inflammation flag similarly demonstrated an association (χ^2 = 10.41, p = 0.001, V = 0.163), with flare rates of 40.6% among visits with evidence of inflammation compared to 24.7% among those without. Previous flare history was significantly associated with next-visit flare (χ^2 = 5.61, p = 0.018, V = 0.120), with patients who had experienced a flare at the preceding visit exhibiting a higher subsequent flare rate (41.2% vs. 28.8%).

Numerous variables demonstrated negligible associations with the target outcome. Gender (V = 0.020, p = 0.689), RF_a status (V = 0.018, p = 0.727), and anti-CCP status (V = 0.044, p = 0.380) showed no meaningful separation between flare and no-flare groups. Among treatment variables, b/tsDMARD usage produced a chi-square statistic of 0.00 (p = 1.000, V = 0.000), with an identical flare rate of 33.0–33.1% across treated and untreated patients. MTX dose category (V = 0.055, p = 0.753) and steroid dose category (V = 0.074, p = 0.345) similarly showed no consistent dose-related pattern in flare rates across their respective categories. Several comorbidity variables warranted caution in statistical interpretation due to low category prevalence. Malignancy (n = 12 positive cases), renal/urological comorbidity (n = 19 positive cases), neurologic/psychiatric (40 positive cases) and autoimmune/other comorbidity (n = 68), were identified as candidates for exclusion on grounds of statistical unreliability.

Table 14: *Association of categorical baseline variables with subsequent RA flare occurrence.*

Variable	Total (n=393)	No Flare (n, row %)	Flare (n, row %)	χ^2	p-value	Cramer's V
DEMOGRAPHICS						
Gender				0.16	0.689	0.020
Male	94	65 (69.1%)	29 (30.9%)			
Female	299	198 (66.2%)	101 (33.8%)			
Age bins				5.35	0.253	0.117
under 40	33	21 (63.6%)	12 (36.4%)			
40-49	53	31 (58.5%)	22 (41.5%)			
50-59	120	76 (63.3%)	44 (36.7%)			
60-69	101	70 (74.5%)	24 (25.5%)			
70 +	93	65 (69.9%)	28 (30.1%)			
CLINICAL VARIABLES AND BIOMARKERS						
RF_a status				0.12	0.727	0.018
No	174	119 (68.4%)	55 (31.6%)			
Yes	216	143 (66.2%)	73 (33.8%)			
Anti-CCP status				0.77	0.380	0.044
No	127	81 (63.8%)	46 (36.2%)			
Yes	263	181 (68.8%)	82 (31.2%)			
Previous Flare				5.61	**0.018**	**0.120**
No	257	183 (71.2%)	74 (28.8%)			
Yes	136	80 (58.8%)	56 (41.2%)			
DAS28 bins				32.49	**< 0.001**	**0.362**
Remission/Low	111	95 (85.6%)	16 (14.4%)			
Moderate	111	67 (60.4%)	44 (39.6%)			
High	26	9 (34.6%)	17 (65.4%)			
Inflammation Flag				10.41	**0.001**	**0.163**
No	186	140 (75.3%)	46 (24.7%)			
Yes	207	123 (59.4%)	84 (40.6%)			
Visit type				1.72	0.189	0.066

Variable	Total (n=393)	No Flare (n, row %)	Flare (n, row %)	χ^2	p-value	Cramer's V
Baseline	106	65 (61.3%)	41 (38.7%)			
Follow-up	287	198 (69.9%)	89 (31.0%)			
TREATMENTS						
MTX dose category				1.20	0.753	0.055
0 mg	175	119 (68.0%)	56 (32.0%)			
1-10 mg	42	29 (69.0%)	13 (31.0%)			
11-20 mg	150	96 (64.0%)	54 (36.0%)			
21-25 mg	26	19 (73.1%)	7 (26.9%)			
Steroid dose category				2.13	0.345	0.074
0 mg	245	169 (69.0%)	76 (31.0%)			
1-10 mg	139	87 (62.6%)	52 (37.4%)			
>10 mg	9	7 (77.8%)	2 (22.2%)			
csDMARD use				3.20	0.074	0.090
No	170	105 (61.8%)	65 (38.2%)			
Yes	223	158 (70.9%)	65 (29.1%)			
b/tsDMARD use				0.00	1.00	0.000
No	233	156 (67.0%)	77 (33.0%)			
Yes	160	107 (66.9%)	53 (33.1%)			
Treatment decision grouped				3.49	0.322	0.094
Baseline-no decision	106	65 (61.3%)	41 (38.7%)			
Escalation/ switch	124	82 (66.1%)	42 (33.9%)			
Stable/no change	105	77 (73.3%)	28 (26.7%)			
Tapering / stop	58	39 (67.2%)	19 (32.8%)			
COMORBIDITIES						
Cardiovascular				2.12	0.145	0.073
No	251	175 (69.7%)	76 (30.3%)			
Yes	142	88 (62.0%)	54 (38.0%)			
Metabolic/Endocrine				2.23	0.136	0.075

Variable	Total (n=393)	No Flare (n, row %)	Flare (n, row %)	χ^2	p-value	Cramer's V
No	241	154 (63.9%)	87 (36.1%)			
Yes	152	109 (71.7%)	43 (28.3%)			
Respiratory				0.82	0.365	0.046
No	334	220 (65.9%)	114 (34.1%)			
Yes	59	43 (72.9%)	16 (27.1%)			
Renal/Urological				0.00	1.000	0.000
No	374	250 (66.8%)	124 (33.2%)			
Yes	19	13 (68.4%)	6 (31.6 %)			
Gastrointestinal/Hepatic				0.60	0.439	0.039
No	285	187 (65.6%)	98 (34.4%)			
Yes	108	76 (70.4%)	32 (29.6%)			
Malignancy				2.37	0.124	0.078
No	381	252 (66.1%)	129 (33.9%)			
Yes	12	11 (91.7%)	1 (8.3%)			
Musculoskeletal				3.09	0.079	0.089
No	278	194 (69.8%)	84 (30.2%)			
Yes	115	69 (60.0%)	46 (40.0%)			
Neurologic/Psychiatric				0.65	0.421	0.041
No	353	239 (67.7%)	114 (32.2%)			
Yes	40	24 (60.0%)	16 (40.0%)			
Other				0.00	1.000	0.000
No	325	217 (66.8%)	108 (33.2%)			
Yes	68	46 (67.6%)	22 (32.4%)			

In summary, univariate statistical screening identified eight variables with statistically significant associations with next-visit flare: DAS28 score, log-transformed ESR, log-transformed CRP, log-transformed lagged ESR, DAS28 change, previous flare history, das28 disease activity category, and inflammation flag. Several variables with borderline associations such as disease duration/disease duration squared, steroid dose, MTX dose, csDMARD use, some comorbidity groups, treatment decision grouping, were retained as

candidates for subsequent evaluation. Variables demonstrating near-zero effect sizes across both statistical tests, specifically treatment count, comorbidity count, gender, RF_a status, MTX dose category, steroid category, and b/tsDMARD use. Among comorbidity variables, respiratory, renal/urological, gastrointestinal/hepatic, malignancy, neurologic/psychiatric and other comorbidities. As univariate tests evaluate each feature independently, they cannot detect multivariate relationships, so the remaining candidate features were therefore carried forward to pairwise correlation analysis and multivariate importance assessment prior to finalizing the feature set.

4.2.2 Correlation Analysis and Redundancy Assessment

Pairwise correlations among continuous and binary features were examined to identify redundant variable pairs arising from mathematical derivation relationships or shared underlying constraints. First, disease duration and its squared transformation exhibited a near perfect relationship (r=0.937), which is expected given that *disease_duration_squared* is calculated from the *disease_duration*. The squared transformation was specifically engineered to capture the hypothesized nonlinear relationship between disease duration and flare risk, where both early and long-lasting disease may carry elevated risk through distinct mechanisms. Therefore, the *disease_duration_squared* was retained and the *disease_duration* was excluded. Second, DAS28 score and binned DAS28 disease activity category demonstrated a Spearman correlation of r=0.905 ($p<0.001$), reflecting the direct association of the categorical variable with the continuous score. The *DAS28_score* was retained in preference to the binned variable, as it preserves the full metric information of the original measurement without information loss through binning. Similarly, age at visit and age bins showed a very strong relationship ($r = 0.971$, $p < 0.001$), resulting in keeping the continuous age variable as it preserves more information.

Among these retained features, several moderate correlations were noted but did not reach the exclusion threshold. Log ESR and log-transformed normalized CRP demonstrated a Pearson correlation of $r = 0.481$ ($p < 0.001$), reflecting their shared role as systemic inflammation markers while retaining sufficient independence to justify simultaneous inclusion. Inflammation flag showed point-biserial correlations of $r = 0.647$ with log ESR ($p < 0.001$) and $r = 0.538$ with log CRP ($p < 0.001$), indicating substantial but not redundant

overlap. Additionally, log ESR and its lagged value from the preceding visit demonstrated a correlation of r=0.640, reflecting the expected persistence of inflammatory state across consecutive visits while remaining sufficiently independent to contribute distinct information. Following the exclusion of near-redundant pairs, the remaining feature set was carried forward for a multivariate importance assessment using RF importance and cross-fold stability analysis, as described in the following section.

4.2.3 Random Forest Importance and Stability

To supplement the univariate statistical analysis and capture non-linear associations not detectable by Mann-Whitney or chi-square tests, a RF classifier was applied to the candidate feature set as a multivariate screening tool. The results are presented in Figure 23 and Table 15.

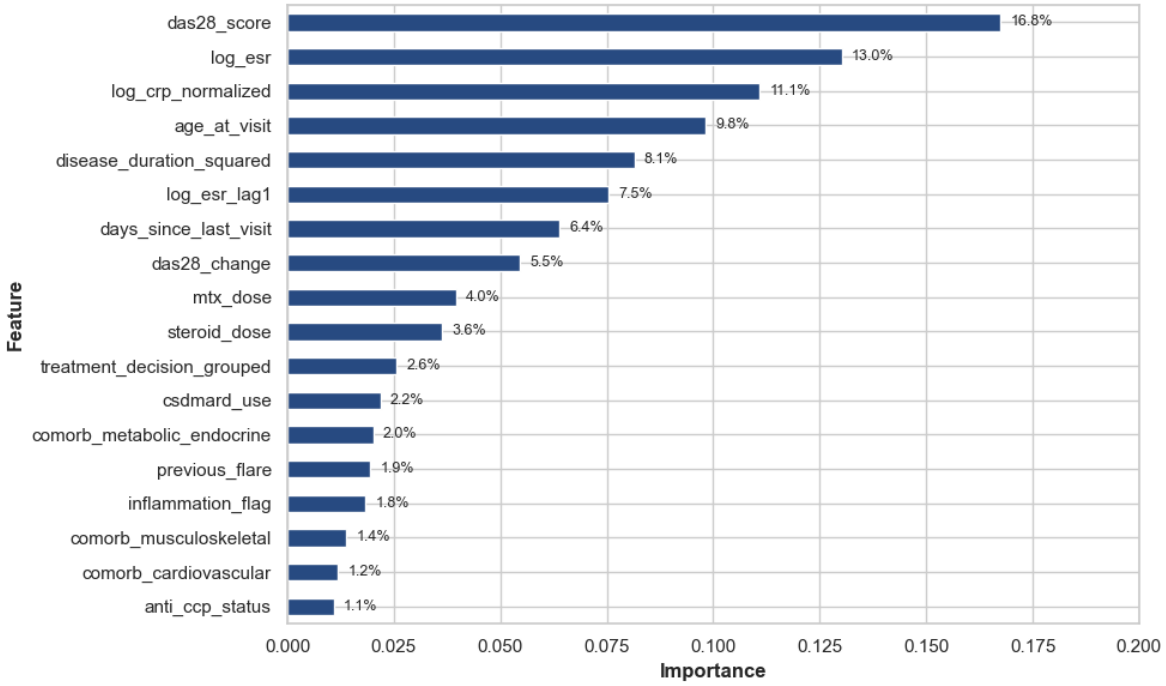


***Figure 23**: Cross-validated RF feature importance.*

***Table 15**: Cross validated RF importance ranking.*

Feature	Importance	Importance std	cv ratio
das28_score	0.168	0.011	0.064

log_esr	0.130	0.005	0.041
log_crp_normalized	0.111	0.011	0.101
age_at_visit	0.098	0.006	0.064
disease_duration_squared	0.081	0.009	0.113
log_esr_lag1	0.075	0.008	0.100
days_since_last_visit	0.064	0.002	0.025
das28_change	0.054	0.007	0.124
mtx_dose	0.040	0.007	0.187
steroid_dose	0.036	0.003	0.090
treatment_decision_grouped	0.026	0.003	0.111
csdmard_use	0.022	0.006	0.256
comorb_metabolic_endocrine	0.020	0.007	0.347
previous_flare	0.019	0.005	0.236
inflammation_flag	0.018	0.006	0.313
comorb_musculoskeletal	0.014	0.002	0.153
comorb_cardiovascular	0.012	0.002	0.163
anti_ccp_status	0.011	0.001	0.072

DAS28 score demonstrated the highest overall importance (mean importance = 0.168, CV ratio = 0.064), consistent with its dominant effect size in univariate screening ($r = 0.475$, $p<0.001$). Log-transformed ESR and log-transformed normalized CRP ranked second and third respectively (mean importance = 0.130 and 0.111), both demonstrating high cross-fold stability (CV ratio = 0.041 and 0.101). These three-disease activity and inflammatory marker features constituted a clear top tier, collectively accounting for most of the model's discriminative capacity. Age at visit (mean importance = 0.098, CV ratio = 0.064) and disease duration squared (mean importance = 0.081, CV ratio = 0.113) ranked fourth and fifth respectively. Lagged ESR from the preceding visit (mean importance = 0.075, CV ratio=0.100) and DAS28 change (mean importance = 0.054, CV ratio=0.124) contributed independently, confirming that temporal features capturing prior inflammatory state and disease trajectory carried additional signal beyond the cross-sectional measurements. Days

since last visit ranked seventh with a relatively high mean importance of 0.064 and the lowest CV ratio of the entire candidate set (0.025), indicating a highly stable importance estimate across folds. MTX dose and steroid dose ranked ninth and tenth respectively (mean importance = 0.040 and 0.036), with moderate stability (CV ratio = 0.187 and 0.090). Both were retained in the candidate feature set on clinical grounds as proxies for physician-assessed disease severity, despite their modest multivariate importance. Treatment decision grouping demonstrated stable low importance (mean = 0.026, CV ratio = 0.111) and was retained based on the clinically meaningful distribution of flare rates across its categories identified in univariate screening.

Several variables demonstrated low importance across all folds. Cardiovascular (mean = 0.012, CV ratio = 0.163), musculoskeletal (mean = 0.014, CV ratio = 0.153), and metabolic/endocrine (mean = 0.020, CV ratio = 0.347) comorbidity groups ranked in the lower third of the importance distribution. Despite their weak univariate significance, these three categories were deliberately kept in the final feature set. This choice was made because they are the most common comorbidity groups in the dataset, making them clinically important to preserve them as baseline controls for the models. The csDMARD use (mean = 0.022, CV ratio = 0.256) and anti-CCP status (mean = 0.011, CV ratio = 0.072) similarly showed limited multivariate contribution. Notably, previous flare (mean = 0.019, CV ratio = 0.236) and inflammation flag (mean = 0.018, CV ratio = 0.313) ranked below their univariate significance suggested. This pattern is consistent with the known tendency of impurity-based RF importance to undervalue binary features relative to continuous variables, which offer more candidate split points and therefore more opportunities to reduce node impurity. Both variables were retained due to their statistically significant univariate associations ($V = 0.120$, $p = 0.018$ and $V = 0.163$, $p = 0.001$ respectively) rather than their RF importance rankings.

Nine features were identified as stable across folds by the automated selection criterion (mean importance above the median and CV ratio below 1.0): DAS28 score, DAS28 change, log ESR, log-transformed CRP, lagged ESR, age at visit, disease duration squared, days since last visit, and MTX dose. The finalized feature set passed forward to the ML models is detailed in Table 16 below.

***Table 16**: The final feature set for modeling.*

anti_ccp_status	*csdmard_use*
das28_score	*mtx_dose*
das28_change	*steroid_dose*
log_esr	*treatment_decision_grouped*
log_crp_normalized	*days_since_last_visit*
log_esr_lag1	*previous_flare*
age_at_visit	*comorb_cardiovascular*
inflammation_flag	*comorb_metabolic_endocrine*
disease_duration_squared	*comorb_musculoskeletal*

4.3 Baseline Models and Imputation Evaluation

Prior to hyperparameter tuning, a baseline evaluation was conducted across all four classifiers (LR, SVM, RF, XGBoost) using default or lightly constrained configurations. This preliminary study served two purposes: to establish a performance benchmark for each algorithm under comparable conditions, and to find which imputation technique is the best for the three disease activity variables with high missingness (*das28_score, log_crp_normalized, log_esr*). Four imputation methods were used: median, KNN, MICE-Bayesian, and MICE-RF, and evaluated within the CV framework, with imputation fitted exclusively on training folds to prevent information leakage. The performance across all metrics (Accuracy, Precision, Recall, F1 score, AP, and ROC-AUC) is reported in Table 17 and the cross-strategy comparison across all baseline classifiers and folds is presented in Figure 24.

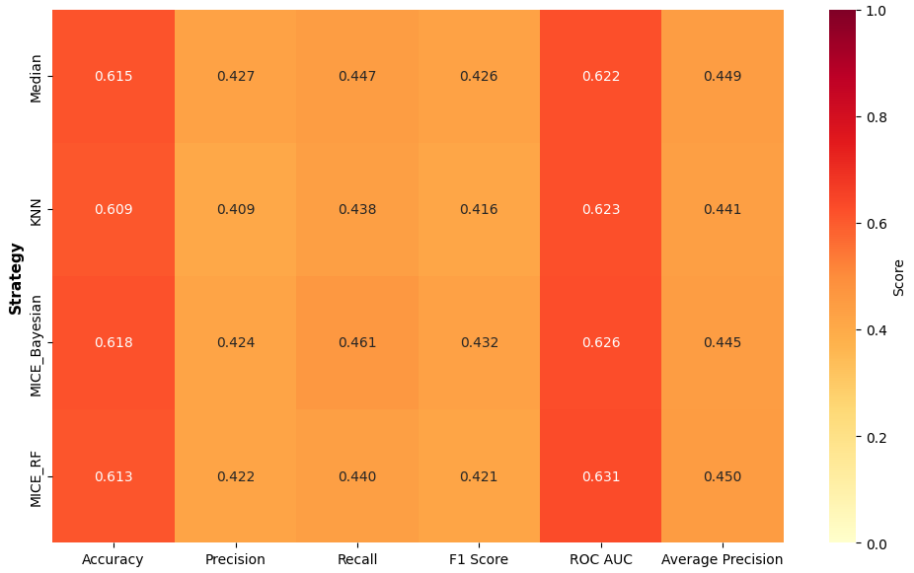


***Figure 24**: Imputation strategy comparison.*

Examining the aggregated cross-strategy comparison first, performance differences across imputation strategies were modest in absolute terms but consistent in direction. MICE-RF achieved the highest ROC-AUC (0.631) and matched median imputation for the highest AP (0.450), while MICE-Bayesian demonstrated the highest Recall (0.461) and F1-score (0.432). Median imputation and KNN produced the weakest overall profiles, with KNN yielding the lowest AP (0.441) and F1-score (0.416) across all classifiers. These results suggest that both MICE variants offered a consistent, if modest, improvement over simpler imputation approaches.

At the individual classifier level, patterns were more nuanced. For LR, MICE-Bayesian achieved the highest AP (0.503) and ROC-AUC (0.670), marginally outperforming median imputation (AP=0.501) and MICE-RF (AP=0.491). For SVM, differences across strategies were minimal. AP ranged narrowly from 0.429 to 0.441, with MICE-Bayesian again producing the highest AP (0.441) and ROC-AUC (0.623). For RF, MICE-Bayesian similarly yielded the highest AP (0.430) and ROC-AUC (0.630), though median imputation produced competitive results (AP = 0.445), and all strategies showed relatively small differences given the high standard deviations across folds. For XGBoost, MICE-RF demonstrated clear superiority, achieving the highest AP (0.446) and ROC-AUC (0.626) and

outperforming MICE-Bayesian (AP = 0.408) and KNN (AP = 0.401), which were the weakest strategies for this classifier.

***Table 17**: Baseline classifier performance across imputation strategies.*

	Accuracy	Precision	Recall	F1-score	AP	ROC-AUC
LR						
Median	0.636 ± 043	0.467 ± 0.056	0.586 ± 0.136	0.511 ± 0.058	0.501 ± 0.066	0.659 ± 0.041
KNN	0.628 ± 031	0.454 ± 0.043	0.579 ± 0.138	0.502 ± 0.060	0.496 ± 0.068	0.661 ± 0.046
MICE-Bayesian	0.649±0.040	0.478 ± 0.060	0.610 ± 0.161	0.528 ± 0.076	0.503 ± 0.076	0.670 ± 0.052
MICE-RF	0.639±0.030	0.466 ± 0.040	0.602 ± 0.136	0.519 ± 0.058	0.491 ± 0.072	0.661 ± 0.050
SVM						
Median	0.596±0.045	0.412 ± 0.062	0.470 ± 0.102	0.431 ± 0.051	0.439 ± 0.052	0.619 ± 0.053
KNN	0.606±0.074	0.424 ± 0.099	0.494 ± 0.121	0.453 ± 0.098	0.439 ± 0.050	0.623 ± 0.064
MICE-Bayesian	0.609±0.064	0.425 ± 0.094	0.509 ± 0.143	0.458 ± 0.100	0.441 ± 0.059	0.623 ± 0.059
MICE-RF	0.599±0.054	0.412 ± 0.076	0.464 ± 0.121	0.429 ± 0.075	0.429 ± 0.053	0.610 ± 0.069
RF						
Median	0.600±0.055	0.395 ± 0.084	0.362 ± 0.126	0.368 ± 0.084	0.445 ± 0.045	0.626 ± 0.024
KNN	0.601±0.039	0.387 ± 0.073	0.363 ± 0.103	0.370 ± 0.080	0.428 ± 0.070	0.621 ± 0.042
MICE-Bayesian	0.613±0.045	0.423 ± 0.065	0.441 ± 0.123	0.423 ± 0.076	0.430 ± 0.052	0.630 ± 0.035
MICE-RF	0.601±0.057	0.392 ± 0.088	0.370 ± 0.108	0.376 ± 0.090	0.432 ± 0.052	0.628 ± 0.039
XGBoost						
Median	0.628±0.029	0.432 ± 0.037	0.370 ± 0.053	0.396 ± 0.035	0.409 ± 0.019	0.584 ± 0.023
KNN	0.600±0.034	0.373 ± 0.053	0.316 ± 0.096	0.338 ± 0.073	0.401 ± 0.071	0.589 ± 0.064
MICE-Bayesian	0.600±0.033	0.369 ± 0.044	0.284 ± 0.050	0.318 ± 0.043	0.408 ± 0.049	0.582 ± 0.046
MICE-RF	0.613±0.071	0.420 ± 0.117	0.323 ± 0.032	0.361 ± 0.058	0.446 ± 0.059	0.626 ± 0.046

Median and KNN were included as reference baselines for comparative purposes and were not considered for adoption in the final modeling pipeline. Median imputation, while computationally efficient, it is too simplistic and disregards the relationships between

variables and introducing systematic bias in derived features that depend on imputed values, such as DAS28 change, and lagged ESR. KNN was also excluded because its reliance on proximity metrics becomes highly unstable under high missingness, which compromises the identification of true nearest neighbors. Furthermore, KNN is sensitive to skewed distributions and differences in feature scale, posing a risk of localized bias even following standardization.

The final selection of an imputation strategy prioritized methodological uniformity and pipeline consistency across all downstream models. MICE-Bayesian demonstrated clear superiority for the linear architectures (LR and SVM), as their decision boundaries benefited from the smooth, regularized distributions produced by Bayesian Ridge Regression. For the RF, the baseline performance metrics between MICE-RF and MICE-Bayesian were exceptionally close, yielding no statistically significant difference. While the un-tuned XGBoost baseline showed a localized preference for MICE-RF, adopting separate imputation pipelines for individual classifiers introduces architectural fragmentation and risks overfitting to baseline parameter noise. Therefore, MICE-Bayesian was selected as the definitive, unified imputation strategy across all four classifiers.

4.4 Hyperparameter Tuning Results

Following the baseline evaluation, each classifier underwent hyperparameter optimization within the nested CV framework described in Section 3.5.1. Performance metrics reported in this section were derived exclusively from out-of-fold probability estimates produced by the outer loop, representing honest generalization to unseen data. The final CV metrics are compared against the corresponding baseline configuration using the imputation strategy selected in Section 4.3.

The tuned LR demonstrated complete hyperparameter stability across all 5 outer folds, with identical parameters selected in every fold: C=0.05, l1_ratio=0.5, and class_weight=None. This uniformity indicates that the inner CV reliably and constantly identified the same optimal regularization configuration regardless of fold composition, providing high confidence in the selected parameters. The l1_ration equals to 0.5, shows that a balance between L1 and L2 regularization was preferred and a strong C reflected the

model's preference for heavily penalized coefficients given the small dataset size. The consistent selection of class_weight=None suggests that the default threshold adjustment was the preferred strategy for handling class imbalance in this model. The tuned model achieved an AP of 0.508 and ROC-AUC of 0.695, representing improvements of 0.005 and 0.025 respectively from the baseline configuration (AP=0.503, ROC-AUC=0.670). The mean predicted probability of 0.330 closely approximated the true flare rate of 0.331, indicating excellent probability calibration. The maximum predicted probability of 0.688 reflects appropriate conservatism. The optimal decision threshold of 0.295 reflects the model's prioritization of recall over precision in the clinical context. The final model changing from class_weight=None to class_weight=balanced was expected given than the final refit operates on the full class distribution rather than individual fold partitions and does not affect the nested CV performance estimates. Inner AP across folds ranged from 0.468 to 0.571, reflecting moderate fold-to-fold variability attributable to the small dataset size rather than model instability.

The tuned SVM demonstrated substantial kernel instability across outer folds, with three different kernels selected across 5 folds: polynomial kernel in folds 1, 4, and 5, RBF kernel in fold 2 and linear kernel in fold 3. This variability indicates that the inner CV did not identify a consistently superior kernel function, reflecting the dataset's position at the boundary between linear and mildly non-linear separability. The class_weight=balanced was selected in all 5 folds, representing the only stable hyperparameter across the SVM configuration. Feature selection via SelectKBest was similarly unstable, with k = 5 selected in 3 folds and k = all selected in one, suggesting the inner CV could not reliably determine the optimal feature subset size at this sample size. Among polynomial kernel folds, low C values (C = 0.01) and high polynomial degrees (degree = 3–4) were preferred, indicating that the inner CV favored boundary flexibility through kernel complexity rather than margin width. The tuned SVM achieved an AP of 0.470 and ROC-AUC of 0.639, representing an improvement of 0.029 in AP and 0.016 in ROC-AUC over the baseline (AP = 0.441, ROC-AUC = 0.623). However, the mean predicted probability of 0.347 was slightly elevated relative to the true flare rate of 0.331, and the maximum predicted probability of 0.817 indicated greater overconfidence than LR. The optimal threshold was identified at 0.288, suggesting a broad but skewed probability distribution that required a sub-default threshold

to achieve adequate flare detection sensitivity. Inner AP ranged from 0.478 to 0.591, with fold 4 notably outperforming other folds. The final model refit selected a linear kernel with C=100, diverging substantially from the low-C polynomial configurations preferred during nested C. This divergence between fold-level selections and the final full-data refit is common when kernel preference is unstable across folds and further underscores the unreliability of kernel selection at this sample size.

RF exhibited moderate hyperparameter stability across outer folds. Maximum depth of 3 was selected in 4 of 5 folds, confirming the model's strong preference for shallow trees. A minimum samples per leaf of 10 was selected in 4 folds, enforcing conservative node sizes that prevent overfitting. However, several parameters showed fold to fold variability. For example, the splitting criterion alternated between Gini impurity and entropy across folds, and class weighting switched between balanced and balanced_subsample, showing that the inner CV could not truly distinguish between these configurations. The min_impurity_decrease parameter varied across its full search range, reflecting similar uncertainty. The tuned RF achieved an AP of 0.414 and ROC-AUC of 0.646, representing an improvement of 0.016 in ROC-AUC but a marginal decrease of 0.016 in AP relative to the baseline (AP = 0.430, ROC-AUC = 0.630). The mean predicted probability of 0.457 substantially exceeded the true flare rate of 0.331, indicating systematic probability overestimation as can been observed from the elevated optimal threshold (0.376). Inner AP ranged from 0.420 to 0.497, with a notably low score in fold 1 (0.420) contributing to the high variance. The final model refit selected entropy criterion with log2 feature sampling and balanced class weighting with max_depth=3, consistent with the dominant fold-level preferences.

XGBoost demonstrated strong stability in several key parameters across outer folds. Maximum depth of 2 was selected in 4 out of 5 folds, with depth 3 selected only in fold 2, confirming the consistent preference for shallow boosting trees observed during baseline evaluation and reflecting the limited training data available per fold. A slow learning rate of 0.005 was selected in 4 out of 5 folds, with 0.01 in fold 1 only, showing a preference for gradual boosting convergence paired with higher tree counts, Subsample was fixed in 0.6 across all folds. Min_child_weight achieved a value of 10 in 4 of 5 folds, suggesting that aggressive subsampling was beneficial as a variance reduction mechanism in this small

dataset. The gamma parameter showed partial stability, with the minimum split gain of 0.1 selected in 4 folds and 0.5 in fold 1 only. The reg_lambda equals to 5 was selected in 4 of 5 folds, representing a preference for strong leaf weight penalization consistent across most fold compositions. The term reg_alpha showed greater variability ranging from 0 to 1 across folds, indicating that the inner CV could not reliably determine the optimal sparsity penalty. The scale_pos_weight values selected across folds ranged between 1.50 and 1.52, consistently below the exact training fold ratio of approximately 2.03, indicating that moderate rather than full class weight correction was preferred. The tuned XGBoost achieved an AP of 0.450 and ROC-AUC of 0.666, representing improvements of 0.042 in AP and 0.084 in ROC-AUC relative to the baseline configuration (AP = 0.408, ROC-AUC = 0.582). The maximum predicted probability was 0.656, indicating a compressed and conservative probability range. The mean predicted probability of 0.413 remained elevated relative to the true flare rate of 0.331. Paired with an optimal threshold of 0.342, this distribution reflects a tendency toward probability that is typical for tree-based models trained on imbalanced data. Inner AP ranged from 0.437 to 0.514, and the final refit selected a depth of 3 with reg_lambda=5, reg_alpha=0.5, and gamma=0.3, partially consistent with the fold level preferences for strong regularization and slow learning. The summary of the results of AP and ROC-AUC, for each classifier is presented in Table 18.

***Table 18**: Cross-model comparison summary.*

Model	Baseline AP	Tuned AP	ΔAP	Baseline ROC-AUC	Tuned ROC-AUC	ΔROC-AUC	Avg Prob	Threshold
LR	0.503	0.508	+0.005	0.670	0.695	+0.025	0.330	0.295
SVM	0.441	0.470	+0.029	0.623	0.639	+0.016	0.347	0.288
RF	0.430	0.414	-0.016	0.630	0.646	+0.016	0.457	0.376
XGBoost	0.408	0.450	+0.042	0.582	0.666	+0.084	0.413	0.342

Across all four classifiers, hyperparameter tuning produced consistent improvements in ROC-AUC. AP results were more mixed: LR, SVM, and XGBoost showed improvements of varying magnitude, while RF showed a marginal decrease attributable to the nested CV framework providing more conservative estimates than the baseline evaluation procedure. The optimal decision thresholds varied systematically across classifiers, ranging from 0.288

for SVM to 0.376 for RF, reflecting each model's characteristic probability output distribution rather than differences in underlying discriminative capacity. All four thresholds fell below the default value of 0.5, consistent with the moderate class imbalance in the dataset and the clinical objective of prioritizing flare detection sensitivity over specificity. XGBoost demonstrated the largest absolute gains following tuning, improving by 0.042 in AP and 0.084 in ROC-AUC. LR achieved the highest AP (0.508) and ROC-AUC (0.695) among all tuned models, alongside the best probability calibration and the most stable hyperparameter selection across folds. These results are examined in the context of model comparison and final model selection in the following section.

4.5 Model Performance Results and Comparison

4.5.1 ML performance

After hyperparameter tuning, the four classifiers were evaluated. Performance was assessed across 6 metrics (Accuracy, Precision, Recall, F1-score, AP, and ROC-AUC) at the F1-optimal decision threshold identified independently for each model. The complete results are presented in Table 19, and the ROC and PR curves for all models are shown in Figure 25.

***Table 19**: ML model's performance metrics at the optimal threshold.*

Model	Accuracy	AP (95% CI)	ROC-AUC (95% CI)	Threshold	Class	Precision	Recall	F1-score
LR	0.60	0.508 (0.432–0.603)	0.695 (0.641–0.752)	0.295	No flare	0.85	0.48	0.62
					Flare	0.44	0.83	0.58
SVM	0.51	0.470 (0.395–0.562)	0.639 (0.581–0.697)	0.288	No flare	0.88	0.30	0.45
					Flare	0.39	0.92	0.55
RF	0.54	0.414 (0.354–0.501)	0.646 (0.591–0.700)	0.376	No flare	0.87	0.38	0.53
					Flare	0.41	0.88	0.56
XGBoost	0.56	0.450 (0.380–0.540)	0.666 (0.613–0.719)	0.342	No flare	0.87	0.40	0.54
					Flare	0.42	0.88	0.57

In Table 19, a consistent pattern emerged across all four classifiers, where high recall for the flare class was achieved at the cost of lower precision and specificity. LR achieved a flare recall of 0.83 and a specificity of 0.48 at a threshold of 0.295, with its flare F_1-score of 0.58 representing the highest among the classifiers. SVM achieved the highest raw flare recall of 0.92 at a threshold of 0.288, but at the greatest cost to precision (0.39) and specificity (0.30), consequently generating the highest number of FP. Both RF and XGBoost achieved a flare recall of 0.88, with XGBoost maintaining a slightly cleaner majority-class profile, showing higher precision (0.42 vs. 0.41) and specificity (0.40 vs. 0.38). These low overall specificity and accuracy values reflect a deliberate clinical decision; the selected operational thresholds explicitly sacrifice majority-class regulation to maximize the sensitivity of flare detection. As detailed in Section 3.5.5, raw accuracy is not an appropriate performance criterion for this highly imbalanced task.

Figure 25 presents the ROC curves (left) and PR curves (right) for all classifiers. On the ROC-curve plot, LR demonstrated the strongest overall discriminative ability, achieving a ROC-AUC of 0.695 (95% CI: 0.641-0.752) and maintaining a consistently higher TPR relative to FPR across most of the operating range. XGBoost followed with ROC-AUC=0.666 (95% CI: 0.613-0.719), RF with 0.646 (95% CI: 0.591-0.700), and SVM with the weakest ROC-AUC of 0.639 (95% CI: 0.581-0.698). All four models outperformed the random classifier baseline (AUC=0.500), confirming that each classifier captured predictive signal. The PR curves provide a more clinically informative perspective given the dataset's class imbalance, where the no information baseline corresponds to a horizontal line at the true flare rate (AP=0.331). All four models exceeded this baseline, confirming again their predictive utility beyond chance. LR demonstrated the clearest advantage, with its PR curve maintaining higher precision across the middle recall range (approximately 0.1-0.6), consistent with its highest AP of 0.508 (95% CI: 0.432-0.603). XGBoost (AP=0.450, 95% CI: 0.380-0.540) and SVM (AP=0.470, 95% CI: 0.395-0.562) showed overlapping curves in the mid-recall region, while RF (AP=0.414, 95% CI: 0.354-0.501) exhibited the weakest PR trade-off, dropping toward the baseline more rapidly at higher recall values. This jagged appearance of the PR curves reflects the relatively small positive class size (n=130 flare visits), where individual threshold crossings produce visible step changes in precision and recall.

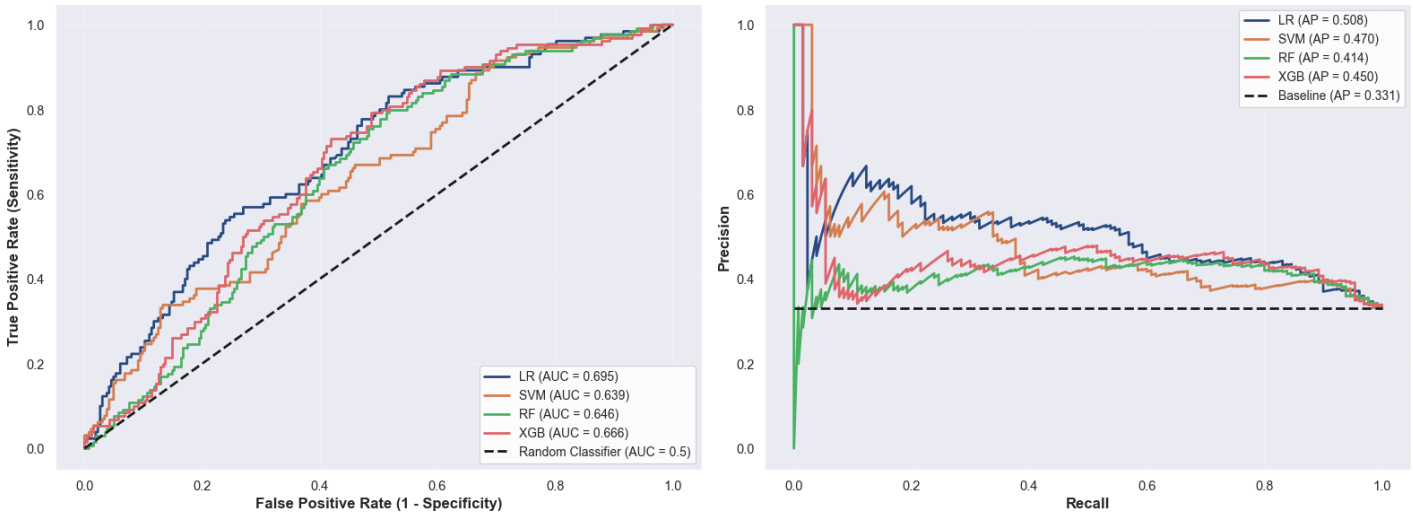


***Figure 25**: ROC (left) and PR (right) curves across all modeling pipelines.*

In addition,

Figure **26** presents the calibration plots for all classifiers. The no-skill baseline (always predicting the true flare rate) yields a Brier score of approximately 0.221. LR demonstrated the best calibration among all classifiers with a Brier score of 0.201 and a calibration curve that closely followed the diagonal across the observable probability range. The predicted probabilities were well-anchored around the true flare rate of 0.331, consistent with the mean predicted probability of 0.330 reported in Section 4.4. A minor overestimation of risk was observed at lower predicted probability values (between 0.20 and 0.30), where the model assigned a slightly higher flare risk than the true observed frequency warranted, though this localized deviation was modest.

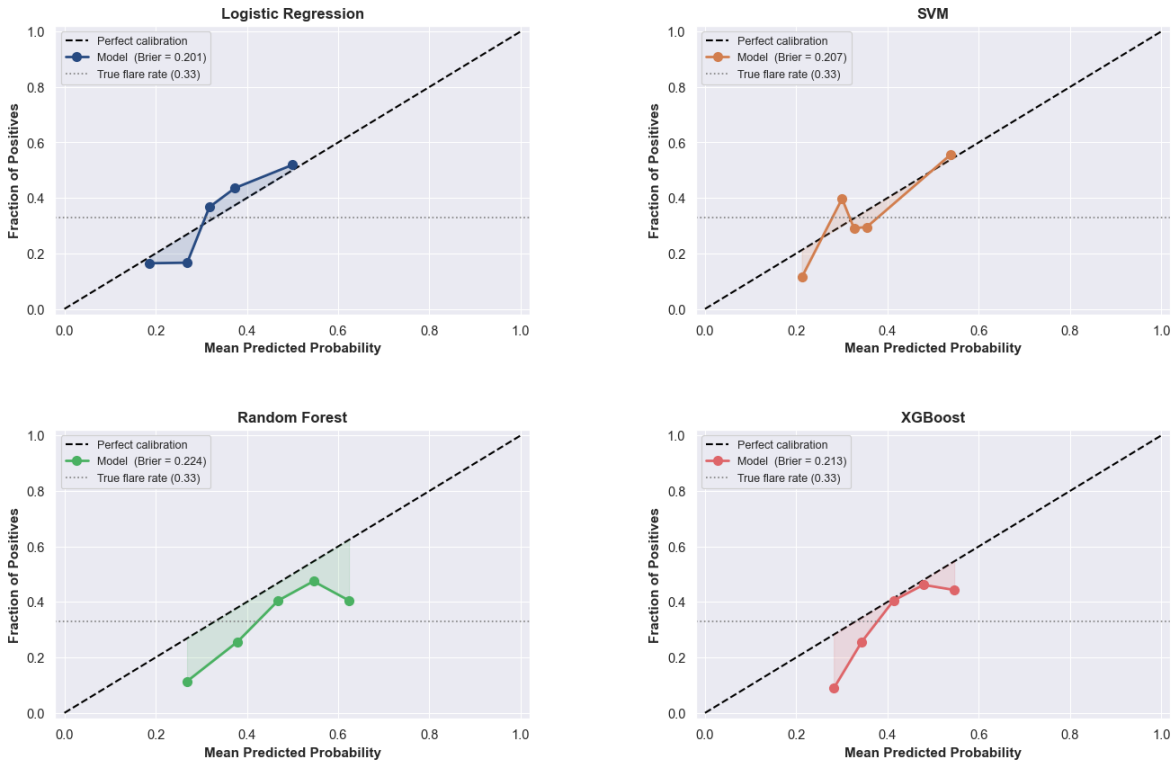


***Figure 26**: Combined calibration plots for LR (top-left), SVM (top-right), RF (bottom-left), XGBoost (bottom-right).*

SVM achieved a competitive Brier score of 0.207, but its calibration curve exhibited structural irregularity and a distinct 'zigzag' pattern across the probability space. This reflects the less stable probability outputs associated with kernel-based models and the variable kernel selection across folds. The tree-based ensembles displayed a vulnerability to probability distortion. RF exhibited the poorest overall calibration, generating a Brier score of 0.224, while XGBoost achieved a Brier score of 0.213. Both curves were positioned entirely under the diagonal line across their entire predicted probability span, signifying a persistent underestimation of flare risks. Overall, three of the four classifiers successfully outperformed the no-skill baseline threshold of 0.221 (RF was the sole exception).

4.5.2 LLM Performance and Comparative Cross-Evaluation

To complement the ML classifiers, an LLM-based classification experiment was conducted using Claude Haiku. Four configurations were evaluated. Two prompting strategies (zero-shot and few-shot) crossed with two serialization formats (List and Text template). The

complete performance metrics are presented in Table 20 and visualized and compared with the ML models in Figure 27.

***Table 20**: LLM performance metrics.*

Strategy	Accuracy	AP (95% CI)	ROC-AUC (95% CI)	Recall (Flare)	Precision (Flare)	Specificity	F1 (Flare)
Zero-shot/List	0.62	0.41 [0.35-0.47]	0.63 [0.58-0.68]	0.64	0.45	0.62	0.53
Zero-shot/Text	0.63	0.40 [0.34-0.46]	0.62 [0.57-0.67]	0.61	0.45	0.63	0.52
Few-shot/List	0.64	0.42 [0.35-0.48]	0.64 [0.59-0.68]	0.64	0.46	0.63	0.54
Few-shot/Text	0.61	0.40 [0.34-0.46]	0.61 [0.56-0.66]	0.62	0.44	0.61	0.51

Across all four LLM configurations, a consistent pattern emerged that differed markedly from the ML classifiers. While the ML models exhibited low specificity at their optimal thresholds (reflecting prioritization of flare recall at the expense of FP), the LLM demonstrated substantially higher specificity (0.61-0.63) at the cost of reduced flare recall (0.61-0.64). This trade-off indicates that the LLM calibrated with an explicit base rate prior of 31% flare frequency, adopted a more conservative prediction strategy than the ML classifiers, flagging fewer visits as high-risk but more accurately identifying stable patients. This behavioral difference reflects the fundamental distinction between the two approaches: the ML classifiers were threshold-optimized to maximize F1 on the flare class, while the LLM produced hard binary predictions without an adjustable threshold.

Comparing zero-shot and few-shot configurations, the impact of context exemplars was highly dependent on the serialization format. For the List template, providing six in-context examples yielded a modest performance gain, shifting the AP from 0.41 to 0.42 and the ROC-AUC from 0.63 to 0.64. Conversely, for the Text template, the introduction of few-shot examples led to a slight degradation in performance: the AP stagnated at 0.40, while the

ROC-AUC decreased from 0.62 to 0.61, and overall accuracy dropped from 0.63 to 0.61. This variance suggests that while in-context examples can partially calibrate the model's predictions toward cohort-specific trends, six examples are structurally insufficient to reliably override the pre-trained semantic biases inherent in dense, narrative text descriptions.

Regarding List and Text formats, they showed that their performance was almost identical in the zero-shot setting, with the List template ahead by a marginal 0.01 in both AP and ROC-AUC. However, the gap widened in the few-shot configuration: the List template achieved an AP of 0.42 and a ROC-AUC of 0.64, while the Text template dropped to an AP of 0.40 and a ROC-AUC of 0.61. This 0.03 difference indicates that structured, bullet lists are less prone to losing important details (context dilution) and work better for showing few-shot examples to the model than long, paragraph-style clinical stories.

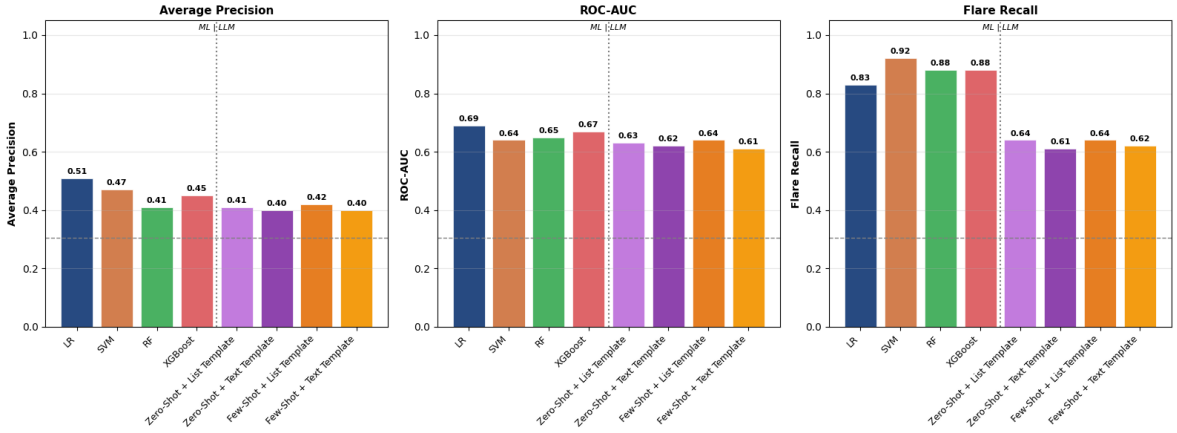


***Figure 27**: ML models vs LLM zero/few-shot classification.*

Comparing LLM configurations against the tuned ML classifiers, all four LLM strategies underperformed on AP ranging through 0.40-0.42 compared to 0.41-0.51 for the ML models. The gap was largest relative to LR (ΔAP=0.09-0.11) and smallest relative to RF, where the LLMs achieved nearly identical performance. ROC-AUC differences were similarly consistent, with LLM configurations achieving 0.61-0.64 compared to 0.64-0.69 for ML models. These results indicate that ML classifiers trained on patient longitudinal data provide superior discriminative performance for this prediction task compared to general medical knowledge applied in zero-shot or few-shot paradigm. The ML models' advantage is attributed to their direct learning of cohort-specific statistical patterns that cannot be

replicated through pretraining on clinical literature alone. Notably, the LLMs achieved higher overall accuracy (0.61–0.64) than the ML models (0.51–0.60), which directly reflects their higher specificity. However, because accuracy is misleading for this imbalanced dataset, this specific advantage does not translate to true clinical superiority.

Among all evaluated approaches, LR remained the best-performing model overall. It achieved the highest AP (0.508) and ROC-AUC (0.695), the best probability calibration (Brier=0.201, mean predicted probability=0.330), the highest F1-score (0.57), and complete hyperparameter stability across all 5 outer fold CV folds.

Figure **28** presents the confusion matrix for the LR model at the optimal threshold of 0.295. Of the 130 flare visits, 108 were correctly identified as flares (TP=108) and 22 were missed (FN=22), meaning that the model successfully detected 83% of all actual flares. Of the 263 non-flare visits, 127 were correctly classified as non-flares (TN=127) and 136 were incorrectly flagged as flares (FP=136). The high FP count shows that approximately 52% of non-flare cases were incorrectly flagged, reflecting the sacrifice of specificity to achieve high flare detection sensitivity.

In clinical practice, a FP rate of 52% would be unsustainable if each flagged visit triggered an immediate high-resource intervention such as an unscheduled rheumatology consultation or immunosuppressive escalation, risking both operational strain and clinician alert fatigue. The model is therefore not intended as an autonomous diagnostic trigger but as a risk-stratification filter within a two-tiered triage protocol. Patients flagged as high-risk would first undergo a low-cost remote assessment (e.g. a brief nurse telephone review or a digital patient-reported outcome questionnaire) to distinguish genuine deterioration from false alarms before any clinical escalation is initiated. This secondary screening step preserves the model's high flare sensitivity while substantially reducing unnecessary clinical burden. Crucially, this decision framework remains highly flexible. In resource-constrained clinics, the decision threshold can be raised to decrease the number of false alarms and ease physician workload, even if it means accepting a slightly higher rate of missed flares. Because the consequences of a missed flare generally outweigh the administrative cost of a brief telephone follow-up, this tunable threshold allows individual clinics to optimize the sensitivity-specificity trade-off based on their specific operational capacity.

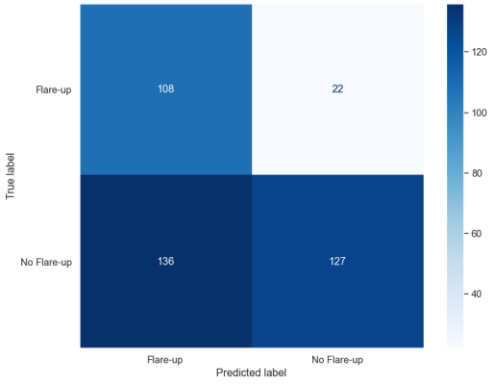


***Figure 28**: Confusion matrix of LR.*

To visually inspect the discriminative power of the LR model, Figure 29 presents the predicted probability distribution by class. The distributions of flare and no flare visits show considerable overlap in the 0.15-0.4 probability range, reflecting the difficulty of the prediction task and the modest but genuine separation between classes. The flare class distribution is shifted rightward relative to the no-flare class, with a heavier tail, extending toward higher probabilities (0.40-0.70), indicating that the model assigns meaningfully higher risk scores to patients who experienced a flare. The no-flare class is more concentrated in the lower probability range (0.10-0.35), with few observations assigned probabilities above 0.5. This distribution separation confirms that the model captures a genuine signal and is not assigning arbitrary probability scores. The overlap between distributions is consistent with the clinical reality that RA flares are difficult to predict with certainty from a single visit's clinical measurements, particularly in small datasets.

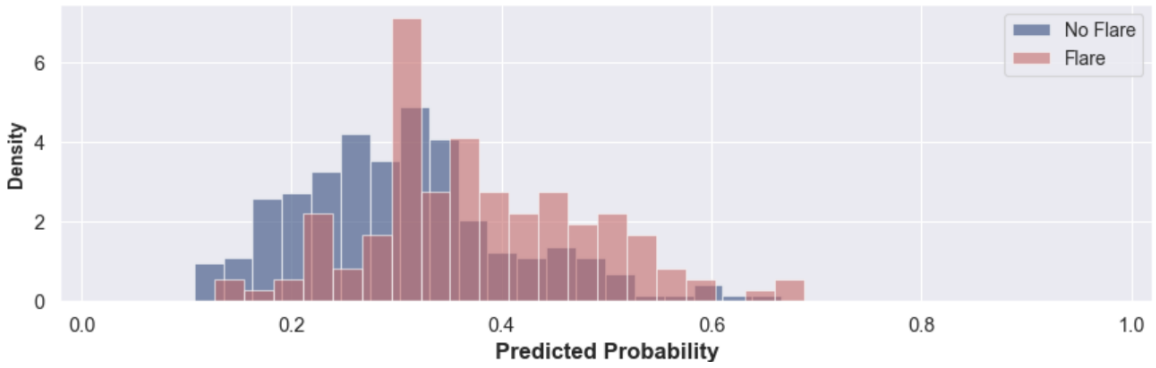


***Figure 29**: Predicted probability distribution by class.*

Lastly, to evaluate the relationship between sample size scaling and model generalization, a learning curve diagnostic was performed on the LR pipeline (

Figure **30**). At small training sizes (approximately 30-60 visits), the training score was moderately elevated (~0.45-0.50) while the CV score was lower and more variable, with wide confidence bands reflecting high instability due to insufficient training data. As the number of visits increased, both curves showed a convergence pattern. The training score gradually declined from its peak of approximately 0.61 at around 100 instances, while the CV score rose incrementally, narrowing the gap between the two. At the maximum available training size of around 315 visits (80% training partition of the 5-fold CV framework), both curves converge to approximately AP=0.55, with substantially reduced confidence band width indicating improved stability. This diminishing gap shows that overfitting is well controlled, while it is suggested that additional training data would likely yield further performance improvements. The fact that the CV curve had not yet plateaued at the maximum available training size is particularly noteworthy, as it implies that the model had not yet reached its performance ceiling and that a larger patient cohort may produce higher predictive performance.

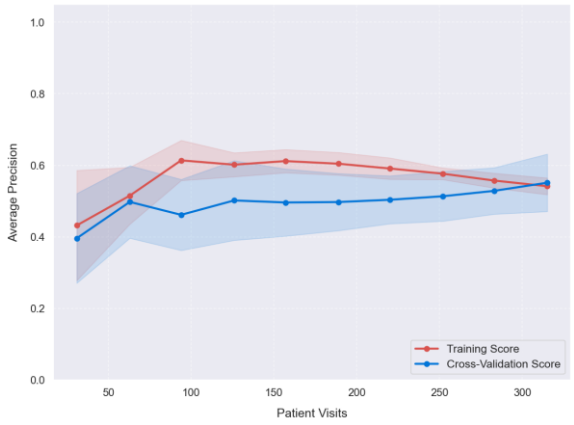


***Figure 30**: Learning curve for the LR model.*

The following section examines the LR model in detail, analyzing the clinical meaning of its learned prediction equation and the contribution of each retained feature to next-visit flare risk.

4.6 Best Model Interpretation

The interpretability of the selected LR model was examined through analysis of its non-zero coefficients, which directly quantify the contribution of each retained feature to the log-odds of next-visit flare. Elastic Net regularization with l1_ratio=0.5 and C=0.05 reduced the feature input space to five non-zero predictors, providing a sparse and clinically interpretable prediction equation. The coefficients and their corresponding odds ratios are presented in Table 21.

***Table 21**: Odds ratio table.*

Feature	Coefficient (β)	Odds Ratio (OR)	Direction
DAS28 score	+0.511	1.667	↑ Risk
Log ESR	+0.053	1.054	↑ Risk
Disease duration squared	+0.031	1.031	↑ Risk
Log CRP	+0.028	1.028	↑ Risk
Age at visit	-0.023	0.977	↓ Risk

Note: Coefficients are on the standardized scale.

The log odds of next-visit flare can be expressed as:

$$\ln\left(\frac{p}{1-p}\right) = -2.015 + 0.511(das28_score) + 0.053(\log_esr) + 0.031(\text{disease_duration}^2) + 0.028(\log_crp_normalized) - 0.023(age_at_visit) \quad (8)$$

The model yielded an intercept (β_0) of -2.015. DAS28 score carried the dominant predictive weight, with a standardized coefficient of 0.511, which is approximately ten times larger than the next stronger predictor, demonstrating that existing clinical disease activity is the primary clinical driver of near-future disease acceleration. A one standard deviation increase in DAS28 score was associated with a 1.67-fold increase in the odds of next-visit flare (OR=1.667). Log-transformed ESR emerged as the second most powerful predictor (β=0.053, OR=1.054). Crucially, log-transformed normalized CRP (β=0.031, OR=1.031) contributed predictive value entirely independent of ESR. This mathematical independence validates the clinical decision to retain both variables during feature selection; because ESR

reflects slow, sustained changes in systemic plasma proteins and CRP responds rapidly to acute inflammatory changes, the model leverages both timelines simultaneously to track inflammatory velocity. Disease duration squared contributed to a positive coefficient, capturing the non-linear relationship between disease chronicity and flare susceptibility. Log-transformed normalized CRP contributed independently of ESR (β=0.028, OR=1.028), consistent with its partial but non-redundant overlap with ESR as an inflammatory marker. Age at visit was the only feature with a negative coefficient (β=-0.023, OR=0.977. This indicates that older age at the time of clinical assessment is marginally associated with a reduced probability of acute flaring once current disease activity and laboratory inflammatory loads have been statistically controlled.

5 Conclusions and Future Work

This thesis was conducted to develop and evaluate a ML framework for predicting disease flare-ups at the next recorded clinical visit in patients with RA, using routine clinical variables collected during standard outpatient care. This study was motivated by the dual observation that existing predictive models in rheumatology have predominantly been developed on demographic narrow cohorts and controlled trial data, and that a clinically applicable flare prediction tool applicable to real-world remains absent from the literature, particularly for Greek and Southern European patient populations. Additionally, the lack of comparative benchmarks between task-specific supervised learning and foundation LLMs left open the question of whether pretrained clinical knowledge could substitute for localized data training. To address these gaps, this analytical pipeline was designed to operationalize and answer three core research questions, evaluating four traditional ML classifiers alongside zero-shot and few-shot LLM configurations.

To achieve this objective, a retrospective longitudinal dataset of 106 patients was compiled from a dual-setting rheumatology cohort and evaluated through an end-to-end analytical pipeline. Various techniques were utilized such as preprocessing, feature engineering, and feature selection, culminating in a final feature set of 18 variables spanning disease activity, inflammatory markers, treatment characteristics, comorbidity burden, and patient demographics. Four imputation strategies were evaluated for the three disease activity variables with elevated missingness, with MICE-Bayesian imputation selected for each classifier. Four ML classifiers (LR, SVM, RF, and XGBoost) were trained and evaluated under a nested CV framework with Stratified Group K-fold splitting, ensuring that patient-level data leakage was eliminated throughout all stages of model development and evaluation. Complementarily, Claude Haiku was evaluated under zero-shot and few-shot inference paradigms using two serialization formats and tested across aligned CV partitions, providing a direct comparison between ML and LLM configurations.

Among all the classifiers, LR achieved the strongest overall performance, with an AP of 0.508 and ROC-AUC of 0.695, alongside complete hyperparameter stability across CV folds and the best probability calibration (Brier score=0.201). The finding that a linear model outperformed ensemble tree-based methods is consistent with the dataset's characteristics. In

this small longitudinal cohort, the primary predictive signal is driven by linear relationships among a few continuous clinical variables, lacking the non-linear complexity required to justify the higher learning capacity of RF and XGBoost. At the F1-optimal decision threshold of 0.295, the model correctly identified 83% of flare visits while having a specificity of 48%. While a 52% false-positive rate risks clinician alert fatigue, this trade-off is operationally manageable. By utilizing the model as an early-stage risk filter within a two-tiered triage protocol, where high-risk flags are first cleared via low-cost remote assessments, clinics can capture most impending patient crises while shielding clinical resources. Furthermore, the decision threshold remains highly tunable, allowing resource-constrained clinical settings to raise the operating cutoff to prioritize workflow efficiency over maximum sensitivity.

The interpretability analysis of the LR model revealed a five-variable prediction equation in which DAS28 score carried dominant weight (OR=1.667), indicating that a one standard deviation elevation in current disease activity was associated with a 66.7% increase in the odds of experiencing a flare at the next visit. As a comprehensive snapshot of joint and systemic burden, its heavy weighting reinforces the clinical principle that patients presenting with higher disease activity at a given visit are substantially more likely to deteriorate further in the short term. Log-transformed ESR contributed independently (OR=1.054), suggesting residual inflammatory signal beyond that captured by DAS28, potentially reflecting sustained systemic inflammation not yet evident in joint assessments. Disease duration squared (OR=1.031) captured a non-linear relationship, confirming that both early and long-standing disease contribute elevated risk through distinct mechanisms and validating the engineered quadratic term. Log-transformed CRP (OR=1.028) also showed an independent positive effect. Its retention alongside ESR, despite moderate correlation, indicates that these markers provide complementary information about inflammation activity. Age at visit associated with a small reduction in risk (OR=0.977), plausibly reflecting more stable disease management, less aggressive phenotypes in older patients. Collectively, the model defines a clinically coherent high-risk profile: patients with high disease activity, elevated ESR and CRP, longer disease duration, and younger age. This alignment with established clinical understanding suggests the model captures meaningful patterns rather than spurious associations. The learning curve analysis indicated that model had not yet reached the

performance ceiling at the available training set size, suggesting that performance gains from an expanded patient cohort remain achievable.

The LLM-based classification experiment demonstrated that Claude Haiku underperformed tuned ML classifiers across primary metrics: AP ranged from 0.40-0.42 (vs 0.41-0.51 for ML) and ROC-AUC from 0.61-0.64 (vs. 0.64-0.69). Due to base-rate prompt calibration, LLMs exhibited higher specificity (0.61-0.63) but lower flare recall (0.61-0.64) than the ML models. Interestingly, the impact of context exemplars depended entirely on layout format; few-shot prompting marginally enhanced performance for the structured List template (moving ROC-AUC from 0.63 to 0.64) but degraded performance across all primary metrics for the narrative Text template, dropping its accuracy, ROC-AUC, and F1-score. These outcomes underscore that pretrained general clinical knowledge cannot fully substitute for patient-specific supervised training. However, the LLMs' higher accuracy (0.61–0.64) relative to the ML models (0.51–0.60) confirms that they offer a stable, training-free baseline for preliminary clinical assessments when labeled data is completely unavailable.

Several limitations of this study must be acknowledged. The primary constraint is the sample size. 106 patients represent a small longitudinal cohort that restricts statistical power, contributes to hyperparameter instability across CV folds and limits the complexity of the models that can be reliably trained. The high FP rate observed at the best model's optimal threshold (136 of 263 non-flare visits incorrectly flagged) reflects both this data limitation and the inherent difficulty of flare prediction from routine clinical measurements. Additionally, the target variable, a physician recorded flare at the next visit, spans a chronologically variable timeframe due to irregular visit intervals. It also may be subjective, introducing label noise that constrains achievable predictive performance. Longitudinal feature engineering, while methodological sound, produced features that could not be computed for first visits, and the relatively low average visit count per patient (4.7) limited the reliability of temporal derivatives. For the LLM experiment, the evaluation was limited to a single commercial model version (claude-haiku-4-5-20251001) and two prompting strategies. Consequently, performance characteristics may vary across alternative frontier model families, larger parameter scales, or open-source architectures, particularly since no formal prompt optimization frameworks or iterative prompt engineering techniques were applied.

A few directions for future research follow naturally from these findings. The most impactful extension would be the expansion of the patient cohort through multi-center data collection within Greek and broader Southern European rheumatology networks, which would address the sample size limitation, and enable more complex modeling architectures. Prospective validation of the LR model on an independent held-out cohort would provide the external validity required before clinical deployment could be considered. Future work could additionally explore RNN architectures or transformer-based sequence models capable of modelling the full temporal trajectory of individual patients, which the visit frequency in the current dataset did not support. Regarding the LLM approach, future work could evaluate fine-tuned clinical language models trained on larger RA datasets, assess open-source locally deployable alternatives that address the data privacy concerns associated with external API transmission, and investigate whether retrieval-augmented generation could further bridge the performance gap with task specific supervised classifiers.

Beyond predicting next-visit flares, this longitudinal dataset could be utilized for alternative ML tasks. First, unsupervised learning techniques, such as clustering algorithms, could be applied to discover hidden patient phenotypes based on their clinical trajectories. This would help identify distinct sub-populations, such as patients with stable, low-grade inflammation versus those with highly volatile, relapsing disease patterns. Second, the analytical focus could expand from binary classification to time-to-event (survival) modeling. Implementing frameworks like Cox Proportional Hazards would allow the pipeline to predict when a flare is likely to occur, rather than restricting the prediction to the next scheduled visit. Finally, regression models could be trained to forecast continuous changes in disease activity scores, providing clinicians with a granular, quantitative warning of disease progression rather than a simple yes-or-no flag.

In conclusion, this study successfully addresses its core objectives by demonstrating that an interpretable proof-of-concept flare prediction model for RA can be developed from routine outpatient clinical data, achieving meaningful predictive capacity within a real-world longitudinal architecture (RQ1). The findings demonstrate that a sparse five variable LR equation (RQ2) reinforces the clinical primacy of disease activity monitoring as the cornerstone of flare risk assessment. Furthermore, the comparison reveals that while pretrained LLM knowledge cannot substitute for patient-specific longitudinal training on

primary metrics, it offers a stable, training-free baseline with unique layout-dependent trade-offs (RQ3). While sample size constraints limit the strength of conclusions that can be drawn, the exploratory framework provides a replicable and extensible foundation for future predictive modeling efforts in rheumatological clinical practice.

6 References

[1] J. Bullock *et al.*, "Rheumatoid Arthritis: A Brief Overview of the Treatment," *Medical Principles and Practice*, vol. 27, no. 6, pp. 501–507, Sep. 2018, doi: 10.1159/000493390.

[2] D. Aletaha and J. S. Smolen, "Diagnosis and Management of Rheumatoid Arthritis: A Review," *JAMA*, vol. 320, no. 13, pp. 1360–1372, Oct. 2018, doi: 10.1001/jama.2018.13103.

[3] M. H. Smith and J. R. Berman, "What Is Rheumatoid Arthritis?," *JAMA*, vol. 327, no. 12, p. 1194, Mar. 2022, doi: 10.1001/jama.2022.0786.

[4] A. Vodencarevic *et al.*, *Predicting Flare Probability in Rheumatoid Arthritis using Machine Learning Methods*. 2018, p. 192. doi: 10.5220/0006930501870192.

[5] Y. Shi *et al.*, "Advancing precision rheumatology: applications of machine learning for rheumatoid arthritis management," *Front. Immunol.*, vol. 15, Jun. 2024, doi: 10.3389/fimmu.2024.1409555.

[6] F. Z. Rahali, M. Tarmidi, R. Hazime, and B. Admou, "Clinical significance of anti-cyclic citrullinated peptide (anti-CCP) antibodies in rheumatoid arthritis: Literature review," *SN Compr. Clin. Med.*, vol. 5, no. 1, p. 272, Nov. 2023, doi: 10.1007/s42399-023-01613-x.

[7] I. Vos *et al.*, "Anti-citrullinated protein antibodies in the diagnosis of rheumatoid arthritis (RA): diagnostic performance of automated anti-CCP-2 and anti-CCP-3 antibodies assays," *Clin Rheumatol*, vol. 36, no. 7, pp. 1487–1492, Jul. 2017, doi: 10.1007/s10067-017-3684-8.

[8] S. C. Shapiro, "Biomarkers in Rheumatoid Arthritis," *Cureus*, vol. 13, no. 5, p. e15063, May 2021, doi: 10.7759/cureus.15063.

[9] P. L. Van Riel and L. Renskers, "The Disease Activity Score (DAS) and the Disease Activity Score using 28 joint counts (DAS28) in the management of rheumatoid arthritis," *Clin Exp Rheumatol*, vol. 34, no. 5 Suppl 101, pp. S40–S44, 2016.

[10] L. Siemons, H. E. Vonkeman, P. M. ten Klooster, P. L. C. M. van Riel, and M. A. F. J. van de Laar, "Interchangeability of 28-joint disease activity scores using the erythrocyte sedimentation rate or the C-reactive protein as inflammatory marker," *Clin Rheumatol*, vol. 33, no. 6, pp. 783–789, Jun. 2014, doi: 10.1007/s10067-014-2538-x.

[11] B. Norgeot *et al.*, "Assessment of a Deep Learning Model Based on Electronic Health Record Data to Forecast Clinical Outcomes in Patients With Rheumatoid Arthritis," *JAMA Netw Open*, vol. 2, no. 3, p. e190606, Mar. 2019, doi: 10.1001/jamanetworkopen.2019.0606.

[12] L. Gumber *et al.*, "Patient-reported outcomes as early warning signs of flare following drug cessation in rheumatoid arthritis," *RMD Open*, vol. 11, no. 2, Apr. 2025, doi: 10.1136/rmdopen-2025-005442.

[13] J. S. Smolen *et al.*, "EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2022 update," *Annals of the Rheumatic Diseases*, vol. 82, no. 1, pp. 3–18, Jan. 2023, doi: 10.1136/ard-2022-223356.

[14] A.-F. Radu and S. G. Bungau, "Management of Rheumatoid Arthritis: An Overview," *Cells*, vol. 10, no. 11, p. 2857, Nov. 2021, doi: 10.3390/cells10112857.

[15] H. Luan *et al.*, "Serum metabolomic and lipidomic profiling identifies diagnostic biomarkers for seropositive and seronegative rheumatoid arthritis patients," *Journal of Translational Medicine*, vol. 19, no. 1, p. 500, Dec. 2021, doi: 10.1186/s12967-021-03169-7.

[16] A. Alarcón-Paredes *et al.*, "Computer-aided diagnosis based on hand thermal, RGB images, and grip force using artificial intelligence as screening tool for rheumatoid arthritis in women," *Med Biol Eng Comput*, vol. 59, no. 2, pp. 287–300, Feb. 2021, doi: 10.1007/s11517-020-02294-7.

[17] M. Wu *et al.*, "A deep learning classification of metacarpophalangeal joints synovial proliferation in rheumatoid arthritis by ultrasound images," *J Clin Ultrasound*, vol. 50, no. 2, pp. 296–301, Feb. 2022, doi: 10.1002/jcu.23143.

[18] K. Üreten, H. Erbay, and H. H. Maraş, "Detection of rheumatoid arthritis from hand radiographs using a convolutional neural network," *Clin Rheumatol*, vol. 39, no. 4, pp. 969–974, Apr. 2020, doi: 10.1007/s10067-019-04487-4.

[19] K.-J. Kim and I. Tagkopoulos, "Application of machine learning in rheumatic disease research," *Korean J Intern Med*, vol. 34, no. 4, pp. 708–722, Jul. 2019, doi: 10.3904/kjim.2018.349.

[20] M. Kalweit *et al.*, "Personalized prediction of disease activity in patients with rheumatoid arthritis using an adaptive deep neural network," *PLOS ONE*, vol. 16, no. 6, p. e0252289, Jun. 2021, doi: 10.1371/journal.pone.0252289.

[21] A. R. Alsaber *et al.*, "Machine learning-based remission prediction in rheumatoid arthritis patients treated with biologic disease-modifying anti-rheumatic drugs: findings from the Kuwait rheumatic disease registry," *Front. Big Data*, vol. 7, Oct. 2024, doi: 10.3389/fdata.2024.1406365.

[22] A. Vodencarevic *et al.*, "Advanced machine learning for predicting individual risk of flares in rheumatoid arthritis patients tapering biologic drugs," *Arthritis Res Ther*, vol. 23, no. 1, p. 67, Feb. 2021, doi: 10.1186/s13075-021-02439-5.

[23] J. Haschka *et al.*, "Relapse rates in patients with rheumatoid arthritis in stable remission tapering or stopping antirheumatic therapy: interim results from the prospective randomised controlled RETRO study," *Annals of the Rheumatic Diseases*, vol. 75, no. 1, pp. 45–51, Jan. 2016, doi: 10.1136/annrheumdis-2014-206439.

[24] L. J. O'Neil *et al.*, "Proteomic Approaches to Defining Remission and the Risk of Relapse in Rheumatoid Arthritis," *Front Immunol*, vol. 12, p. 729681, 2021, doi: 10.3389/fimmu.2021.729681.

[25] H. Labinsky *et al.*, "An AI-Powered Clinical Decision Support System to Predict Flares in Rheumatoid Arthritis: A Pilot Study," *Diagnostics*, vol. 13, no. 1, p. 148, 2023, doi: 10.3390/diagnostics13010148.

[26] L. Gossec *et al.*, "Detection of Flares by Decrease in Physical Activity, Collected Using Wearable Activity Trackers in Rheumatoid Arthritis or Axial Spondyloarthritis: An Application of Machine Learning Analyses in Rheumatology," *Arthritis Care & Research*, vol. 71, no. 10, pp. 1336–1343, Oct. 2019, doi: 10.1002/acr.23768.

[27] J. Gandrup, D. A. Selby, and W. G. Dixon, "Classifying Self-Reported Rheumatoid Arthritis Flares Using Daily Patient-Generated Data From a Smartphone App: Exploratory Analysis Applying Machine Learning Approaches," *JMIR Formative Research*, vol. 8, no. 1, p. e50679, May 2024, doi: 10.2196/50679.

[28] L. Austin *et al.*, "Providing 'the bigger picture': benefits and feasibility of integrating remote monitoring from smartphones into the electronic health record: Findings from the Remote Monitoring of Rheumatoid Arthritis (REMORA) study," *Rheumatology (Oxford)*, vol. 59, no. 2, pp. 367–378, Feb. 2020, doi: 10.1093/rheumatology/kez207.

[29] Scikit-learn developers, "sklearn.model_selection.StratifiedGroupKFold," scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html

[30] D. Chicco, A. Sichenze, and G. Jurman, "A simple guide to the use of Student's t-test, Mann-Whitney U test, Chi-squared test, and Kruskal-Wallis test in biostatistics," *BioData Min*, vol. 18, p. 56, Aug. 2025, doi: 10.1186/s13040-025-00465-6.

[31] O. Troyanskaya *et al.*, "Missing value estimation methods for DNA microarrays," *Bioinformatics*, vol. 17, no. 6, pp. 520–525, Jun. 2001, doi: 10.1093/bioinformatics/17.6.520.

[32] D. Mongin *et al.*, "Imputing missing data of function and disease activity in rheumatoid arthritis registers: what is the best technique?," *RMD Open*, vol. 5, no. 2, p. e000994, 2019, doi: 10.1136/rmdopen-2019-000994.

[33] D. Gianola, "Priors in whole-genome regression: the bayesian alphabet returns," *Genetics*, vol. 194, no. 3, pp. 573–596, Jul. 2013, doi: 10.1534/genetics.113.151753.

[34] L. L. Doove, S. Van Buuren, and E. Dusseldorp, “Recursive partitioning for missing data imputation in the presence of interaction effects,” *Computational Statistics & Data Analysis*, vol. 72, pp. 92–104, Apr. 2014, doi: 10.1016/j.csda.2013.10.025.

[35] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, “AI in health and medicine,” *Nat Med*, vol. 28, no. 1, pp. 31–38, Jan. 2022, doi: 10.1038/s41591-021-01614-0.

[36] “Introduction to the Logistic Regression Model,” in *Applied Logistic Regression*, John Wiley & Sons, Ltd, 2013, pp. 1–33. doi: 10.1002/9781118548387.ch1.

[37] E. W. Steyerberg, *Clinical Prediction Models*. in Statistics for Biology and Health. New York, NY: Springer, 2009. doi: 10.1007/978-0-387-77244-8.

[38] H. Zou and T. Hastie, “Regularization and Variable Selection Via the Elastic Net,” *J. R. Stat. Soc. Ser. B. Stat. Methodol.*, vol. 67, no. 2, pp. 301–320, Apr. 2005, doi: 10.1111/j.1467-9868.2005.00503.x.

[39] C. Cortes and V. Vapnik, “Support-vector networks,” *Mach Learn*, vol. 20, no. 3, pp. 273–297, Sep. 1995, doi: 10.1007/BF00994018.

[40] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, Oct. 2001, doi: 10.1023/A:1010933404324.

[41] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, in KDD ’16. New York, NY, USA: Association for Computing Machinery, Aug. 2016, pp. 785–794. doi: 10.1145/2939672.2939785.

[42] “LogisticRegression,” scikit-learn. Accessed: Jun. 21, 2026. [Online]. Available: https://scikit-learn/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

[43] “Learning with Support Vector Machines [1 ed.] 3031004248, 9783031004247,” dokumen.pub. Accessed: Jun. 14, 2026. [Online]. Available: https://dokumen.pub/learning-with-support-vector-machines-1nbsped-3031004248-9783031004247.html

[44] “1.4. Support Vector Machines,” scikit-learn. Accessed: Jun. 21, 2026. [Online]. Available: https://scikit-learn/stable/modules/svm.html

[45] “RandomForestClassifier,” scikit-learn. Accessed: Jun. 21, 2026. [Online]. Available: https://scikit-learn/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

[46] “XGBoost Parameters — xgboost 3.3.0 documentation.” Accessed: Jun. 21, 2026. [Online]. Available: https://xgboost.readthedocs.io/en/stable/parameter.html

[47] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, "TabLLM: Few-shot Classification of Tabular Data with Large Language Models," Mar. 17, 2023, *arXiv*: arXiv:2210.10723. doi: 10.48550/arXiv.2210.10723.

[48] K. Ono and S. A. Lee, "Text Serialization and Their Relationship with the Conventional Paradigms of Tabular Machine Learning," Jun. 19, 2024, *arXiv*: arXiv:2406.13846. doi: 10.48550/arXiv.2406.13846.

[49] T. B. Brown *et al.*, "Language Models are Few-Shot Learners," Jul. 22, 2020, *arXiv*: arXiv:2005.14165. doi: 10.48550/arXiv.2005.14165.

[50] "Prompting best practices," Claude Platform Docs. Accessed: Jul. 08, 2026. [Online]. Available: https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/claude-prompting-best-practices

[51] "Claude Haiku." Accessed: Jul. 08, 2026. [Online]. Available: https://www.anthropic.com/claude/haiku

[52] A. A. Vallejo *et al.*, "SAT0099 Polypharmacy is associated with an increased risk of adverse outcomes in patients with rheumatoid arthritis," *Annals of the Rheumatic Diseases*, vol. 76, no. Suppl 2, pp. 806–806, Jun. 2017, doi: 10.1136/annrheumdis-2017-eular.2262.

[53] J. S. Smolen *et al.*, "EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2022 update," *Annals of the Rheumatic Diseases*, vol. 82, no. 1, pp. 3–18, Jan. 2023, doi: 10.1136/ard-2022-223356.

[54] A. A. Drosos, A. A. Venetsanopoulou, E. Pelechas, and P. V. Voulgari, "Exploring Cardiovascular Risk Factors and Atherosclerosis in Rheumatoid Arthritis," *European Journal of Internal Medicine*, vol. 128, pp. 1–9, Oct. 2024, doi: 10.1016/j.ejim.2024.07.016.

7 Appendix

1. System Prompt Context

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You are an expert rheumatologist specializing in Rheumatoid Arthritis. You will be given a clinical visit summary and asked to predict whether the patient will experience a disease flare at their next visit. In this specific clinical cohort, approximately 31% of visits are followed by a disease flare at the next appointment. The majority of patients (69%) remain stable between visits. A disease flare is a clinically meaningful increase in disease activity at the subsequent visit, characterized by worsening joint symptoms and elevated inflammatory markers. You must weigh risk factors against protective factors carefully. Do not default to predicting flare unless the clinical evidence clearly supports it.
Structure your response as:
<prediction>Yes</prediction> or <prediction>No</prediction>
Write only one prediction tag, and do not include any text outside the tags.
```

2. Tabular data serialization formats

2.1 List Template Format

```
Age: 41 years
Disease duration: 25 years
Gender: Female
RF status: Negative
Anti-CCP status: Negative
DAS28 score: 2.04 (remission)
DAS28 change from previous visit: -0.16 (decreased)
CRP: 1.90 mg/L (within normal range; laboratory upper limit: 5.00 mg/L)
ESR: 6.0 mm/hr
MTX dose: Not in use at this visit
Steroid dose: Not in use at this visit
csDMARD use: Yes
b/tsDMARD use: Yes
Recent treatment decision: stable
Follow-up visit. Days since last assessment: 84 days
Flare at previous visit: No
Comorbidities: No known comorbidities
```

2.2 Text Template Format

```
A 41-year-old female patient with 25 years of rheumatoid arthritis, who is RF-negative and anti-CCP-negative. The DAS28 score at this visit is 2.04, indicating remission. This represents a decrease of 0.16 from the previous visit. Inflammatory markers: CRP is 1.90 mg/L
```

(within normal range; laboratory upper limit: 5.00 mg/L). ESR is 6.0 mm/hr. The patient is currently receiving a csDMARD, a b/tsDMARD. The treatment decision at this visit was: stable. This is a follow-up visit, 84 days since the last assessment. At the previous visit, the patient did not have a flare. Comorbidities: No known comorbidities.

3. Zero-shot inference template

```
Clinical Visit Summary:
[Dynamically generated patient serialization profile here]
Based on this clinical profile, will this patient experience a disease flare at their next visit?
Return your answer exactly as:
<prediction>Yes</prediction> or <prediction>No</prediction>
```

4. Few-Shot Context Template

```
Here are 6 example cases with known outcomes:
Example 1:
[Serialized profile of training patient 1]
<prediction>Yes</prediction>
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Example 2:
[Serialized profile of training patient 2]
<prediction>No</prediction>
---
Now predict for this new patient:
Clinical Visit Summary:
[Serialized profile of the active test patient]
Provide your prediction using the XML format specified in the system instructions.
```